Rows and Columns: A Theory of Returns to Data*

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November 2026

Abstract

A firm learns how covariates affect an outcome, then predicts it. Some covariates, like a user's next search, arrive too late to use but help the firm learn. With enough of them, each observation adds more than the last in small samples; without them, each adds less. Each covariate helps the firm learn the others' effects; how fast prediction error falls with sample size measures these spillovers. This reconciles conflicting evidence on returns to data. Suppliers of covariates with rights over data claim more than the data are worth; rights for individuals avoid this unless enough covariates arrive late.

JEL Classification: C55, C78, D23, D24, D83, L41, L86.
Keywords: value of data, returns to data, learning multiplier, post-prediction covariates, data-driven competition, rights over data, anticommons.

*First public draft: "Increasing Returns to Data for a Bayesian Platform," June 2023. I am grateful to Özlem Bedre-Defolie, Jad Beyhum, Michele Bisceglia, Emilio Calvano, Doruk Cetemen, Zhijun Chen, Alexandre de Cornière, Jacques Crémer, Matteo Escudé, Jean-Pierre Florens, Eric Gautier, Johannes Gessner, Andrei Hagiu, Johannes Hörner, Doh-Shin Jeon, Garrett Johnson, Bruno Jullien, Hiroaki Kaido, Gastón Llanes, Simon Loertscher, Friedrich Lucke, Leonardo Madio, Thierry Magnac, Nour Meddahi, Giovanni Morzenti, Juan Ortner, Alessio Ozanne, Patrick Rey, Andrew Rhodes, Christoph Riedl, François Salanié, Enrico Mattia Salonia, Maximilian Schäfer, Sara Shahanaghi, Tim Simcoe, Alex Smolin, Marshall Van Alstyne, Nikhil Vellodi, and Davide Viviano for helpful comments and discussions. I thank seminar participants at the Toulouse School of Economics IO workshop (2022–2026), the 2022 CRC TR 224–MaCCI Summer School on Platform Economics, Boston University (Micro Theory Workshop and Questrom Platforms Seminar, 2025), EARIE 2025 (Valencia), and LUISS (2026). I gratefully acknowledge financial support from the Horizon Europe – European Research Council (ERC) under Grant Agreement No. 101055280-DIGECON.

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1 Introduction

Do additional data improve a firm's predictions more when the firm already holds more data? If so, returns to data increase: a data-rich firm values additional data more and will pay more for them, so its lead grows. Policymakers see data concentrated in a few firms as a threat to competition and to democracy.[1] Yet the evidence on returns to data appears contradictory. Demand forecasts and image classifiers improve at a diminishing rate with the number of observations they learn from, and the UK Competition and Markets Authority (CMA) found that the returns to click-and-query data diminish rapidly.[2] By contrast, in medical applications, adding covariates to a prediction model can raise accuracy at an increasing rate; in search, the gain from additional searches rises with the length of user histories, and a US court credited expert testimony that a search engine with little data would "keep getting better at a faster and faster rate up to some point."[3]

On closer scrutiny, the debate mixes the effect of adding rows to a dataset (the *observations* of the individuals in a sample) with the effect of adding columns (the *covariates* observed on each individual). Moreover, some evidence comes from domains in which important covariates become available only after the prediction, such as a user's next search or a later X-ray, and some from domains in which the firm observes its covariates before it predicts, such as demand forecasting and image classification. Machine learning shows that this distinction matters: covariates available only in training speed up learning.[4] This paper reconciles the evidence, developing a model that distinguishes rows from columns, and covariates available at prediction from those available only after it, which I call *post-prediction covariates*.

In my model, a Bayesian firm learns the effects of many independent covariates on an outcome and predicts it for a new individual. The *value* of a dataset is the fall in the firm's expected squared prediction error relative to predicting without data. When the firm learns one effect, the effects it has not learned act as noise, and with many covariates this noise equals the *out-of-sample error*, the error of its model on past individuals it did not learn from. Each effect the firm learns lowers the noise against which it learns the others, so it learns them better, which lowers noise again. The fixed point of this feedback loop determines the noise, and with it the value of the firm's dataset, and a *learning multiplier*, $1 + \alpha$, measures how much the feedback amplifies the effect of each additional observation or covariate on the noise. I show that the amplification α is the percentage fall in out-of-sample error when the

[1] Crémer et al. (2019); Digital Competition Expert Panel (2019); Stigler Committee on Digital Platforms (2019), which warns that "the concentration of economic, media, data and persuasion, and political power is incredibly dangerous for our democracies."

[2] Bajari et al. (2019); Varian (2019), who argues that data have diminishing returns; Competition and Markets Authority (2020).

[3] Carballa-Smichowski et al. (2025); Schäfer and Sápi (2023); United States v. Google LLC (2024, ¶ 106). In 2025 the court ordered Google to share search data with competing search engines and AI chatbots (United States v. Google LLC 2025b,a), and the Digital Markets Act requires gatekeeper search engines to share query, click and view data with rival search engines (European Union 2022, Article 6(11)).

[4] Vapnik and Vashist (2009).

sample grows by one percent, so a firm can measure it on its own data.[5]

The model yields two results on returns to data. The first is that post-prediction covariates can make returns to observations increase. In search, a user's next search explains much of what she clicks but is unavailable when the engine ranks results. The engine learns its large effect from few observations; once it has, that part of the click is no longer noise, and further observations reveal more about the weaker effects of the query terms. The first observations thus mostly improve an estimate that the engine cannot use, and later ones the estimates that it does use, so each observation can raise the accuracy of the rankings more than the last. Returns to observations increase in small samples when the post-prediction covariates explain enough of the outcome; without post-prediction covariates, returns diminish. The result reconciles the evidence. Demand forecasts and image classifiers use every covariate they observe, and their returns diminish. Search logs include each user's next search; when it explains enough of the click, returns to search logs increase in small samples, as in the testimony that the US court credited, and diminish in large ones, as the CMA found.

The second result is that covariates observed on the same individuals are complements at every sample size. In search, learning the effect of the next search helps the engine learn the effects of the query terms, and every covariate does the same for every other. Each covariate available at prediction is worth the part of its explanatory power that the firm learns, plus a *spillover*, the value it adds by lowering the noise against which the firm learns every other effect; a post-prediction covariate is worth only its spillover. The learning multiplier amplifies every spillover, so α measures both the strength of the complementarity and the share of each covariate's value that is spillover. Observations and covariates are complements in small samples, as in search, and can be substitutes in large samples, where a new covariate does part of the work of additional observations.

A natural objection is that a firm that can choose which covariates to acquire takes the strongest first, so that each new covariate adds less value than the last. Section 6 addresses it with *returns to scope*, which compare the value of the next covariate with that of the last along the order of acquisition. Returns to scope increase unless explanatory power declines fast along the order. In a random order, explanatory power does not decline on average, and returns to scope increase at every sample size; Carballa-Smichowski et al. (2025), who add health covariates to a prediction model in random order, find that they increase at first. If the firm acquires strong covariates first, returns to scope diminish in small and in large samples, yet its covariates remain complements, because complementarity does not depend on the order.

I apply the model to rights over data. The European Commission estimates that 80 percent of industrial data is never used, and European law has addressed this by giving users rights to access the data that their connected products generate and to grant third parties access to

[5]The results are exact when covariates are many and the sample grows in proportion.

them.[6] Who controls access to data does not matter if holders bargain efficiently.[7] When each holder bargains alone, however, complementarity can make the cumulative payments for the complete dataset exceed its value; the firm then acquires only part of the data, or none, a textbook case of the tragedy of the anticommons.[8] How fast out-of-sample error falls with sample size measures the excess of payments over value. Suppose that out-of-sample error falls by about 2 percent when the sample grows by 10 percent, so that $\alpha = 0.2$, and that the out-of-sample R^2 is 0.5. Holders that each have few covariates on every individual, paid what their covariates add, claim 120 percent of the value of the data, and bargaining with them breaks down once their bargaining weight exceeds about 0.83. Individuals who each hold their own observation claim 20 percent. Portability and access rights let each individual decide whether a firm of her choice obtains her observation. Such rights integrate the complementary covariates within each observation, and the firm can buy every observation whatever the individuals' bargaining weight, unless post-prediction covariates make returns to observations increase, as may occur in search.

A firm that generates complementary data in house avoids the anticommons that an entrant faces when it assembles the parts from separate holders. The model thus offers an explanation for the rise of ecosystems, such as digital conglomerates, that hold many types of data.[9] In small samples, the same holds for observations with increasing returns: a search engine that records its users' searches itself keeps their value, while an entrant that buys them from individuals pays more than they are worth.

The theory also predicts the effects of data and privacy regulation. A purpose limitation, which lets a firm learn from a covariate but bars its use at prediction, reduces value by the part of the covariate's explanatory power that the firm has learned; a ban on collecting the covariate reduces value by its spillover as well. A regulator that mandates access to data can set the payment to each holder at the increase in out-of-sample error when the firm removes the holder's covariates from its estimated model, without re-estimating it. These payments sum to the value of the data. Some platforms that license content for generative AI already divide revenue among content owners in proportion to measured contributions;[10] when each holder is small, dividing a fixed sum in proportion to marginal contributions approximates

[6] European Commission (2022); European Parliament and Council (2023, Articles 4 and 5). The Commission traces the shortfall in data sharing between firms in part to "imbalances in negotiating power" (European Commission 2020). The right to data portability of the earlier General Data Protection Regulation, adopted to protect individuals, has a similar effect for personal data (European Parliament and Council 2016, Article 20).

[7] Coase (1960).

[8] Heller (1998); Buchanan and Yoon (2000). The bargaining model is stylized: each holder controls access to its part and bargains with the firm separately, by the Nash-in-Nash rule, as in the holdout problems of land assembly (Menezes and Pitchford 2004). With all holders at one table, a stable division exists (Section 5.2). How the assignment of rights shapes the incentives to generate data is outside the scope of the model (Grossman and O. D. Hart 1986; O. Hart and Moore 1990).

[9] On digital ecosystems and the role of data in them, see Crémer et al. (2019).

[10] ProRata.ai divides half of its revenue among publishers according to their contribution to each answer (Salazar 2024), and Bria pays the owners of its training data according to their influence on each generated output (Bria 2025).

this rule. Finally, the Digital Markets Act requires a gatekeeper to give rival search engines access to its search data, anonymized where they are personal data.[11] Each shared search is an observation, and the user's next search is a post-prediction covariate. If the shared data link each query to the same user's next search, and that search explains enough of the click, the shared observations have increasing returns in small samples, so they are worth more to a rival that already has many observations of this kind than to an entrant. If anonymization removes this link, they have diminishing returns, and the first ones shared are worth the most.

Contribution and related literature. The paper contributes to the economics of data by deriving the returns to data, along both dimensions of a dataset, from how a firm learns. Models of data-driven competition take the value of data, or the relation between data and prediction quality, as given (Prüfer and Schottmüller 2021; Hagiu and Wright 2023; de Cornière and Taylor 2025). In Montiel Olea et al. (2022), agents predict with different sets of covariates, and the sample size determines which set wins; here the sample size determines whether observations and covariates are complements. The model explains the complementarity between the number of users and the length of their histories that Schäfer and Sápi (2023) estimate in search, and predicts that it can reverse in large samples. Observations here come from a fixed population; when the population changes over time, a flow of recent observations can be worth more than a larger stock of old ones (Valavi et al. 2022).

The paper contributes to the theory of the value of information in two ways. First, to my knowledge, it is the first to show that, in prediction under squared loss, independent and identically distributed observations can have increasing returns when each has positive marginal value. The value of information is nonconcave when the marginal value of a small amount of it is zero (Radner and Stiglitz 1984), and the marginal value of samples eventually falls (Moscarini and Smith 2002); here returns increase over a range of sample sizes because observations reveal the effects of covariates that arrive after the prediction, and they diminish in large samples. In economics, the paper is also the first to model such covariates. Second, it shows that learning makes independent covariates complements. Models of information treat data as signals about one unknown state (Bergemann et al. 2018; Farboodi and Veldkamp 2026), and Börgers et al. (2013) characterize when two signals are complements in every decision problem, for a decision maker who knows how they relate to the state. Liang and Mu (2020) show that complementarities among correlated information sources determine whether a community learns efficiently. Here covariates are independent, and each effect the firm learns lowers the noise against which it learns the others. To my knowledge, the paper is the first to identify these learning spillovers. In independent work with a different setup, Schäfer (2025) obtains increasing returns to covariates from their correlation when the weights are known; here they arise from learning, with independent covariates.

The analysis brings to economics tools from high-dimensional statistics. The fixed point

[11] European Union (2022, Article 6(11)).

that defines learning noise is the Silverstein–Bai equation, which underlies the asymptotics of ridge regression (Silverstein and Bai 1995; Dobriban and Wager 2018). I prove it by a leave-one-out argument that allows post-prediction covariates and makes the multiplier explicit (Appendix A). Theories of scaling laws use the same covariates to learn and to predict (Bahri et al. 2024), so returns to observations diminish there; here post-prediction covariates can make them increase.

The application contributes to the economics of rights over data and of data markets in three ways. First, it shows which assignment of rights lets a firm acquire a complete dataset. Jones and Tonetti (2020) show that firms that own data may hoard them and that consumer ownership can bring allocations close to the optimum, and Dosis and Sand-Zantman (2023) show that a firm and its users disagree on who should own data; in these analyses, an owner holds its data as one block. Here the parts of a dataset have different holders, and how the firm learns decides whether they are complements. To my knowledge, the paper is the first to give the anticommons in data statistical microfoundations; in patents, the anticommons arises when research inputs are complements (Llanes and Trento 2011), and pools of complementary patents relieve it (Lerner and Tirole 2004). Rhodes et al. (2025) study an ecosystem that uses data from one product to improve its others and compare data sharing, data cooperatives and compensation schemes; here learning determines what combining data across products is worth, and the anticommons explains why an ecosystem that generates complementary data in house has an advantage over an entrant. Second, it derives which data are complements and which are substitutes, which models of data markets take as given. Data brokers are more likely to share data that are worth less together than apart (Gu et al. 2022), and suppliers of substitutes gain by bargaining together while suppliers of complements lose (Horn and Wolinsky 1988), so the case for data unions (Arrieta-Ibarra et al. 2018) holds only for observations without post-prediction covariates. In Dasaratha et al. (2025), firms sell prediction models valued by their bias and variance, and prices equal marginal contributions and support efficient entry when models are substitutes; here the marginal contributions of holders of covariates exceed value, by the factor $1 + \alpha$ when holdings are small. Third, it shows how privacy rules change what data are worth. Models of privacy study externalities among individuals whose data are correlated (Choi et al. 2019; Acemoğlu et al. 2022; Bergemann et al. 2022; Ichihashi 2021); here observations are independent, yet each changes the marginal value of the others through learning. Consent-based regulation can favor large firms (Campbell et al. 2015; Johnson et al. 2023); here individual rights harm most the small firms whose observations include post-prediction covariates. Under the GDPR, a data intermediary observed fewer consumers, each over a longer history (Aridor et al. 2023), a trade of observations for covariates whose effect on value the model determines.

Roadmap. Section 2 presents the model and Section 3 the value of every dataset and the learning multiplier. Section 4 studies the returns to observations, the economies of scope

among groups of covariates and the interaction of scale and scope, and Section 5 applies these results to rights over data. Section 6 extends the analysis to returns to scope along an order of acquisition. The Appendix contains the proofs; the Online Appendix contains further results and connections to statistics.

2 Model

A firm predicts an outcome for a new individual using outcomes and covariates observed in a sample of other individuals. Some covariates are also observed for the new individual; others become available only after the prediction. The firm chooses the sample size and which covariates of each kind to observe. I ask how these choices affect its expected prediction error.

2.1 Data-Generating Process

The firm learns from a sample of N individuals, indexed by $i = 1, \dots, N$, and predicts the outcome for a new individual, indexed by $i = 0$. For each individual i, the outcome is a linear combination of K covariates,

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta},$$

where $\boldsymbol{x}_i \in \mathbb{R}^K$ is the covariate vector of individual i and $\boldsymbol{\beta} \in \mathbb{R}^K$ is a vector of unknown weights common to all individuals.[12]

Weights. The weights are independent of the covariates, and they are mutually independent zero-mean Gaussians, $\boldsymbol{\beta} \sim \mathcal{N}\left(0, \sigma^2 \cdot \boldsymbol{I}_K\right)$. The variance $\sigma^2 > 0$ measures prior uncertainty about how covariates affect the outcome.

Covariates. Covariates are independent across individuals and across coordinates: for all i,

$$x_{ik} \sim \mathcal{N}\left(0, \frac{\omega_k}{K}\right), \qquad \omega_k \geq 0, \qquad \frac{1}{K}\sum_{k=1}^{K} \omega_k = 1.$$

The outcome therefore has variance $\mathbb{V}(y_i) = \sigma^2$, and covariate k explains the fraction ω_k/K of it. I call ω_k the *signal* of covariate k.[13] The firm knows each covariate's signal, which reflects its prior information about relative predictive importance. Without prior information distinguishing covariates, it assigns them equal signal: $\omega_k = 1$ for every $k = 1, \dots, K$.

[12] The model omits a separate idiosyncratic error. In the many-covariate limit, the aggregate contribution of unobserved covariates acts as Gaussian noise (Step 2 of the proof of Theorem 1).

[13] Heterogeneous prior variances do not change the results. If $x_{ik} \sim \mathcal{N}(0, a_k/K)$ and $\beta_k \sim \mathcal{N}(0, c_k)$, rescaling covariate k by the known constant $\sqrt{c_k}$ yields this model with signal proportional to $a_k c_k$ and leaves value unchanged.

Unknown weights, independent covariates. Weights have zero mean, so the firm cannot tell whether the outcome rises or falls with a covariate.[14] With independent covariates, the firm cannot use one covariate as a proxy for another, so all interactions between covariates arise from learning about the weights. Simulations with correlated covariates reproduce the patterns of the returns in Section 4 (Online Appendix OA.2.5).

2.2 Datasets

Covariates become available at different moments. The first $\overline{S}$ covariates, $k = 1, \dots, \overline{S}$, are available when the firm predicts, so it can observe them both in the sample and for the new individual. I call them *prediction covariates*. The next $\overline{P}$ covariates, $k = \overline{S}+1, \dots, \overline{S}+\overline{P}$, become available only afterward, so the firm can observe them in the sample but not for the new individual. I call them *post-prediction covariates*. In search, a user's next search is a post-prediction covariate: the engine records it for past queries, but it is unavailable when the engine ranks results for a new one. I assume $\overline{S}+\overline{P} < K$, so some covariates are never observed.

The firm chooses a dataset $D \equiv (N, \mathcal{S}, \mathcal{P})$: a sample size $N \in \mathbb{N}_0$, a set $\mathcal{S} \subseteq \{1, \dots, \overline{S}\}$ of prediction covariates and a set $\mathcal{P} \subseteq \{\overline{S}+1, \dots, \overline{S}+\overline{P}\}$ of post-prediction covariates. It observes the outcome and the covariates in $\mathcal{S} \cup \mathcal{P}$ for each of the N individuals in the sample, and the covariates in $\mathcal{S}$ for the new individual:

$$\boldsymbol{d}_D \equiv \Big(\big(y_i,\, (x_{ik})_{k\in\mathcal{S}\cup\mathcal{P}}\big)_{i=1}^{N},\; (x_{0k})_{k\in\mathcal{S}} \Big),$$

which are empty for the empty dataset $D_\varnothing \equiv (0, \varnothing, \varnothing)$.[15] Figure 1 depicts a dataset.

2.3 Prediction and the Value of a Dataset

Predictors and expected loss. A *predictor* for dataset D is a measurable function $\hat{y}$ of the data $\boldsymbol{d}_D$. The firm incurs quadratic loss $(y_0 - \hat{y})^2$. Write $R(D)$ for the lowest expected loss that any predictor for D achieves,

$$R(D) \equiv \inf_{\hat{y}} \mathbb{E}\big[(y_0 - \hat{y}(\boldsymbol{d}_D))^2\big].$$

Value. The *value* of dataset D is the reduction in expected loss relative to predicting without data,

$$V(D) \equiv R(D_\varnothing) - R(D).$$

[14] With a known nonzero prior mean, value gains a term independent of the sample, and the noise that the firm faces when it learns the weights rises. I measure the value that data add to prior knowledge.

[15] The firm could also observe for the new individual a covariate absent from its sample. Such a covariate adds no value: weights and covariates are symmetric about zero, so the posterior mean of its weight is zero (Remark OA.1 in the Online Appendix).

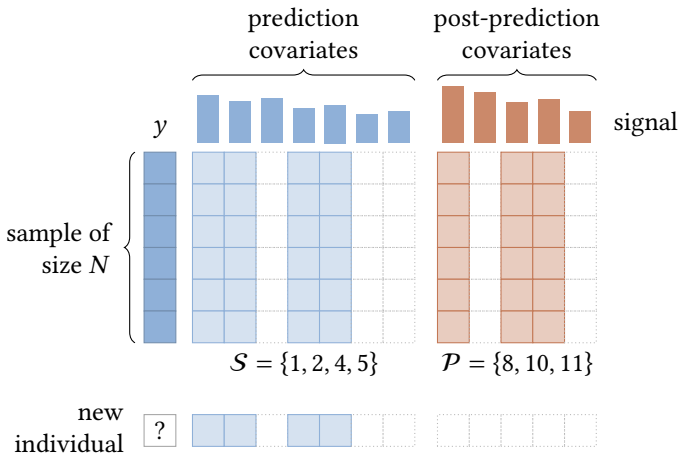


Figure 1. A dataset $D = (N, \mathcal{S}, \mathcal{P})$ with $S = 7$ prediction covariates, 1 to 7, and $P = 5$ post-prediction covariates, 8 to 12; bars show their signal. The covariates $k > 12$, which are never observed, are not drawn. Shaded cells are observed: the outcome y and the covariates in $\mathcal{S} \cup \mathcal{P}$ for each of the N individuals in the sample, and the covariates in $\mathcal{S}$ for the new individual, whose outcome the firm predicts.

Order on datasets. $D \leq D'$ if and only if $N \leq N'$, $\mathcal{S} \subseteq \mathcal{S}'$ and $\mathcal{P} \subseteq \mathcal{P}'$: under a larger dataset, the firm observes everything it observes under a smaller one.

3 Value and the Learning Multiplier

This section characterizes the value of every dataset. With many covariates, the firm learns every weight against a common noise that a fixed point determines, and value is the learned signal of the covariates that the firm applies. The slope of the fixed-point map determines a learning multiplier that amplifies the effect of every margin.[16] Subscripts denote partial derivatives, $f_x \equiv \partial f / \partial x$. Proofs are in the Appendix.

3.1 Learning and Prediction

Under quadratic loss, the best predictor is the posterior mean, $\hat{y}^\star_D \equiv \mathbb{E}[y_0 \mid \boldsymbol{d}_D]$, and the value of a dataset is the part of the variance of the outcome that this predictor explains:

$$V(D) = \mathbb{V}(y_0) - \mathbb{E}\big[\mathbb{V}(y_0 \mid \boldsymbol{d}_D)\big] = \mathbb{V}(\hat{y}^\star_D). \tag{1}$$

Value rises with the dataset, because a larger dataset reveals everything a smaller one does.[17] Write $\boldsymbol{y}_N \in \mathbb{R}^N$ and $\boldsymbol{X}_D \in \mathbb{R}^{N \times |\mathcal{S} \cup \mathcal{P}|}$ for the outcomes and the covariates that the firm observes for the sample, and $\boldsymbol{m}_D \equiv \mathbb{E}[\boldsymbol{\beta} \mid \boldsymbol{y}_N, \boldsymbol{X}_D]$ for the posterior mean of the weights, with $\boldsymbol{m}_D = \boldsymbol{0}$ when $N = 0$.

[16] Table OA.1 in the Online Appendix lists every object with the page on which it is defined.

[17] Both facts hold for any square-integrable outcome (Online Appendix OA.3.1), so the optimal predictor does not overfit: its expected error is nonincreasing in the sample size and in the number of covariates the firm observes.

Lemma 1 (Learning and prediction). *For every dataset $D = (N, \mathcal{S}, \mathcal{P})$,*

$$\hat{y}_D^\star = \sum_{k \in \mathcal{S}} x_{0k}\, m_{D,k} \qquad \textit{and} \qquad V(D) = \frac{1}{K} \sum_{k \in \mathcal{S}} \omega_k\, \mathbb{V}(m_{D,k}),$$

and each $\mathbb{V}(m_{D,k})$ is nondecreasing in D.

The variance of a learned weight, as a share of the prior variance, $\mathbb{V}(m_{D,k})/\sigma^2 \in [0,1]$, is the *fraction learned* of the weight. The sample, post-prediction covariates included, determines the fractions learned; the covariates observed for the new individual determine which learned weights the firm applies. A post-prediction covariate therefore adds value as an observation does, by improving what the firm learns about the weights it applies.[18]

3.2 Learning with Many Covariates

At finite K, what the firm learns about one weight depends on what it learns about every other. When covariates are many, the part of the outcome that the firm has not learned acts on each weight as Gaussian noise, and the firm learns each weight as it would if that weight were the only unknown. The variance of this noise, learning noise, is the same for every weight.

Units, positions and datasets. From now on, $\sigma^2 = 1$, so value is the share of the variance of the outcome that the firm's predictions explain, and the fraction learned of weight k is $\mathbb{V}(m_{D,k})$.[19] As K grows, I hold fixed the shares $\overline{s}$ and $\overline{p}$ of prediction and post-prediction covariates and how signal varies across covariates. Covariate k sits at position k/K of $[0,1]$, and a *signal function* ω on $[0,1]$ gives the signal of the covariate at each position, $\omega_k \approx \omega(k/K)$ (Appendix A states the construction precisely). Prediction covariates occupy the positions in $[0, \overline{s}]$, post-prediction covariates those in $(\overline{s}, \overline{s} + \overline{p}]$, and the remaining positions belong to covariates that are never observed. A dataset in the limit consists of a *scale* n, which approximates N/K, and two sets of positions, each a finite union of intervals: a set $\mathcal{S} \subseteq [0, \overline{s}]$ of prediction covariates and a set $\mathcal{P} \subseteq (\overline{s}, \overline{s} + \overline{p}]$ of post-prediction covariates. The set $\mathcal{T} \equiv \mathcal{S} \cup \mathcal{P}$ collects the covariates in the sample. For a set of positions $\mathcal{X}$, write $\int_{\mathcal{X}} \omega \equiv \int_{\mathcal{X}} \omega(u)\, du$ for its *signal*, the share of the variance of the outcome that its covariates explain.

Assumption 1 (Many covariates). *The firm chooses a dataset $D_K = (N, \mathcal{S}_K, \mathcal{P}_K)$ for each K. As $K \to \infty$, $N/K \to n \in (0, \infty)$; the sample contains the covariates at the positions in $\mathcal{S}$ and $\mathcal{P}$, $\mathcal{S}_K = \{k \leq \overline{S} \,:\, k/K \in \mathcal{S}\}$ and $\mathcal{P}_K = \{\overline{S} < k \leq \overline{S} + \overline{P} \,:\, k/K \in \mathcal{P}\}$; and $\int_{\mathcal{T}} \omega < 1$.*

[18] Lemma 1 uses only that weights and covariates are independent, centered and square-integrable, and it holds for every set of covariates in the sample and every subset of them observed for the new individual.

[19] Dividing the outcome by σ divides every weight by σ. The firm knows σ, so the rescaled data contain the same information, and value, a variance, is divided by σ^2.

The condition $\int_{\mathcal{T}} \omega < 1$ states that the sample leaves out some signal. It keeps the noise that the firm faces when it learns the weights away from zero.

Learning a single weight. Suppose that the firm observes the outcome and one covariate, of signal ω, for $N = nK$ individuals, and that everything else in the outcome is Gaussian noise of variance E. Each observation provides precision x_{ik}^2/E about the weight, which averages $\omega/(KE)$, so the sample provides precision $\tau\omega$, where $\tau \equiv n/E$ is the *precision* per unit of signal (the prior has unit precision). The fraction learned of the weight is then $\ell(\tau\omega)$, where $\ell(x) \equiv x/(1+x)$ for $x \geq 0$ is the share of the posterior precision that comes from the data. At a given precision, the firm learns the weights of covariates with more signal better. Once the firm applies its learned weight, the covariate explains the share $\omega\,\ell(\tau\omega)$ of the variance of the outcome, its *learned signal*.

Learning noise. The noise on each weight consists of the signal outside the sample and the signal of the covariates in the sample that the firm has not learned. Each covariate has a vanishing share of the signal, so this noise is the same for every weight. If the firm learned every weight at precision τ, the covariates in a set $\mathcal{X}$ would explain

$$v(\tau, \mathcal{X}) \equiv \int_{\mathcal{X}} \omega(u)\,\ell\big(\tau\omega(u)\big)\,du$$

of the variance of the outcome once the firm applied their weights: their learned signal, which rises with precision and with the set. The noise on each weight would be

$$e(\tau; \mathcal{T}) \equiv \underbrace{1 - \int_{\mathcal{T}} \omega}_{\text{misspecification}} + \underbrace{\int_{\mathcal{T}} \omega(u)\big[1 - \ell\big(\tau\omega(u)\big)\big]\,du}_{\text{unlearned signal}} = 1 - v(\tau, \mathcal{T}), \tag{2}$$

the prediction error of a model whose weights the firm learns at precision τ, for an individual for whom the firm observes every covariate in the sample. Misspecification is the signal that no model of the sample can learn. The error falls as precision rises, from 1 toward misspecification; write $e_\tau = -v_\tau(\tau, \mathcal{T}) < 0$. Precision is scale per unit of noise, so noise and precision must satisfy $E = e(\tau; \mathcal{T})$ and $\tau = n/E$.

Theorem 1 (Value and learning noise)**.** *Under Assumption 1:*

(i) the system

$$\tau^\star = n/E^\star, \qquad E^\star = e(\tau^\star; \mathcal{T}), \qquad \mathcal{V} = v(\tau^\star, \mathcal{S}) \tag{3}$$

has a unique solution $\big(\tau^\star(n, \mathcal{T}), E^\star(n, \mathcal{T}), \mathcal{V}(n, \mathcal{S}, \mathcal{P})\big)$*, at which* $E^\star \in \big[1 - \int_{\mathcal{T}} \omega, 1\big]$*;*

(ii) as $K \to \infty$*, the fraction learned of every covariate in the sample converges, uniformly, to* $\ell(\tau^\star\omega_k)$*, and* $V(D_K) \to \mathcal{V}(n, \mathcal{S}, \mathcal{P})$*.*

I call $E^\star$ the *learning noise* and $\tau^\star$ the *learning precision*. When the margins are fixed, I suppress their arguments; a star marks an object evaluated at learning noise.[20] Learning depends only on the sample: learning noise and learning precision depend on the covariates in $\mathcal{T}$, not on which of them the firm applies. Precision therefore acts as an intermediate input: the firm produces it from observations and, by lowering noise, from covariates of either kind, and it uses it to learn the weights of the covariates it applies. Value is bounded above by $\int_S \omega$, the signal of the prediction covariates.

Because learning depends only on the sample, the firm learns the same weights whichever covariates it can apply. For disjoint sets $\mathcal{A}$ and $\mathcal{B}$, write $\mathcal{V}(n, \mathcal{A}, \mathcal{B}) \equiv v\big(\tau^\star(n, \mathcal{A} \cup \mathcal{B}), \mathcal{A}\big)$ for the value of a firm that learns from the covariates in $\mathcal{A} \cup \mathcal{B}$ and applies those in $\mathcal{A}$. Since learned signal adds up across covariates:

Corollary 1 (Value decomposition)**.**

$$\mathcal{V}(n, \mathcal{S}, \mathcal{P}) = \underbrace{\mathcal{V}(n, \mathcal{S} \cup \mathcal{P}, \varnothing)}_{\textit{full value } = 1-E^\star} - \underbrace{\mathcal{V}(n, \mathcal{P}, \mathcal{S})}_{\textit{reflected value}}. \tag{4}$$

The *full value* is the value of a firm that could apply every covariate in its sample; it equals $1 - E^\star$, the out-of-sample R^2 (Section 3.3). The *reflected value* is the value of a firm that learns from the same sample but could apply only the post-prediction covariates: their learned signal. Value is the full value minus the reflected value. The decomposition holds at every K, by Lemma 1: the three datasets share their sample, so they share the learned weights, and differ only in which of them the firm applies. Figure 2(a) draws the three values as areas; mirrored around the vertical axis, the two kinds of covariates grow independently.

3.3 The Learning Multiplier

From here on, I fix $n > 0$, $\mathcal{S}$ with positive length and $\mathcal{P}$ with $\int_{\mathcal{T}} \omega < 1$.[21] Every margin acts on learning noise through a feedback loop: lower noise means more precision, which raises the fraction learned of every weight in proportion to the part still to learn, and the newly learned signal leaves the noise. The loop multiplies the effect of every margin, and one statistic measures it: the elasticity of the firm's out-of-sample error with respect to its sample size.

Write $\Psi(E; n, \mathcal{T}) \equiv e(n/E; \mathcal{T})$ for the error at the precision that noise E implies. Learning noise is the fixed point of this map, $E^\star = \Psi(E^\star; n, \mathcal{T})$ (Figure 2(b)). When the noise at which the firm learns falls by one unit at fixed scale, precision rises by $n/E^{\star 2}$, and each unit of precision

[20] I treat the limit as an approximation to a large but finite environment; in the simulations of Online Appendix OA.2.2, it is close to the simulated value already with 100 covariates.

[21] The signal function is positive, and it is continuous at every position at which I add covariates.

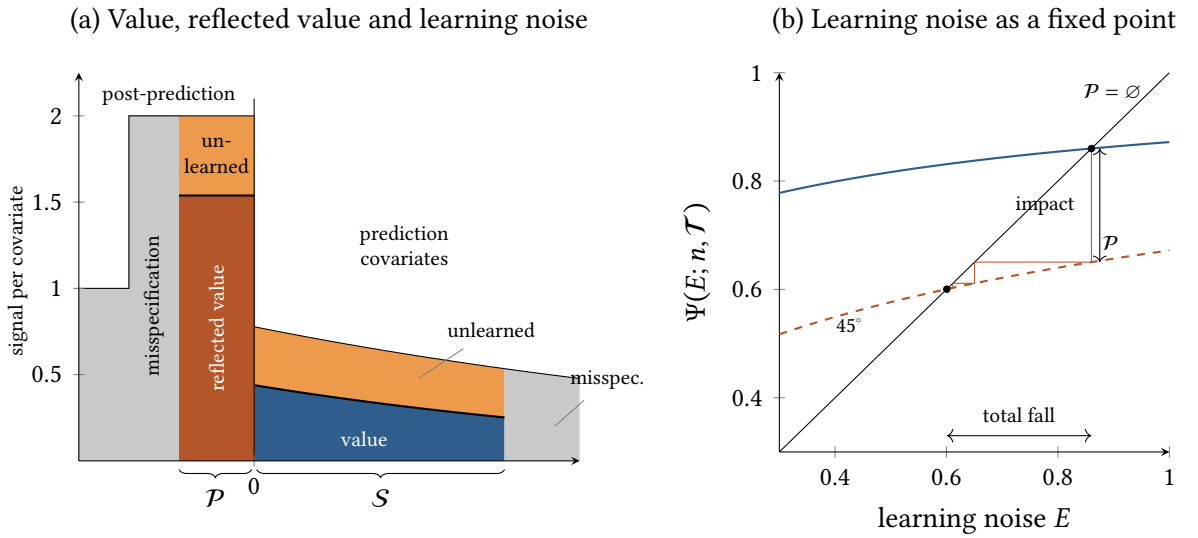


Figure 2. Value and learning noise. In the example, prediction covariates, to the right of the vertical axis, have signal $\omega(u) = c\, e^{-0.75u}$ on $[0, 0.65]$; post-prediction covariates, to the left, have signal 2 on an interval of length 0.25; never-observed covariates have signal 1 on the remaining 0.1; and $c = 0.7775$ sets the total signal to one. Here $n = 1$, $\mathcal{S} = [0, 0.5]$, and $\mathcal{P}$ contains post-prediction covariates of length 0.15. (a) Learned signal (thick line) and signal (thin line) per covariate. Value is the learned signal of the prediction covariates and the reflected value that of the post-prediction covariates; together they make the full value, $1 - E^\star$. Learning noise is the unlearned signal plus misspecification. (b) The map Ψ without and with the post-prediction covariates, its fixed points, and the iterations that follow when the firm adds them.

lowers the error by $-e_\tau$. The error therefore falls by the slope of the map,

$$\Psi' \equiv \frac{\partial \Psi(E^\star; n, \mathcal{T})}{\partial E} = \frac{n}{E^{\star 2}}(-e_\tau) = \frac{1}{E^\star}\int_{\mathcal{T}} \omega(u)\, \ell\big(\tau^\star \omega(u)\big)\big[1 - \ell\big(\tau^\star \omega(u)\big)\big]\, du.$$

The learned signal of a covariate of signal ω rises with the log of precision at the rate $\omega\ell(\tau^\star\omega)[1 - \ell(\tau^\star\omega)]$, its *speed of learning*, which is greatest when the firm has learned half of the weight. Summing the speeds over the sample and dividing by $E^\star$ gives Ψ', the *strength* of the feedback loop (Lemma 2).

The strength of the loop determines the *learning multiplier* $1 + \alpha$, where $\alpha \equiv \Psi'/(1 - \Psi') \in [0, \infty)$ is the *amplification*: a unit fall in noise lowers noise by a further Ψ', then by Ψ'^2, and the falls sum to $1 + \Psi' + \Psi'^2 + \cdots = 1 + \alpha$.[22] The total effect of a margin on learning noise, and through $\tau^\star = n/E^\star$ on precision, is its *impact*, its effect at given noise, times the multiplier. The staircase of Figure 2(b) shows these falls after the firm adds post-prediction covariates. Write $v_\tau(\mathcal{X}) \equiv \partial v(\tau^\star, \mathcal{X})/\partial\tau$ for the *value of precision* for the weights in $\mathcal{X}$, so that $-e_\tau = v_\tau(\mathcal{T})$. Figure 3 draws the system as a graph.

Write

$$\alpha_s \equiv \alpha \frac{v_\tau(\mathcal{S})}{v_\tau(\mathcal{T})} \leq \alpha$$

for the amplification times the share of the value of precision that accrues to the weights the

[22] The sum is finite because $\Psi' < 1$: the right side of $n = \tau e(\tau; \mathcal{T})$ rises with precision, at the rate $E^\star(1 - \Psi')$ at $\tau^\star$.

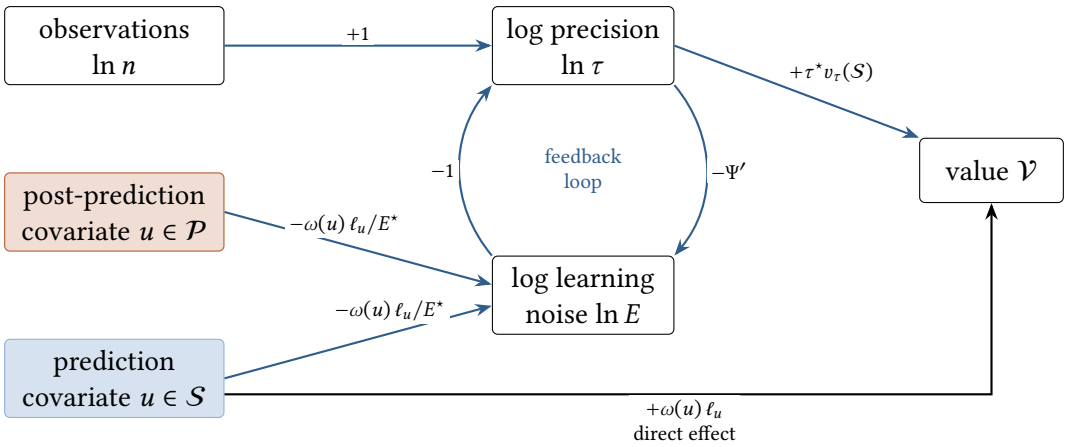


Figure 3. How data affect value. The graph represents the system of Theorem 1, with scale, noise and precision in logs. Each arrow is the partial derivative of the variable at its head with respect to the variable at its tail, at given values of the others; for a covariate at position u, per unit length added, with $\ell_u = \ell(\tau^\star \omega(u))$. Observations raise precision. Covariates of either kind lower noise by their learned signal, and prediction covariates also raise value directly. Noise and precision form a feedback loop of strength Ψ', which multiplies the effect of every margin on noise and precision by $1 + \alpha = 1/(1 - \Psi')$ (Section 3.3).

firm applies; $\alpha_s = \alpha$ when $\mathcal{P} = \varnothing$. For a position u outside the sample, write $\ell_u \equiv \ell(\tau^\star \omega(u))$, and let the *marginal value* $\mathcal{V}_u$ of the covariate at u be the increase in value, per unit length, when the sample gains the covariates in $[u, u + \delta)$, as $\delta \to 0$.

Proposition 1 (Marginal values)**.**

(i) The marginal value of an observation is

$$\mathcal{V}_n = \underbrace{\frac{1}{E^\star}}_{\text{impact}} \underbrace{(1+\alpha)}_{\text{multiplier}} \underbrace{v_\tau(S)}_{\text{value of precision}} = \frac{\alpha_s E^\star}{n}. \tag{5}$$

(ii) The marginal value of a covariate is its direct effect, if the firm applies it, plus its spillover:

$$\mathcal{V}_u = \underbrace{\mathbf{1}\{u \le \bar{s}\}\, \omega(u)\, \ell_u}_{\text{direct effect}} + \underbrace{\alpha_s\, \omega(u)\, \ell_u}_{\text{spillover}}. \tag{6}$$

The direct effect of a prediction covariate is its learned signal. Its spillover is the value of the precision it adds: its learned signal leaves the noise, the multiplier amplifies the fall, and the extra precision is worth $v_\tau(S)$ on the weights the firm applies. The coefficient α_s is the same for both kinds of covariates, so a post-prediction covariate is all spillover. Since $\alpha_s = \tau^\star \mathcal{V}_n$, a covariate's spillover equals the value of $\tau^\star \omega(u) \ell_u = n\, \omega(u) \ell_u / E^\star$ additional observations: the sample size times the share of learning noise that the covariate's learned signal removes. That share is the proportional increase in out-of-sample error when the firm removes the covariate from its estimated model without re-estimating it, so the firm can measure any spillover in observations without knowing the signals (Corollary OA.2(i) in Online Appendix OA.5.4).

A purpose limitation, which lets a firm learn from a covariate but bars its use at prediction, moves the covariate from $\mathcal{S}$ to $\mathcal{P}$. It leaves the sample, and with it learning noise, unchanged, so it reduces value by the direct effect of the covariate; a ban on collecting the covariate reduces value by its spillover as well. A restriction on use therefore reduces value by the fraction $1/(1+\alpha_s)$ of the reduction under a ban on collection. The comparison is marginal: it concerns one covariate.

Measurement. The firm can estimate learning noise by cross-validation: it estimates its model on part of the sample and measures its prediction error on the remaining individuals, using every covariate in the sample. This *out-of-sample error* is the lowest expected loss for the dataset $(N, \mathcal{S}\cup\mathcal{P}, \varnothing)$, so the out-of-sample R^2 is the full value, and its limit is $1-E^\star$. Ridge regression with a cross-validated penalty and approximate message passing attain $\mathcal{V}$ (Online Appendix OA.4.3). Once the firm observes the outcomes of new individuals, it also observes its prediction error, $1-\mathcal{V}$; the gap with out-of-sample error is the reflected value.

Hestness et al. (2017) study empirically how out-of-sample error falls with sample size, and Bahri et al. (2024) and Bordelon et al. (2024) study it theoretically. In Figure 3, the path from scale to noise has arrows 1 and $-\Psi'$. With the multiplier, a sample one percent larger lowers learning noise by $(1+\alpha)\Psi' = \alpha$ percent:

$$\alpha = -\frac{\partial \ln E^\star}{\partial \ln n}$$

is the elasticity of out-of-sample error with respect to sample size, in absolute value. The firm can estimate it by re-estimating its model on nested subsamples and regressing the log of cross-validated error on the log of the subsample size near the full sample. Hestness et al. (2017) estimate that out-of-sample error falls as a power of sample size, with exponents between -0.07 and -0.35 across translation, language, image and speech tasks. If their measured losses corresponded to learning noise, these estimates would imply α between 0.07 and 0.35, a learning multiplier between 1.07 and 1.35, and spillovers worth 7 to 26 percent of the marginal value of a covariate in these tasks, which apply every input. The multiplier is a sufficient statistic in the sense of Chetty (2009): it can be estimated without the primitives and determines the effects of interest whatever they are; for every signal function, precision rises by $1+\alpha$ percent when the sample grows by one percent. It sets the spillover of each covariate (Proposition 1), the complementarity among covariates (Sections 4 and 6), and the excess of marginal contributions over value and the price of mandated access (Section 5).

4 Returns to Data

This section studies complementarity among observations, among covariates, and between the two. Observations have diminishing returns whenever the firm applies every weight it

learns, and post-prediction covariates can make them increase in small samples. Covariates are complements at every scale, and the learning multiplier bounds the economies of scope. Observations and covariates are complements in small samples and can be substitutes in large ones. These results hold in every order of acquisition; Section 6 studies returns along the order itself.

4.1 Returns to Observations

Returns to observations *increase* at scale n if value is strictly convex in scale there, $\mathcal{V}_{nn} > 0$, so that the next observation adds more value than the last, and they *diminish* if $\mathcal{V}_{nn} < 0$. The value decomposition shows when each happens.

By Corollary 1, value is the full value minus the reflected value, $\mathcal{V} = v(\tau^\star, \mathcal{T}) - v(\tau^\star, \mathcal{P})$, and observations act on both only through learning precision. Two forces shape the curvature. Precision accelerates, $\tau^\star_{nn} > 0$: each observation lowers noise, which raises the precision that the next one adds. The value of precision decays, $v_{\tau\tau}(\mathcal{X}) < 0$ for every set of weights: one more unit of precision raises the learned signal of a weight by the square of its signal still to learn, $\omega^2(1-\ell)^2$, which falls as the firm learns the weight.

Proposition 2 (Returns to observations)**.**

(i) The curvature of value in scale is

$$\mathcal{V}_{nn} = \underbrace{\tau^\star_{nn}\left[v_\tau(\mathcal{T}) - v_\tau(\mathcal{P})\right]}_{\text{acceleration of precision } (+)} + \underbrace{\tau^{\star 2}_{n}\left[v_{\tau\tau}(\mathcal{T}) - v_{\tau\tau}(\mathcal{P})\right]}_{\substack{\text{decay of the} \\ \text{value of precision } (-)}}. \tag{7}$$

(ii) For small enough n, returns to observations increase if

$$\underbrace{\int_S \omega^2 + \int_P \omega^2}_{\text{speed-up of learning}} > \underbrace{\frac{\int_S \omega^3}{\int_S \omega^2}}_{\substack{\text{decay of the} \\ \text{value of precision}}} \tag{8}$$

and diminish if the reverse inequality holds; for large enough n, they diminish. Without post-prediction covariates, they diminish at every n.

Each term of part (i) nets the reflected value out of the full value, so post-prediction covariates enter the two terms asymmetrically. They add to the acceleration, because the firm learns their weights from the same observations and their learned signal leaves the noise; the decay of their own value of precision falls on the reflected value, which the firm does not use.

Without post-prediction covariates, the reflected value is zero and the decay always wins, so returns to observations diminish at every scale. The decay falls hardest on the covariates with most signal, whose weights the firm learns first, while part of the acceleration goes

to averaging out misspecification, which does not increase value (Appendix B). The same argument shows that the full value, the value to a firm that applies every covariate in its sample, is concave in scale. Observations without post-prediction covariates are therefore substitutes.

With post-prediction covariates, value is convex in scale where the reflected value is more concave than the full value: where the firm learns the post-prediction weights fast, so that the learned signal it cannot use rises steeply and then flattens. Inequality (8) says when this happens in small samples, where noise is close to one and precision close to scale (Appendix B). Its left side is the acceleration: the firm learns a weight of signal ω at the speed $n\omega^2$, so each observation speeds up learning by the squared signal of the sample. Its right side is the decay: the average signal of the prediction covariates, weighted by their value of precision. Post-prediction covariates add to the left side and leave the right side unchanged, so enough of them make the inequality hold.

The inequality requires post-prediction covariates that are stronger, on average, than the prediction covariates: the firm learns a strong covariate fast, so it speeds up learning more than weak covariates with the same total signal. They must be stronger by enough to overcome two forces: misspecification, which slows the acceleration, and the dispersion of signal among the prediction covariates, which speeds up the decay. Figure 4 shows a stark case, with prediction covariates of equal signal and a few strong post-prediction covariates: the inequality holds once these have length above 0.013, and with all of them the marginal value of an observation nearly doubles up to $n = 0.25$ before it falls.

The inequality is likely to hold when the prediction covariates are many, individually weak and similar, the post-prediction covariates individually strong, and the sample explains most of the outcome. Search fits this description: each term of a query has little signal about which result the user will click, while her next search often reveals what she wanted. So may credit scoring for applicants without a credit history, where each item of an application has little signal about default and later repayments of other debts have much. The inequality fails when a few prediction covariates have most of the signal, as a long payment history does in credit scoring, because dispersion is then high. Under the reverse ranking, with every prediction covariate at least as strong as every post-prediction covariate, returns to observations diminish at every scale (Lemma 4(i)).[23]

In search, the next search is correlated with the query, so in the model the post-prediction covariates stand for the part of the user's later behavior that the query does not predict; residualizing the covariates in this way can correlate their weights, which the model rules out. Search logs include post-prediction covariates; whether they satisfy Inequality (8) in a given

[23]Without a ranking, returns to observations can change sign more than once. In the example of Online Appendix OA.3.5, a few strong prediction covariates make Inequality (8) fail, yet returns increase at intermediate scales, once the firm has learned their weights. If every post-prediction covariate has at least as much signal as every prediction covariate, returns change sign at most once, so they diminish at every scale when Inequality (8) fails (Online Appendix OA.5.3).

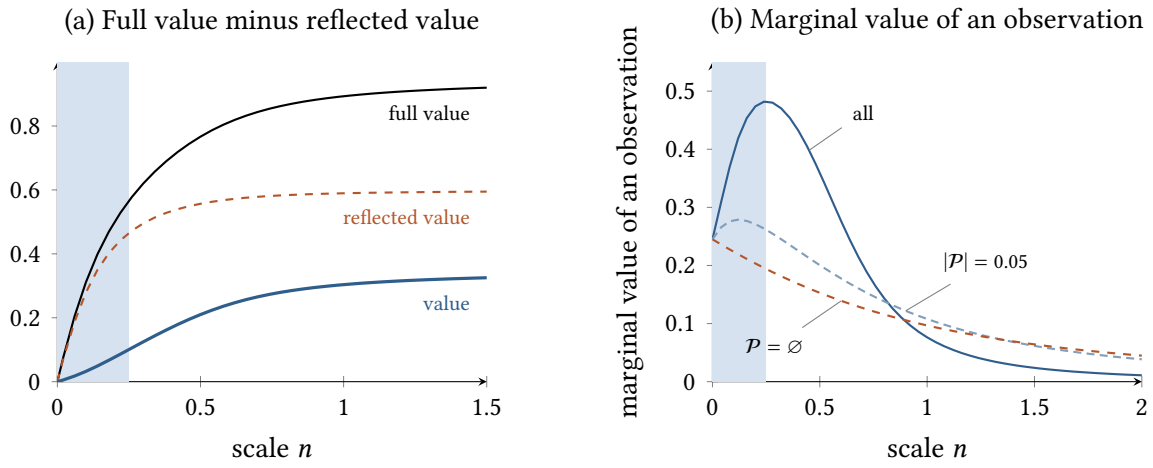


Figure 4. Returns to observations. Prediction covariates $\mathcal{S} = [0, 0.5]$ of signal 0.7 and post-prediction covariates of signal 6 on a set of length up to 0.1, so that misspecification is 0.05 with all of them. (a) Full value, reflected value and value, their difference, with all post-prediction covariates. (b) Marginal value of an observation without post-prediction covariates, with a set $\mathcal{P}$ of length 0.05 and with all of them. Shaded, the scales at which returns to observations increase with all post-prediction covariates.

market is a quantitative question, which data on how prediction error for new individuals falls with sample size answer. In simulations, increasing returns to observations appear already with a few hundred covariates and hold with non-Gaussian data (Online Appendix OA.2.4). Observations whose post-prediction covariates satisfy Inequality (8) are therefore complements while they are scarce (Section 5).

The evidence on returns to data. Proposition 2 reconciles the evidence on returns to observations by the class of data; Sections 4.3 and 6 do the same for the evidence on the complementarity between scale and scope and on returns to scope. Demand forecasts and image classifiers apply every input, and their diminishing returns (Bajari et al. 2019; Varian 2019) are what part (ii) predicts without post-prediction covariates. Search logs include each user's next search: part (ii) then allows increasing returns in small samples, as in the expert testimony on search engines with little data that the US court credited (United States v. Google LLC 2024, ¶ 106), and predicts diminishing returns in large samples, as the UK competition authority found for click-and-query data (Competition and Markets Authority 2020). Models of data-driven competition derive tipping and competitive advantage from a relation between data and prediction quality that they take as given (Prüfer and Schottmüller 2021; Hagiu and Wright 2023); Proposition 2 ties increasing returns in this relation to a class of data and confines them to small samples.

4.2 Economies of Scope

Covariates are complements: a group of covariates adds more value the more covariates the firm already holds. Write $|\mathcal{S}|$, $\int_{\mathcal{S}} \omega$ and $\int_{\mathcal{S}} \omega^2$ for the *length*, *signal* and *squared signal* of a

group $\mathcal{S}$. Two disjoint groups $\mathcal{S}_1$ and $\mathcal{S}_2$ of prediction covariates exhibit *economies of scope* at scale n if they add more value together than apart (Baumol et al. 1982), and the difference

$$\mathcal{V}(n, \mathcal{S}_1 \cup \mathcal{S}_2, \varnothing) - \mathcal{V}(n, \mathcal{S}_1, \varnothing) - \mathcal{V}(n, \mathcal{S}_2, \varnothing)$$

measures them. The comparison concerns finite groups, as when a firm buys a broker's file.

Proposition 3 (Economies of scope)**.** *Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be disjoint groups of prediction covariates.*

(i) For every K and N, the value that $\mathcal{S}_2$ adds to $\mathcal{S}_1$ is at least its value alone:

$$V(N, \mathcal{S}_1 \cup \mathcal{S}_2, \varnothing) - V(N, \mathcal{S}_1, \varnothing) \geq V(N, \mathcal{S}_2, \varnothing).$$

(ii) In the limit, the value that $\mathcal{S}_2$ adds strictly increases with the covariates the firm already holds, so the two groups exhibit economies of scope at every scale.

(iii) The economies of scope are less than α times the value of the two groups together, where α is the amplification for the dataset that contains both.

Part (i) holds exactly, in every finite sample: a firm that observes more covariates estimates each weight at least as precisely, so no weight loses value when the groups are combined. In the limit, the gain is strict, because combining the groups removes the learned signal of each from the noise against which the firm learns the weights of the other. Part (ii) says more than economies of scope: value is supermodular in the set of covariates, so every covariate the firm adds raises the value that any other group adds. Section 5 rests on this property.

Part (iii) bounds the economies of scope by the learning multiplier. They arise from spillovers, and the spillovers of all the covariates in a sample sum to $\alpha\mathcal{V}$ (Proposition 1). The bound is measurable: α is the elasticity of out-of-sample error on the combined data. Since α tends to zero in small and in large samples, so do the economies of scope relative to value, and complementarity among covariates matters most at intermediate scales. Which groups come closest to the bound differs at the two ends (Lemma 7 in Appendix B). In small samples, the spillover of one group on the other is the fall in noise that it causes, about scale times its squared signal, times the value of lower noise on the other group's weights, about scale times that group's squared signal; the economies of scope reach at most half the bound, when the two groups have equal squared signal. In large samples, the firm learns each weight it applies up to an error of misspecification over scale, as least squares does, so a group gains in proportion to its length times the signal that the other group removes from the misspecification. The economies of scope then approach the bound when a long group of weak covariates joins a short group of strong ones, as when a broker's file of many weak covariates joins a firm's few strong ones.

Groups of post-prediction covariates, and a group of each kind, are complements too (Online Appendix OA.5.4 and Appendix D). The result uses the independence of the covariates: a group perfectly correlated with the covariates a firm holds does not increase its value.

4.3 Scale and Scope

Returns to observations compare observations with one another, and economies of scope compare groups of covariates. The interaction of scale and scope compares observations with covariates. Scale and the covariate at a position u outside the sample are *complements* if its marginal value rises with scale, $\partial \mathcal{V}_u / \partial n > 0$, so that the covariate raises the marginal value of an observation, and *substitutes* if its marginal value falls. For a set $\mathcal{X}$ of covariates in the sample, write $\varepsilon_{\mathcal{X}} \equiv -\tau^\star v_{\tau\tau}(\mathcal{X}) / v_\tau(\mathcal{X})$ for the elasticity of the value of precision for the weights in $\mathcal{X}$: a one percent increase in precision lowers it by $\varepsilon_{\mathcal{X}}$ percent. The elasticity is high when the firm has nearly learned the weights of the covariates with most signal, whose value of precision is largest.

Proposition 4 (Scale and scope)**.** *Let the covariate at u lie outside the sample.*

(i) Its marginal value changes with scale as

$$\frac{\partial \mathcal{V}_u}{\partial n} = \mathcal{V}_n \frac{(1+\alpha)\,\omega(u)\,\ell_u}{E^\star} \Bigg(\underbrace{\mathbf{1}\{u \le \overline{s}\} \frac{1-\ell_u}{\alpha_s}}_{\text{new weight}} + \underbrace{2-\ell_u}_{\text{larger impact}} + \underbrace{\alpha(2-\varepsilon_{\mathcal{T}}) - \varepsilon_{\mathcal{S}}}_{\substack{\text{returns to} \\ \text{observations}}} \Bigg), \tag{9}$$

where the last term has the sign of $\mathcal{V}_{nn}$. Scale and the covariate are therefore substitutes only if returns to observations diminish and $\varepsilon_{\mathcal{S}} > 1$.

(ii) For large enough n, scale and a post-prediction covariate are substitutes. Scale and a prediction covariate are substitutes if misspecification, $1 - \int_{\mathcal{T}} \omega$, is below $|\mathcal{S}|\,\omega(u)$, and complements if it is above.

By part (i), a covariate changes the marginal value of an observation in three ways. A prediction covariate adds a weight that the firm learns from every additional observation; a covariate of either kind lowers noise, which raises the precision that one more observation adds; and it raises precision, along which the marginal value of an observation rises or falls with the returns to observations. The value of precision is inelastic, $\varepsilon_{\mathcal{S}} \le 1$, while the firm has learned, on average, at most about half of the weights it applies, so scale and covariates of both kinds are complements in small samples. Schäfer and Sápi (2023) document such a complementarity in search, where the gain from additional searches rises with the length of user histories.

In large samples, the firm learns each weight it applies up to an error of misspecification over scale, and an additional observation is worth the fall in these errors, summed over the

$|S|$ weights. A new prediction covariate adds one error to the sum, and it removes its signal from the misspecification in each of the $|S|$ errors already there. When the second effect is larger, the covariate does part of the work that additional observations would do, and the two are substitutes. A post-prediction covariate adds no weight that the firm applies, so in large samples it is always a substitute for observations.

5 Application: Rights over Data

A firm rarely holds all the data it learns from. Individuals hold their own observations, the user of a connected product holds rights over the data her use generates, and other firms hold covariates on the firm's users, such as a broker's demographic file or a bank's transaction history. This section asks how the assignment of rights over the parts of a dataset determines whether bargaining over them closes. When the firm bargains with each holder separately, the answer depends on whether the parts are substitutes or complements (Horn and Wolinsky 1988), and Section 4 says which they are. Groups of covariates are complements, and they exhibit economies of scope (Proposition 3) in every order of acquisition (Section 6). Observations without post-prediction covariates are substitutes, because returns to observations then diminish (Proposition 2(ii)). Observations whose post-prediction covariates satisfy Inequality (8) are complements while they are scarce, because returns to observations then increase. Post-prediction covariates are often recorded with the outcome, so they are recorded with the observation: a search session records the query, the result the user clicked and the user's next search.[24]

5.1 Holders and Bargaining

I work in the limit of Section 3.2 under the assumptions of Section 3.3. There are $m \geq 2$ holders, and the firm earns one unit per unit of value.[25] Write $R^2 \equiv 1 - E^\star$ for the *out-of-sample* R^2, its expected value for the estimated model (Section 3.3). It is the full value; without post-prediction covariates, value equals it and $\alpha_s = \alpha$.

Three assignments of rights. A right over a part of the dataset is control over access to it: the firm can use the part only with the holder's agreement. Portability and access rights give an individual or a user control over whether a firm of her choice obtains her observation, while the source that generated it keeps its copy; the model describes the firm that obtains the data through her. Rights can be vested in holders of observations, in holders of covariates, or

[24] Today a lender learns about repayments to other lenders from a credit bureau, which holds these post-prediction covariates; rights vested in the borrower over all the data on her would make them part of her observation.

[25] If the firm serves a mass μ of new individuals and earns θ per unit of value, every value, payment and payoff below is multiplied by $\theta\mu$. Data are nonrival, so the firm uses one dataset for every new individual.

in both. In the *economy of observations*, the firm observes the covariates of the new individual and obtains its sample from *m holders of observations*: holder h holds the outcomes, covariates and post-prediction covariates of a set of individuals of scale $n_h > 0$, with $\sum_h n_h = n$, as an individual holds her own observation, the user of a connected product the data her use generates, and a hospital the observations of its patients. In the *economy of covariates*, the firm holds the outcome of individuals at scale n, without post-prediction covariates, and obtains covariates from *m holders of covariates*: holder h holds, for the individuals in the sample and the new individual, the covariates in a set $\mathcal{S}_h$, a finite union of intervals, as a broker holds a file on the firm's customers. The sets $\mathcal{S}_h$ are disjoint, each has positive length, and their union is $\mathcal{S}$. In the *joint economy*, without post-prediction covariates, holders of observations hold the outcomes of their individuals and holders of covariates hold the covariates; the firm can use individual i's covariate k only if it reaches agreement both with the holder of i's observation and with the holder of k, as when data-protection law requires an individual's consent for the use of a broker's file on her.

Value and marginal contributions. A coalition $\mathcal{H}$ of holders supplies a dataset. Its value $W(\mathcal{H})$ is $\mathcal{V}(n_{\mathcal{H}}, \mathcal{S}, \mathcal{P})$, where $n_{\mathcal{H}} \equiv \sum_{h\in\mathcal{H}} n_h$, in the economy of observations; $\mathcal{V}\left(n, \bigcup_{h\in\mathcal{H}} \mathcal{S}_h, \varnothing\right)$ in the economy of covariates; and, in the joint economy, the value of the covariates in $\mathcal{H}$ at the scale of the observations in $\mathcal{H}$. The *marginal contribution* of holder h is

$$\Delta_h \equiv W(\mathcal{H}_{\text{all}}) - W(\mathcal{H}_{\text{all}} \setminus \{h\}),$$

where $\mathcal{H}_{\text{all}}$ is the set of all holders. Holdings are *atomistic* in the limit as the largest holding, $\max_h n_h$ or $\max_h \int_{\mathcal{S}_h} \omega$, tends to zero, taken after the large-K limit.

Bargaining. The firm negotiates with every holder simultaneously and bilaterally. As in Collard-Wexler et al. (2019), each pair splits the gains from its own agreement by the generalized Nash bargaining solution, taking the other agreements as given. The weight on the holder is $\varphi \in (0, 1]$, and its disagreement payoff is zero, because data are nonrival and costless to supply.[26] Holder h therefore receives $\varphi\Delta_h$, and the firm's payoff is $\Pi \equiv W(\mathcal{H}_{\text{all}}) - \varphi \sum_h \Delta_h$. With $\varphi = 1$, every holder captures its full marginal contribution, as when it makes a take-it-or-leave-it offer. Payment by marginal contribution is also how machine-learning practice values data (Ghorbani and Zou 2019). Bargaining *closes* when the payments for the complete dataset do not exceed its value, $\varphi \sum_h \Delta_h \leq W(\mathcal{H}_{\text{all}})$. Otherwise, under this bargaining rule, the firm does not acquire the complete dataset: it prefers to buy from a subset of holders, or from none, and part of the data goes unused.

[26] A holder that also sells its data to other firms keeps those sales whether or not it agrees, so they cancel from its gains. A cost c_h of supplying the data, such as a privacy cost, raises the payment to $c_h + \varphi(\Delta_h - c_h)$.

Scope of the analysis. The bargaining model is stylized. Nash-in-Nash bargaining describes holders that each control access to their own part and a firm that bargains with each separately, as in the holdout problems of land assembly (Menezes and Pitchford 2004). With all holders at one table, a stable division exists (footnote 27), so the failures below are failures of bilateral bargaining. I take the data as given: the assignment of rights also shapes the incentives to generate data (Grossman and O. D. Hart 1986; O. Hart and Moore 1990), which is outside the scope of the model. The criterion throughout is whether the firm can acquire the complete dataset under this rule.

5.2 Complements, Substitutes and the Anticommons

Under constant returns, factors paid their marginal products exactly exhaust output (Wicksteed 1894). Data do not have constant returns, and the direction in which they depart from them depends on who holds the data and on whether observations include post-prediction covariates. When the post-prediction covariates satisfy Inequality (8), write $\hat{n}$ for a scale up to which returns to observations increase, which exists by Proposition 2(ii).

Proposition 5 (Holders of observations)**.**

(i) Without post-prediction covariates, marginal contributions fall short of value:

$$\alpha E^\star < \sum_{h=1}^{m} \Delta_h < \mathcal{V}.$$

(ii) If the post-prediction covariates satisfy Inequality (8)*, then at every $n < \hat{n}$ marginal contributions exceed value:*

$$\mathcal{V} < \sum_{h=1}^{m} \Delta_h < \alpha_s E^\star.$$

Without post-prediction covariates, observations are substitutes: each is worth less to a firm that holds more of them, so a holder's marginal contribution falls short of its average contribution, and the payments leave the firm a rent. Below $\hat{n}$, observations with increasing returns are complements: each is worth more to a firm that holds more of them, and the payments exceed value. A search query with its clicked result is a substitute for other queries, and a session that also records the user's next search is a complement while sessions are scarce, if the next search explains enough of the click. The bounds $\alpha E^\star$ and $\alpha_s E^\star$ equal $n\,\partial\mathcal{V}/\partial n$, which is what holders receive as holdings become atomistic (Proposition 7).

Proposition 6 (Holders of covariates)**.** *Marginal contributions exceed value:*

$$\mathcal{V} < \sum_{h=1}^{m} \Delta_h < (1+\alpha)\,\mathcal{V}.$$

A group of covariates is worth more to a firm that holds more groups (Proposition 3), so marginal contributions sum to more than value, and with $\varphi = 1$ the payments to holders of covariates exceed value however many holders there are. With two holders, the excess is the economies of scope, and the upper bound is that of Proposition 3(iii). Section 6 shows that the order in which covariates are acquired leaves them complements, so the result holds when firms acquire strong covariates first.[27]

Proposition 7 (Atomistic holdings)**.** *As holdings become atomistic, marginal contributions sum to*

(i) $\alpha E^\star$*, the share* $\alpha(1-R^2)/R^2 < 1$ *of value, for holders of observations without post-prediction covariates;*

(ii) $\alpha_s E^\star > \mathcal{V}$ *for holders of observations whose post-prediction covariates satisfy Inequality* (8)*, at* $n < \hat{n}$*;*

(iii) $(1+\alpha)\mathcal{V}$ *for holders of covariates;*

(iv) $\alpha E^\star + (1+\alpha)\mathcal{V} = \mathcal{V} + \alpha$*, the share* $1 + \alpha/R^2$ *of value, in the joint economy.*

Atomistic holders of observations receive $n\,\partial\mathcal{V}/\partial n$, so their share of value is the elasticity of value with respect to scale, and the learning multiplier determines it: when the sample grows by one percent, out-of-sample error falls by α percent, and value rises by $0.01\,\alpha E^\star$. The share tends to one as $n \to 0$, where value is proportional to scale, and to zero as $n \to \infty$: an individual's observation is worth little at the margin of a large sample, although all observations together are worth all of the value. With post-prediction covariates, value rises by $0.01\,\alpha_s E^\star$, the fall in the prediction error for new individuals, and below $\hat{n}$ the elasticity exceeds one.

By Proposition 1(ii), the marginal contribution of an atomistic group of covariates is its learned signal plus its spillover. Each group claims the spillovers it creates, yet the firm learns every weight at the precision that all groups raise together, so the claims add up to $(1+\alpha)\mathcal{V}$, and the excess, $\alpha\mathcal{V}$, is the sum of the spillovers. The excess vanishes at small scale, where a group barely raises precision, and at large scale, where $\alpha \to 0$ (Appendix B), so it is largest at intermediate scale. In the joint economy, each holder of observations faces all covariates and each holder of covariates faces all observations, so the two sums add up, and marginal contributions exceed value by α: the elasticity of out-of-sample error with respect to sample size, times the variance of the outcome, which is the firm's prediction error without data.

[27] With $\varphi = 1$, the payments are those of the Vickrey–Clarke–Groves mechanism. By the procurement analogue of Ausubel and Milgrom (2006), these payments lie in the core if and only if value is submodular in holders, as it is for observations without post-prediction covariates and is not for covariates. The game among the firm and the holders of covariates is convex, so its core is nonempty and contains the Shapley value (Shapley 1971) (Online Appendix OA.5.5). A stable division exists.

Table 1. Complements, substitutes and the division of value

	Rights over observations		Rights over	Rights over
	no post-prediction	post-prediction, $n < \hat{n}$	covariates	both
Holdings are	substitutes	complements	complements	see notes
Contributions / value	$\alpha(1-R^2)/R^2 < 1$	$\alpha_s E^\star/\mathcal{V} > 1$	$1+\alpha > 1$	$1+\alpha/R^2 > 1$
Firm's share of value	$1-\varphi\,\alpha(1-R^2)/R^2$	$1-\varphi\,\alpha_s E^\star/\mathcal{V}$	$1-\varphi(1+\alpha)$	$1-\varphi(1+\alpha/R^2)$
Bargaining closes for	every φ	$\varphi \leq \mathcal{V}/(\alpha_s E^\star)$	$\varphi \leq 1/(1+\alpha)$	$\varphi \leq R^2/(R^2+\alpha)$
Payment after a merger	rises	falls	falls	see notes

Notes: The second to fourth rows concern atomistic holdings, and the second gives the sum of marginal contributions relative to value. The fifth gives the change in the payment to two holders when they merge. α is the elasticity of out-of-sample error with respect to sample size, in absolute value, R^2 the out-of-sample R^2 and φ the holders' bargaining weight. "Post-prediction" refers to observations whose post-prediction covariates satisfy Inequality (8), at scales below $\hat{n}$; $\alpha_s E^\star$ is the fall in the firm's prediction error for new individuals per unit of log scale. When rights are vested in both, holders of observations behave among themselves as in the first column and holders of covariates as in the third.

The firm keeps the share $1-\varphi\sum_h \Delta_h/\mathcal{V}$ of value. Under constant returns it would keep $1-\varphi$; it keeps more when holdings are substitutes and less when they are complements. When $\varphi(1+\alpha) > 1$, the payments that bilateral bargaining assigns to atomistic holders of covariates exceed the value they divide, so under this rule the firm cannot acquire the complete dataset and acquires part of it, or none. This is the anticommons of Heller (1998) and Buchanan and Yoon (2000), which Cournot (1838) first identified for complementary monopolists; here α measures it. A firm that buys observations with increasing returns faces it in small samples, once $\varphi\,\alpha_s E^\star > \mathcal{V}$. If the exponents that Hestness et al. (2017) estimate, between 0.07 and 0.35 in absolute value, measured α (Section 3.3), bargaining with atomistic holders of covariates would close only if φ were below a threshold between 0.74 and 0.93. At $\alpha = 0.2$ and $R^2 = 0.5$, the firm keeps $1-0.2\,\varphi$ of value when it buys observations without post-prediction covariates and $1-1.2\,\varphi$ when it buys covariates: 90 and 40 percent at equal bargaining weights. Table 1 summarizes.

5.3 Assignments of Rights

Corollary 2 compares the three assignments of rights: to individuals, each holding her own observation, a row of the dataset, as rights to data portability would give her; to sources, each holding its group of covariates, a set of columns; and to both.

Corollary 2 (Property rights over data)**.** *With atomistic holdings and without post-prediction covariates, bilateral bargaining closes*

(i) for every φ when rights are vested in holders of observations;

(ii) if and only if $\varphi \leq 1/(1+\alpha)$ when they are vested in holders of covariates;

(iii) if and only if $\varphi \leq R^2/(R^2+\alpha)$ when they are vested in both.

Since $1 > 1/(1+\alpha) > R^2/(R^2+\alpha)$*, each assignment closes whenever the next one does.*

An individual who controls access to all the covariates on her integrates complements at the level of the individual, as common ownership does at the level of the firm, and bargaining with holders of observations without post-prediction covariates always closes. Portability rights, such as Article 20 of the GDPR and Article 6(9) of the Digital Markets Act, and the access rights of the Data Act (European Parliament and Council 2023, Articles 4 and 5) move control over access from the sources that generate covariates toward the individuals and users that the covariates describe. Regimes that require both the individual's consent and the source's license are the most prone to breakdown: at $\alpha = 0.2$ and $R^2 = 0.5$, bargaining closes for every φ when rights are vested in individuals, for $\varphi \leq 0.83$ when they are vested in sources, and for $\varphi \leq 0.71$ when they are vested in both.

Where individual rights fail. When observations have increasing returns, rights vested in individuals fail as well: below $\hat{n}$, bargaining breaks down once φ exceeds $\mathcal{V}/(\alpha_s E^\star)$ (Proposition 7(ii)). By Section 4.1, Inequality (8) is easier to satisfy when the covariates available at prediction are many and individually weak, and the individual's later behavior reveals much about the outcome. Search is the clearest case, and Section 4.1 maps it into the model: each term of a query reveals little about which result the user will click, and her next search often reveals what she wanted. Three other markets share this structure: recommendation, where what a user watches or buys next reveals her taste; clinical prediction, where later test results reveal the diagnosis; and credit scoring for applicants without a credit history, where later repayments of other debts reveal creditworthiness. In these markets, a firm that buys observations from individuals faces the anticommons while its sample is small, so individual rights harm small firms and entrants most. Evidence that the GDPR raised concentration among web technology vendors (Johnson et al. 2023) fits this prediction, although it concerns the cost of consent. Whether returns to observations increase in a given market is a quantitative question, which the firm's own data answer: they increase where prediction error for new individuals falls at an increasing rate with sample size.

5.4 Unions, Intermediaries and Ecosystems

Holders can also change the terms of bargaining by bargaining together, and firms can avoid bargaining by generating data in house.

Corollary 3 (Unions and intermediaries)**.** *When two holders merge, the firm's payoff changes by the same amount as their payment, in the opposite direction.*

(i) Without post-prediction covariates, the payment of two holders of observations rises strictly when they merge. If all of them bargain as one, they receive $\varphi\mathcal{V}$*, which exceeds what atomistic holders receive by the factor* $R^2/[\alpha(1-R^2)]$*.*

(ii) If the post-prediction covariates satisfy Inequality (8)*, the payment of two holders of observations falls strictly when they merge at $n < \hat{n}$.*

(iii) The payment of two holders of covariates falls strictly when they merge. If all of them bargain as one, they receive $\varphi\mathcal{V}$, a fraction $1/(1+\alpha)$ of what atomistic holders receive, and the firm gains $\varphi\alpha\mathcal{V}$.

Corollary 3 is the result of Horn and Wolinsky (1988) applied to data: suppliers of substitutes gain by bargaining together, and suppliers of complements lose. Holders of observations gain from a union by a factor that is largest when the firm has learned much: at $\alpha = 0.1$ and $R^2 = 0.8$, a union multiplies their payment by 40. The result gives an economic rationale for proposals for data unions and for collective licensing of training data (Arrieta-Ibarra et al. 2018). A union receives $\varphi\mathcal{V} \leq \mathcal{V}$, so collective management of individual rights redistributes value and bargaining still closes. The case for a union holds only for observations without post-prediction covariates, or in large samples: below $\hat{n}$, holders of observations with increasing returns lose by merging (part (ii)), as holders of covariates do. Holders of covariates gain by staying apart, unless $\varphi(1+\alpha) > 1$, in which case separate bargaining fails and merging lets them trade.

Ecosystems. An intermediary that buys covariates from separate holders faces the same excess of claims over value as the firm, so integration avoids it only when a single owner generates the data. A firm that provides several services to the same users, as Google does with Search, Gmail, Maps, Android and Google Pay, can generate complementary covariates in house, where the law allows it to combine them,[28] and a search engine records each user's next search in its own logs. Such an ecosystem holds every part itself, and Proposition 7 and Corollary 3(iii) show what this changes. When $\varphi(1+\alpha) > 1$, bargaining with separate holders breaks down, and integration lets the data be used. When $\varphi(1+\alpha) \leq 1$, bargaining closes, and integration leaves value unchanged and changes only who receives it: an intermediary that merges the holders keeps the spillovers, $\varphi\alpha\mathcal{V}$, that separate holders would claim, and an ecosystem that generates the data keeps all of the value. In both cases, integration gives the ecosystem an advantage over an entrant that must assemble the parts from separate holders and faces the anticommons that the ecosystem avoids. The advantage follows from the complementarity of the parts. It is largest where observations have increasing returns: below $\hat{n}$, an entrant that buys search sessions with each user's next search from individuals pays more than they are worth, while the first sessions it records itself are worth less than later ones. Increasing returns to observations thus raise the cost of entering by acquiring data, and individual rights leave that cost in place; whether it deters entry depends on how firms compete, which is outside the scope of the model.

[28] Article 5(2) of the Digital Markets Act bars gatekeepers from combining personal data across their services without the user's consent (European Union 2022).

5.5 The Price of Mandated Access

Rules that oblige holders to make data available need a compensation rule. The Data Act requires that the compensation for data a holder must make available be "non-discriminatory and reasonable" and allows it to "include a margin" (European Parliament and Council 2023, Article 9(1)). When several holders hold covariates on the same products or users, paying each its marginal contribution stacks claims, as patent royalties stack when many holders own complementary patents (Lemley and Shapiro 2007). A rule that deflates by the learning multiplier avoids the stacking.

Proposition 8 (Payment of learned signal)**.** *In the economy of covariates, pay holder h the amount $\lambda\, v(\tau^\star, \mathcal{S}_h)$, the learned signal of its group times some $\lambda \in [0, 1]$.*

(i) Payments sum to $\lambda\mathcal{V}$ for every partition.

(ii) The allocation lies in the core of the game among the firm and the holders.

(iii) As holdings become atomistic, each payment approaches $\lambda/(1+\alpha)$ times the holder's marginal contribution: $\sum_h \left|\lambda\, v(\tau^\star, \mathcal{S}_h) - \lambda\Delta_h/(1+\alpha)\right| \to 0$.

A group's learned signal is the increase in out-of-sample error when the firm removes the group's covariates from its estimated model without re-estimating it. The rule is the top-down approach to fair, reasonable and non-discriminatory royalties: cap the aggregate and apportion it, here by the model. Some platforms that license content for generative AI already divide revenue among content owners in proportion to measured contributions.[29] When holdings are small, dividing a fixed sum $P \leq \mathcal{V}$ in proportion to marginal contributions approximates the rule with $\lambda = P/\mathcal{V}$: by part (iii), marginal contributions sum to approximately $(1+\alpha)\mathcal{V}$, so holder h receives approximately $P\Delta_h/[(1+\alpha)\mathcal{V}]$, and $\Delta_h/(1+\alpha)$ approximates $v(\tau^\star, \mathcal{S}_h)$. For observations without post-prediction covariates, marginal contributions already lie in the core, because these observations are substitutes.

The Digital Markets Act requires a gatekeeper to give rival search engines access to its query, click and view data, anonymized where they are personal data (European Union 2022, Article 6(11)). Each shared search is an observation: the query provides the prediction covariates, the clicked result is the outcome, and the user's next search is a post-prediction covariate. If the shared data link each query to the same user's next search, and the next search explains enough of the click to satisfy Inequality (8), the shared observations have increasing returns below $\hat{n}$, so they are worth more to a rival that already has many observations of this kind than to an entrant. If anonymization removes this link, the shared observations have no post-prediction covariate and diminishing returns, so the first ones shared are worth the most. A mandate that keeps the link thus favors established rivals over entrants.

[29] ProRata.ai keeps half of its revenue and divides the other half among publishers according to their contribution to each answer (Salazar 2024); Bria pays the owners of its training data according to their influence on each generated output (Bria 2025).

6 Extension: Returns to Scope and the Order of Acquisition

Section 4.2 compared fixed groups of covariates. Returns to scope compare the next covariate with the last along the order in which the firm acquires covariates. Firms acquire the covariates with most signal first: a search engine records its users' queries before it buys their histories from a broker, and a lender reads an application before it buys a bureau file. The next covariate is then weaker than the last, and returns to scope can diminish. This section shows that the order decides which covariate comes next and leaves unchanged how each covariate raises the value of the others: covariates are complements in every order, which is all that the results on rights over data in Section 5 require. When a firm acquires strong covariates first, returns to scope diminish in small and in large samples and can increase in between.

Orders and profiles. Let the firm acquire covariates of each kind in the order of their positions, so that it observes $\mathcal{S} = [0, s]$ and $\mathcal{P} = (\overline{s}, \overline{s} + p]$. *Scope* s and *post-prediction scope* p are their lengths, and the *signal profiles* $\omega_s(u) \equiv \omega(u)$ on $[0, \overline{s}]$ and $\omega_p(u) \equiv \omega(\overline{s} + u)$ on $[0, \overline{p}]$ give the signal of the covariate at position u of each order. A decreasing profile describes a firm that acquires strong covariates first, and a step profile describes groups of covariates of similar signal. A firm free to choose the order acquires covariates in decreasing order of signal, which weakly raises the value of every dataset (Proposition OA.1 in Online Appendix OA.3.2), so the order that this section studies is the one a firm would choose. Write $\mathcal{V}(n, s, p)$ for value and, for $x \in \{s, p\}$, $\ell_x \equiv \ell(\tau^\star \omega_x(x))$ for the fraction learned of the marginal covariate.[30] By Proposition 1(ii), the marginal value of each kind of scope is

$$\mathcal{V}_x = \big(\mathbf{1}\{x = s\} + \alpha_s\big)\, \omega_x(x)\, \ell_x, \tag{10}$$

so a post-prediction covariate is all spillover. Returns to x *increase* at scale n if $\mathcal{V}_{xx} > 0$, so that the next covariate of that kind adds more value than the last, and they *diminish* if $\mathcal{V}_{xx} < 0$.

Proposition 9 (Returns to scope)**.** *Let* $x \in \{s, p\}$.

(i) The curvature of value in x *is*

$$\begin{aligned} \mathcal{V}_{xx} = {} & \underbrace{\big(\mathbf{1}\{x = s\} + \alpha_s\big)\ell_x(2 - \ell_x)\, \omega_x'(x)}_{\text{composition: change in signal}} \\ & + \underbrace{\big(\mathbf{1}\{x = s\} + \alpha_s\big)\omega_x(x)^2(1 - \ell_x)^2\, \tau_x^\star}_{\text{complementarity: improved learning } (+)} + \underbrace{\omega_x(x)\, \ell_x\, \frac{\partial \alpha_s}{\partial x}}_{\substack{\text{complementarity:} \\ \text{stronger spillovers } (+)}} . \end{aligned} \tag{11}$$

[30] Whenever I differentiate with respect to a margin, its profile is continuously differentiable near the marginal position; a step profile satisfies this away from its jumps.

(ii) Scope and post-prediction scope are complements:

$$\mathcal{V}_{sp} = (1+\alpha_s)\,\omega_s(s)^2(1-\ell_s)^2\,\tau_p^\star + \omega_s(s)\,\ell_s\,\frac{\partial \alpha_s}{\partial p} > 0.$$

Part (i) separates the two ways in which moving along the order changes the marginal value of scope. The first term is composition: the next covariate has the signal of the last plus its change, and each unit of signal is worth $\ell(2-\ell)$, because the firm also learns a stronger covariate better. The other two terms are complementarity: the last covariate lowers noise, so the firm learns the weight of the next covariate better ($\tau_x^\star > 0$), and it adds its speed to the loop and lowers noise, which speeds up the learning of every weight and raises the spillover coefficient. These two terms are the increase in the marginal value of a covariate of given signal when scope grows. They are positive in every order, and only composition depends on how signal changes along the order.[31]

Corollary 4 (Returns to scope)**.** *For $x \in \{s, p\}$:*

(i) if signal does not decline at x, $\omega_x'(x) \geq 0$, returns to x increase at every scale;

(ii) if signal declines at x, $\omega_x'(x) < 0$, there is a positive function $d_x(n)$, which does not depend on $\omega_x'(x)$ and tends to zero as $n \to 0$ and as $n \to \infty$, such that returns to x increase at scale n if and only if $-\omega_x'(x) < d_x(n)$.

The function d_x is complementarity per unit of decline. In small samples, a new covariate barely lowers noise; in large samples, the firm has nearly learned the weights it applies, and the spillover coefficient tends to zero. Complementarity therefore vanishes at both ends, while composition remains: a firm that acquires strong covariates first faces diminishing returns to scope in small and in large samples, and increasing returns in between where signal declines slowly enough. With equal signals, value is strictly convex in scope at every scale.[32]

What the order leaves unchanged. Diminishing returns to scope along an order thus reveal declining signal, and covariates remain complements. Groups of covariates exhibit economies of scope at every scale, whatever their signal (Proposition 3), and the spillover of every prediction covariate is the same share, $\alpha_s/(1+\alpha_s)$, of its marginal value, whether the covariate is strong or weak. Propositions 6 and 7 rest on this complementarity and hold for every partition of the covariates among holders, so they hold when firms acquire strong

[31] Ricardo's extensive margin has the same structure: diminishing returns in agriculture arise because the best land is cultivated first (Ricardo 1817), a composition effect independent of how the lands in use interact.

[32] Composition also vanishes if the firm adds covariates in random order: each new covariate is then a random draw from the same distribution of signal, and value is strictly convex in s at every scale (Online Appendix OA.3.3). Carballa-Smichowski et al. (2025) add health covariates in random order and find returns to scope that first increase and then diminish. Spillovers can account for the increase; under a random order, the model predicts no decline.

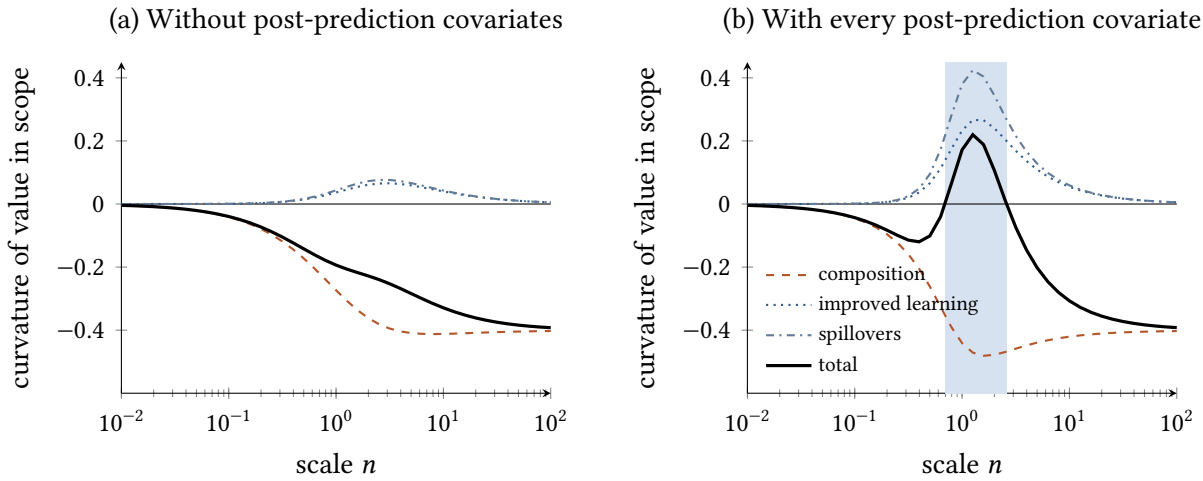


Figure 5. Returns to scope and their three effects. Example of Figure 2, $s = 0.5$. Signal declines along the order. The curvature of value in scope (thick) is the sum of composition (dashed), improved learning (dotted) and stronger spillovers (dash-dotted) of Proposition 9(i). Shaded, returns to scope increase.

covariates first. In the example of Figure 2, signal declines along the order: without post-prediction covariates, returns to scope diminish at every scale (Figure 5(a)), yet any two groups of covariates add more value together than apart.

By part (ii), every factor of Equation (10) for a post-prediction covariate rises with scope: a new prediction covariate adds a weight that gains from lower noise, it lowers noise, which raises the learned signal of the post-prediction covariate, and it raises the spillover coefficient. A search engine with long user histories gains more from recording what each user searched for next. Post-prediction covariates in turn raise complementarity in returns to scope: in the example, at $s = 0.5$, returns to scope increase for n between 0.70 and 2.6 with all post-prediction covariates, although signal declines along the order (Figure 5(b)). Returns to post-prediction scope increase more easily than returns to scope (Corollary OA.3 in the Online Appendix): composition scales the whole marginal value of a prediction covariate, with weight $1 + \alpha_s$, and only the spillover of a post-prediction covariate, with weight α_s. The marginal value of post-prediction scope vanishes in small and in large samples, so a post-prediction covariate is worth most at intermediate scale (Corollary OA.2(ii) in the Online Appendix).

7 Conclusion

This paper develops a theory of the value of data along both dimensions of a dataset, observations and covariates, for a firm that learns many effects from a sample and predicts for a new individual. The paper has two results. Returns to observations increase in small samples when post-prediction covariates explain enough of the outcome, and diminish when the firm applies every covariate it learns from. Covariates on the same individuals are complements at every scale, in every order of acquisition. The learning multiplier, one plus the elasticity of out-of-sample error with respect to sample size, sets the spillovers, the strength of the

complementarity, the excess of marginal contributions over value and the price of mandated access.

The results reconcile evidence on returns to data that appeared contradictory. Returns to observations diminish in demand forecasting and image classification, which apply every input, and increase in small samples of search data when each user's next search explains enough of the click. Returns to scope increase unless signal declines fast along the order of acquisition, and scale and scope are complements in small samples and can be substitutes in large ones. The application rests on the second result. Sources that each hold some covariates claim more than the value of the data, while individuals who each control access to their own observation claim less. Access and portability rights therefore let bilateral bargaining close, except where observations have increasing returns.

The analysis leaves three questions open: how these returns shape market structure when better predictions attract users who supply further observations; how the assignment of rights shapes the incentives to generate data; and which multilateral mechanisms reach the core where bilateral bargaining fails. Estimates of how out-of-sample error falls with sample size, across prediction tasks, would measure the learning multiplier and test a prediction of the theory: prediction error for new individuals falls at an increasing rate only where observations include post-prediction covariates.

Appendix: Proofs

This appendix proves, in order of appearance, the results of Sections 3 to 6 that the text does not prove, and states the supporting results that the proofs use where they first need them. Online Appendix OA.5 proves the claims that the text makes in passing and states and proves further results on post-prediction covariates. Every limit result that concerns sets of covariates depends on a set only through the distribution of signal across its covariates, so it suffices to prove it for initial segments of the two orders of Section 6, $\mathcal{S} = [0, s]$ and the first p post-prediction covariates, with profiles ω_s and ω_p that rearrange the signals. Below, $\mathcal{V}(n, s, p)$ is the value at these sets, and $p = 0$ means $\mathcal{P} = \varnothing$, in which case $\alpha_s = \alpha$. Table OA.1 in the Online Appendix collects the notation. Throughout, $\bar{\omega}$ is the bound on the signal of every potential covariate stated at the start of Appendix A.

A Proofs for Section 3

Signal function. In Section 3.2, $\overline{S} = \lfloor \overline{s}K \rfloor$ and $\overline{P} = \lfloor \overline{p}K \rfloor$, the signal function ω is measurable and bounded by some $\bar{\omega}$, with $\int_0^1 \omega = 1$, and covariate k has signal $K \int_{(k-1)/K}^{k/K} \omega(u)\, du$, the average of ω over the k-th cell of width $1/K$. The profiles of Section 6 are $\omega_s(u) = \omega(u)$ and $\omega_p(u) = \omega(\overline{s} + u)$, and $\underline{E}(s, p) \equiv 1 - \int_0^s \omega_s - \int_0^p \omega_p$ denotes misspecification.

Equalities between random variables hold almost surely.

Proof of Lemma 1. The predictor. Write $y_0 = \sum_k \beta_k x_{0k}$ and take expectations given the data term by term. For $k > S$, the firm does not observe x_{0k}, which has mean zero and is independent of β_k and of the data, so the term contributes zero. For $k \leq S$, the firm observes x_{0k}, and the covariates of the new individual are independent of the weights and of the sample, so $\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}_D] = x_{0k}\, \mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_D] = x_{0k} m_{D,k}$. Summing gives the predictor.

Value. By Equation (1) and the predictor, $V(D) = \mathbb{V}\big(\sum_{k=1}^{S} x_{0k} m_{D,k}\big)$. The covariates of the new individual are mutually independent, independent of the sample, with mean zero and $\mathbb{E}[x_{0k}^2] = \omega_k / K$, so the cross terms vanish and

$$V(D) = \frac{1}{K} \sum_{k=1}^{S} \omega_k\, \mathbb{E}\big[m_{D,k}^2\big] = \frac{1}{K} \sum_{k=1}^{S} \omega_k\, \mathbb{V}(m_{D,k}),$$

because $\mathbb{E}[m_{D,k}] = \mathbb{E}[\beta_k] = 0$. For monotonicity, let $D \leq D'$. The sample under D is a subvector of that under D'. The expected posterior variance of β_k is the lowest mean squared error of any estimate of β_k that is a function of the sample, and the lowest error over a larger set of functions is weakly smaller. Since $\mathbb{V}(m_{D,k}) = \sigma^2 - \mathbb{E}\big[\mathbb{V}(\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_D)\big]$, $\mathbb{V}(m_{D,k}) \leq \mathbb{V}(m_{D',k})$.

The argument uses the covariates in the sample and the covariates observed for the new individual only through the independence of the primitives, never through their order, so it holds for every set of covariates in the sample and every subset $\mathcal{A}$ of them observed for the new individual, with the sum over $\mathcal{A}$ in place of the sum over $k \leq S$. □

Proof of Theorem 1. At $p = 0$, $e(\tau; s, 0) = 1 - v(\tau, s)$, so the limit of value, $v(\tau^\star, s)$, equals $1 - E^\star$, and out-of-sample error equals the lowest expected loss R.

Notation. Fix K and suppress it when no confusion arises. Write C for the set of the $S + P$ covariates in the sample, the first S covariates and the first P post-prediction covariates, indexed $k \in C$ in matrices, $\boldsymbol{X} \equiv \boldsymbol{X}_{D_K}$, $\boldsymbol{G} \equiv \boldsymbol{X}^\top \boldsymbol{X}$, $\boldsymbol{x}_k$ for the column of $\boldsymbol{X}$ of covariate k, $\boldsymbol{X}_{-k}$ for $\boldsymbol{X}$ without that column, and $\boldsymbol{z}_C \equiv (z_k)_{k \in C}$ for the coordinates of a vector $\boldsymbol{z}$ in C. Let $n_K \equiv N/K$, and let $\bar{c} \equiv \frac{1}{K} \sum_{k \notin C} \omega_k$ be the signal outside the sample at finite K. Step 5 shows that $\bar{c} \to \underline{E}(s, p) > 0$, so $\bar{c} \geq c_0 \equiv \underline{E}(s, p)/2$ and $n_K \in [n/2, 2n]$ for K large. The letter C denotes a positive constant that depends only on σ^2, c_0, $\bar{\omega}$ and n; its value may change from line to line. Steps 2 to 4 keep σ^2 general; the limits of value and out-of-sample error in part (ii) take $\sigma^2 = 1$.

The proof is a leave-one-out argument, the rigorous counterpart of the cavity method of statistical physics (Mézard et al. 1987): Step 3 removes one covariate and shows that the firm learns its weight against a noise common to all weights, and Step 4 closes the system by counting degrees of freedom.

Step 1: the limit equation has a unique solution, and $\Psi' < 1$ there. Write $\int_C f(\omega)\, du \equiv \int_0^s f\big(\omega_s(u)\big)\, du + \int_0^p f\big(\omega_p(u)\big)\, du$ within this proof. The error $e(\tau; s, p) = \underline{E}(s, p) + \int_C \omega\big(1 - \ell(\tau\omega)\big)\, du$ is continuously differentiable in τ on $[0, \infty)$, with $e_\tau = -\int_C \omega^2/(1 + \tau\omega)^2\, du \leq 0$, $e(0; s, p) = 1$ and $e(\tau; s, p) \to \underline{E}(s, p)$ as $\tau \to \infty$ by dominated convergence, where $\underline{E}(s, p) > 0$ by Assumption 1; so e maps into $[\underline{E}(s, p), 1]$. Since $\tau\omega\big(1 - \ell(\tau\omega)\big) = \ell(\tau\omega)$,

$$\tau\, e(\tau; s, p) = \underline{E}(s, p)\, \tau + \int_C \ell(\tau\omega)\, du,$$
$$\frac{d}{d\tau}\big[\tau\, e(\tau; s, p)\big] = \underline{E}(s, p) + \int_C \omega\big(1 - \ell(\tau\omega)\big)^2 du \geq \underline{E}(s, p),$$

so $\tau e(\tau; s, p)$ rises strictly and continuously from zero without bound, and $n = \tau e(\tau; s, p)$ has a unique solution $\tau^\star(n, s, p) > 0$. Then $E^\star(n, s, p) \equiv n/\tau^\star = e(\tau^\star; s, p)$ lies in $[\underline{E}(s, p), 1]$, and the system of Theorem 1(i) has the unique solution $\big(\tau^\star, E^\star, v(\tau^\star, s)\big)$. In terms of noise, $\Psi(E; n, s, p) = e(n/E; s, p)$ for $E > 0$ is continuously differentiable, with

$$\Psi'(E; n, s, p) = \frac{n}{E^2}\big(-e_\tau\big) = \int_C \frac{n\omega^2}{(n\omega + E)^2}\, du \geq 0,$$
$$\frac{\partial^2 \Psi(E; n, s, p)}{\partial E^2} = -\int_C \frac{2n\omega^2}{(n\omega + E)^3}\, du \leq 0,$$

and it extends continuously to $E = 0$ by $\Psi(0; n, s, p) = \underline{E}(s, p)$, the limit of e as $\tau \to \infty$; so $\Psi(\cdot; n, s, p)$ is nondecreasing and concave on $[0, \infty)$. At $\tau^\star$, the derivative of $\tau e(\tau; s, p)$ equals $e + \tau e_\tau = E^\star\big(1 - \Psi'(E^\star; n, s, p)\big)$, which is positive, so $\Psi'(E^\star; n, s, p) < 1$. Since $E < E^\star$ if and only if $n/E > \tau^\star$, that is, $(n/E)\, e(n/E; s, p) > n$,

$$E < \Psi(E; n, s, p) \ \text{ for } E < E^\star(n, s, p), \qquad E > \Psi(E; n, s, p) \ \text{ for } E > E^\star(n, s, p). \tag{12}$$

The step uses only that the profiles are bounded and nonnegative and that the signal outside the sample is positive.

Step 2: the covariates outside the sample act as Gaussian noise whose variance is their signal. The outcome of each individual in the sample is the part that the covariates in the sample explain plus a residual that collects the covariates outside $\mathcal{C}$: covariates and post-prediction covariates outside the sample and never-observed ones,

$$y_i = \boldsymbol{x}_{i\mathcal{C}}^\top \boldsymbol{\beta}_{\mathcal{C}} + w_i, \qquad w_i \equiv \sum_{k \notin \mathcal{C}} \beta_k x_{ik}.$$

Given the weights outside $\mathcal{C}$, the residuals are i.i.d. $\mathcal{N}(0, \sigma^2 c)$, independent of the weights in $\mathcal{C}$ and of $\boldsymbol{X}$, with

$$c \equiv \frac{1}{K} \sum_{k \notin \mathcal{C}} \omega_k \frac{\beta_k^2}{\sigma^2}, \qquad \mathbb{E}[c] = \bar{c}.$$

A firm that knew c would face a Gaussian linear regression with known noise variance, and its posterior mean of the weights in $\mathcal{C}$ would be ridge regression with penalty c: for $\lambda > 0$, let $\tilde{\boldsymbol{m}}(\lambda) \equiv (\lambda \boldsymbol{I} + \boldsymbol{G})^{-1} \boldsymbol{X}^\top \boldsymbol{y}_N$ and $f_k(\lambda) \equiv \left[(\boldsymbol{I} + \boldsymbol{G}/\lambda)^{-1}\right]_{kk}$; then

$$\mathbb{E}[\boldsymbol{\beta}_{\mathcal{C}} \mid \boldsymbol{y}_N, \boldsymbol{X}, c] = \tilde{\boldsymbol{m}}(c), \qquad \mathbb{V}(\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c) = \sigma^2 f_k(c).$$

The firm does not know c, but c averages $K - |\mathcal{C}|$ independent terms and concentrates at $\bar{c}$: $c - \bar{c} = \frac{1}{K} \sum_{k \notin \mathcal{C}} \omega_k (\beta_k^2/\sigma^2 - 1)$ averages independent centered terms with bounded moments, so $\mathbb{E}[(c - \bar{c})^4] \le C/K^2$, and the event $\mathcal{G} \equiv \{c \ge \bar{c}/2\}$ has $\Pr(\mathcal{G}^c) \le 16\, \mathbb{E}[(c - \bar{c})^4]/\bar{c}^4 \le C/K^2$. By the law of total variance, the expected posterior variance of β_k given the sample exceeds that given the sample and c by $\varepsilon_k \equiv \mathbb{E}\left[(\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c] - m_{D,k})^2\right]$. The posterior mean $m_{D,k}$ is the best estimate of $\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c]$ among functions of the sample, and $\tilde{m}_k(\bar{c})$ is one such function, so $\varepsilon_k \le \mathbb{E}\left[(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2\right]$.

On $\mathcal{G}$, both penalties are at least $\bar{c}/2 \ge c_0/2$. The derivative of the ridge estimate with respect to the penalty is $\partial_\lambda \tilde{\boldsymbol{m}}(\lambda) = -\boldsymbol{q}(\lambda)$, with $\boldsymbol{q}(\lambda) \equiv (\lambda \boldsymbol{I} + \boldsymbol{G})^{-2} \boldsymbol{X}^\top \boldsymbol{y}_N$. Given $(\boldsymbol{X}, c)$, $\boldsymbol{y}_N \sim \mathcal{N}\left(0, \sigma^2(\boldsymbol{X}\boldsymbol{X}^\top + c\boldsymbol{I}_N)\right)$, so $q_k(\lambda)$ is centered Gaussian with variance at most

$$\sigma^2 \max_{x \ge 0} \frac{x^2 + cx}{(\lambda + x)^4} \le \sigma^2 \left(\frac{1}{16\lambda^2} + \frac{27c}{256\lambda^3} \right) \le C(1 + c) \qquad \text{for } \lambda \ge c_0/2.$$

By the Cauchy–Schwarz inequality along the segment between c and $\bar{c}$, $(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2 \le |c - \bar{c}| \left| \int_{\bar{c}}^{c} q_k(\lambda)^2 d\lambda \right|$, so, taking expectations given $(\boldsymbol{X}, c)$ and then unconditionally,

$$\mathbb{E}\left[(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2 \mathbf{1}_{\mathcal{G}}\right] \le C\, \mathbb{E}\left[(c - \bar{c})^2 (1 + c)\right] \le C \left(\mathbb{E}[(c - \bar{c})^4]\right)^{1/2} \left(\mathbb{E}[(1 + c)^2]\right)^{1/2} \le \frac{C}{K}.$$

Off $\mathcal{G}$, $(\tilde{m}_k(c) - \tilde{m}_k(\bar{c}))^2 \le 2\tilde{m}_k(c)^2 + 2\tilde{m}_k(\bar{c})^2$. Since $\tilde{m}_k(c) = \mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}, c]$, Jensen's inequality gives $\mathbb{E}[\tilde{m}_k(c)^4] \le \mathbb{E}[\beta_k^4] = 3\sigma^4$. Given $(\boldsymbol{X}, c)$, $\tilde{m}_k(\bar{c})$ is centered Gaussian with variance at

most $\sigma^2 \max_{x\ge 0}(x^2+cx)/(\bar c+x)^2 \le \sigma^2(1+c/\bar c)$, so $\mathbb{E}[\tilde m_k(\bar c)^4] \le C$. By the Cauchy–Schwarz inequality, both terms contribute at most $C\Pr(\mathcal{G}^c)^{1/2} \le C/K$, so $\varepsilon_k \le C/K$. For the posterior variance, $f_k(\lambda) \in [0,1]$ and $f_k'(\lambda) = \big[\boldsymbol{G}(\lambda \boldsymbol{I}+\boldsymbol{G})^{-2}\big]_{kk} \in [0, 1/(4\lambda)]$, so $|\mathbb{E}f_k(c) - \mathbb{E}f_k(\bar c)| \le \mathbb{E}|c-\bar c|/(2c_0) + \Pr(\mathcal{G}^c) \le C/\sqrt{K}$. Since $\mathbb{V}(m_{D,k})/\sigma^2 = 1 - \mathbb{E}[f_k(c)] - \varepsilon_k/\sigma^2$,

$$\max_{k\in\mathcal{C}} \big|\mathbb{V}(m_{D,k})/\sigma^2 - \ell_k^\circ\big| \le \frac{C}{\sqrt{K}}, \qquad \ell_k^\circ \equiv 1 - \mathbb{E}\big[f_k(\bar c)\big]. \tag{13}$$

The same bounds show that ridge regression with penalty $\bar c$ is within $CK^{-1/2}$ of the posterior mean of each weight in $\mathcal{C}$ in root mean square.

Step 3: every weight in $\mathcal{C}$ faces the same learning noise. Let $\boldsymbol{R} \equiv (\bar c \boldsymbol{I}_N + \boldsymbol{X}\boldsymbol{X}^\top)^{-1}$ and $\boldsymbol{R}_{-k} \equiv (\bar c \boldsymbol{I}_N + \boldsymbol{X}_{-k}\boldsymbol{X}_{-k}^\top)^{-1}$, both with operator norm at most $1/\bar c$. By the Schur complement and the push-through identity,

$$f_k(\bar c) = \frac{1}{1+\tau_k}, \qquad \tau_k \equiv \boldsymbol{x}_k^\top \boldsymbol{R}_{-k}\boldsymbol{x}_k, \qquad \text{so} \qquad \ell_k^\circ = \mathbb{E}[\ell(\tau_k)].$$

The column $\boldsymbol{x}_k \sim \mathcal{N}(0, \frac{\omega_k}{K}\boldsymbol{I}_N)$ is independent of $\boldsymbol{R}_{-k}$, so τ_k concentrates around its conditional mean: with $g_{-k} \equiv \frac{1}{N}\operatorname{tr}\boldsymbol{R}_{-k}$,

$$\mathbb{E}[\tau_k \mid \boldsymbol{X}_{-k}] = \omega_k\, n_K\, g_{-k}, \qquad \mathbb{V}(\tau_k \mid \boldsymbol{X}_{-k}) = 2\Big(\frac{\omega_k}{K}\Big)^2 \operatorname{tr}\boldsymbol{R}_{-k}^2 \le \frac{2\bar\omega^2 N}{K^2\bar c^2} \le \frac{C}{K}.$$

The normalized trace $g \equiv \frac{1}{N}\operatorname{tr}\boldsymbol{R}$ is common to all weights, and it concentrates at its mean $\bar g_K \equiv \mathbb{E}[g]$. Removing one column changes $\operatorname{tr}\boldsymbol{R}$ by at most $1/\bar c$, so $|g - g_{-k}| \le 1/(N\bar c)$. Replacing one column by an independent copy changes $\operatorname{tr}\boldsymbol{R}$ by at most $2/\bar c$, so the Efron–Stein inequality gives $\mathbb{V}(g) \le 2|\mathcal{C}|/(N^2\bar c^2) \le C/K$. Since ℓ is 1-Lipschitz on $[0,\infty)$, the three bounds give, uniformly in $k \in \mathcal{C}$,

$$\big|\ell_k^\circ - \ell(n_K\omega_k/E_K)\big| \le \frac{C}{\sqrt{K}}, \qquad E_K \equiv \frac{1}{\bar g_K}. \tag{14}$$

Since $\boldsymbol{R} \preceq \bar c^{-1}\boldsymbol{I}$, $E_K \ge \bar c$. Since $x \mapsto 1/(\bar c + x)$ is convex, Jensen's inequality, applied to the eigenvalues of $\boldsymbol{X}\boldsymbol{X}^\top$ and then to the expectation, gives $\bar g_K \ge 1/\big(\bar c + \mathbb{E}[\frac{1}{N}\operatorname{tr}\boldsymbol{X}\boldsymbol{X}^\top]\big) = 1/\big(\bar c + \frac{1}{K}\sum_{k\in\mathcal{C}}\omega_k\big) = 1$, so $E_K \le 1$.

Step 4: counting degrees of freedom determines the learning noise. Since $(\bar c\boldsymbol{I}_N + \boldsymbol{X}\boldsymbol{X}^\top)\boldsymbol{R} = \boldsymbol{I}_N$, taking traces gives $\bar c \operatorname{tr}\boldsymbol{R} + \sum_{j\in\mathcal{C}} \boldsymbol{x}_j^\top\boldsymbol{R}\boldsymbol{x}_j = N$. By the Sherman–Morrison formula, $\boldsymbol{R} = \boldsymbol{R}_{-j} - \boldsymbol{R}_{-j}\boldsymbol{x}_j\boldsymbol{x}_j^\top\boldsymbol{R}_{-j}/(1+\tau_j)$, so $\boldsymbol{x}_j^\top\boldsymbol{R}\boldsymbol{x}_j = \ell(\tau_j)$: the degrees of freedom spent on weight j equal the fraction of it that the firm learns. Taking expectations and dividing by N gives the exact identity

$$\frac{\bar c}{E_K} = 1 - \frac{1}{N}\sum_{j\in\mathcal{C}} \ell_j^\circ. \tag{15}$$

Its limit is Equation (21) below: multiplied by n_K, it reads $\bar c\, n_K/E_K + \frac{1}{K}\sum_{j\in\mathcal{C}}\ell_j^\circ = n_K$. Substi-

tuting (14), multiplying by $E_K \le 1$, and using $\ell(\tau\omega)/\tau = \omega\big(1-\ell(\tau\omega)\big)$ gives

$$\big|E_K - \Psi_K(E_K)\big| \le \frac{C}{\sqrt{K}}, \qquad \Psi_K(E) \equiv \bar{c} + \frac{1}{K}\sum_{j\in C}\omega_j\big(1-\ell(n_K\omega_j/E)\big).$$

Step 5: the finite fixed point converges to the limit one. It remains to show that Ψ_K converges to Ψ uniformly and that $\bar{c} \to \underline{E}(s,p)$. Let $\omega_{s,K}$ be the step function equal on the j-th cell to the signal of the j-th covariate, and define $\omega_{p,K}$ likewise for the post-prediction covariates, so that $\frac{1}{K}\sum_{j=1}^{S} f(\omega_j) = \int_0^{S/K} f(\omega_{s,K}(u))\,du$ for every f, and likewise for the post-prediction covariates. Averaging over the cells of a uniform partition is a contraction in L^1 and converges uniformly for continuous functions, which are dense in L^1; hence $\|\omega_{s,K} - \omega_s\|_{L^1} \to 0$ and $\|\omega_{p,K} - \omega_p\|_{L^1} \to 0$. With $f(\omega) = \omega$, the signal of the covariates in the sample converges to $\int_0^s \omega_s(u)\,du + \int_0^p \omega_p(u)\,du$, and the normalization gives $\bar{c} \to \underline{E}(s,p)$. The functions $f_{E,m}(\omega) \equiv \omega\big(1-\ell(m\omega/E)\big) = \omega E/(m\omega+E)$ satisfy $|\partial_\omega f_{E,m}| \le 1$ and are Lipschitz in (E,m) uniformly over $\omega \in [0,\bar{\omega}]$, $E \in [c_0,1]$ and $m \in [n/2, 2n]$. Therefore

$$\sup_{E\in[c_0,1]}\left|\int_0^{S/K} f_{E,n_K}\big(\omega_{s,K}(u)\big)\,du - \int_0^{s} f_{E,n}\big(\omega_s(u)\big)\,du\right| \le \|\omega_{s,K}-\omega_s\|_{L^1} + \bar{\omega}\left|\frac{S}{K}-s\right| + C|n_K-n| \longrightarrow 0,$$

and likewise for the post-prediction covariates. With $\bar{c} \to \underline{E}(s,p)$, this gives

$$\sup_{E\in[c_0,1]} |\Psi_K(E) - \Psi(E;n,s,p)| \to 0,$$

and hence $\Phi(E_K) \to 0$. Since $E_K \in [c_0,1]$ for K large and Φ is continuous with the unique zero $E^\star(n,s,p)$, every limit point of (E_K) equals $E^\star(n,s,p)$, so $E_K \to E^\star(n,s,p)$.

Part (ii). The convergence of the fractions learned follows from (13), (14), $E_K \to E^\star(n,s,p)$ and $n_K \to n$, because $(\omega,m,E) \mapsto \ell(m\omega/E)$ is uniformly Lipschitz on $[0,\bar{\omega}]\times[n/2,2n]\times[c_0,1]$. For value, where $\sigma^2 = 1$, Lemma 1 gives $V(D_K) = \frac{1}{K}\sum_{k=1}^{S}\omega_k \mathbb{V}(m_{D_K,k})$. By the convergence of the fractions learned,

$$\Big|V(D_K) - \frac{1}{K}\sum_{k=1}^{S}\omega_k\,\ell(\tau^\star\omega_k)\Big| \le \bar{\omega}\max_{k\in C}\big|\mathbb{V}(m_{D_K,k}) - \ell(\tau^\star\omega_k)\big| \longrightarrow 0.$$

Since $|\partial_\omega[\omega\,\ell(\tau\omega)]| \le 1$, the argument of Step 5 shows that the sum converges to $\int_0^s \omega_s(u)\,\ell(\tau^\star\omega_s(u))\,du$. The same argument applies to any set of covariates observed at prediction whose positions form a finite union of intervals within the positions of the sample, such as the whole sample, which the firm observes for an individual left out of the sample. By Lemma 1 for that set, which holds for every subset of the covariates in the sample observed for the individual whose outcome the firm predicts, one minus out-of-sample error is $\frac{1}{K}\sum_k \omega_k \mathbb{V}(m_{D_K,k})$, the sum over the $S+P$ covariates in the sample, so it converges to $1-\Psi(E^\star;n,s,p) = 1-E^\star$. □

B Proofs for Section 4

From here on, $\sigma^2 = 1$, and the assumptions of Section 3.3 hold unless stated otherwise; when $p > 0$, the post-prediction profile is also positive and continuously differentiable near p whenever I differentiate with respect to p. The results hold for every p; at $p = 0$, the averages $\langle\cdot\rangle$ and $\langle\cdot\rangle_s$ below coincide, $\Psi'_s = \Psi'$ and $\alpha_s = \alpha$. In integrals over positions, ω stands for $\omega_s(u)$ on $[0, s]$ and for $\omega_p(u)$ on $[0, p]$. In proofs, $M \equiv 1 + \alpha$ denotes the learning multiplier. Then $\mathcal{V}(n, s, p) = \int_0^s \omega\, \ell\big(n\omega/E^\star(n, s, p)\big)\, du$ is continuously differentiable in n, and continuously differentiable in (n, s, p) on every open set of margins on which ω_s is continuous at s and ω_p at p (proof of Lemma 3). The next lemma collects the derivatives of the fraction learned and of the learned signal at noise E, and that of Ψ'; in it, ℓ stands for $\ell(n\omega/E)$.

Lemma 2 (Derivative identities)**.** *For $n > 0$ and $E > 0$, with $\omega \geq 0$ in (a), (c) and (e) and $\omega > 0$ in (b):*

(a) $\dfrac{\partial \ell}{\partial E} = -\dfrac{\ell(1-\ell)}{E}$ *and* $\dfrac{\partial \ell}{\partial n} = \dfrac{\ell(1-\ell)}{n}$;

(b) $\dfrac{\partial \ln(\omega\ell)}{\partial \ln \omega} = 2 - \ell$ *and* $\dfrac{\partial \ln(\omega\ell)}{\partial E} = -\dfrac{1-\ell}{E}$;

(c) $\dfrac{\ell^2}{n} = \dfrac{\omega\,\ell(1-\ell)}{E}$;

(d) $\dfrac{\partial \Psi'(E; n, s, p)}{\partial E} = -\dfrac{2}{nE}\Big[\int_0^s \ell\Big(\dfrac{n\omega}{E}\Big)^2\Big(1-\ell\Big(\dfrac{n\omega}{E}\Big)\Big)du + \int_0^p \ell\Big(\dfrac{n\omega}{E}\Big)^2\Big(1-\ell\Big(\dfrac{n\omega}{E}\Big)\Big)du\Big]$;

(e) the learned signal satisfies

$$\frac{\partial(\omega\ell)}{\partial E} = -\frac{\ell^2}{n}, \qquad \frac{\partial(\omega\ell)}{\partial n} = \frac{E}{n}\cdot\frac{\ell^2}{n}. \tag{16}$$

Proof. With $\tau = n/E$, $d\ell(\tau\omega)/d\tau = \omega/(1+\tau\omega)^2 = \ell(1-\ell)/\tau$, and $\partial \ln\tau/\partial E = -1/E$, $\partial \ln\tau/\partial n = 1/n$, which gives (a). For (b), $\ln(\omega\ell) = \ln\tau + 2\ln\omega - \ln(1+\tau\omega)$. For (c), both sides equal $n\omega^2/(n\omega+E)^2$. For (d), $\Psi'(E; n, s, p)$ is $1/n$ times the integrals of ℓ^2 over the two kinds of covariates; differentiate under the integrals and use (a). For (e), $\partial(\omega\ell)/\partial E = -n\omega^2/(n\omega+E)^2$ and $\partial(\omega\ell)/\partial n = \omega^2 E/(n\omega+E)^2$. □

The next lemma collects the marginal values of the three margins.

Lemma 3 (Marginal values)**.**

(i) Precision rises with every margin:

$$\frac{\partial \ln\tau^\star}{\partial \ln n} = 1+\alpha, \qquad \frac{\partial \ln\tau^\star}{\partial s} = (1+\alpha)\,\frac{\omega_s(s)\,\ell_s}{E^\star}, \qquad \frac{\partial \ln\tau^\star}{\partial p} = (1+\alpha)\,\frac{\omega_p(p)\,\ell_p}{E^\star}.$$

(ii) Value rises with every margin:

$$\frac{\partial \mathcal{V}}{\partial \ln n} = \alpha_s \, E^\star, \qquad \frac{\partial \mathcal{V}}{\partial s} = \omega_s(s) \, \ell_s \, (1 + \alpha_s), \qquad \frac{\partial \mathcal{V}}{\partial p} = \omega_p(p) \, \ell_p \, \alpha_s.$$

Proof of Lemma 3. (i). The map $\Psi(E; n, s, p)$ is continuously differentiable in (E, n) for $E > 0$, with $\partial\Psi/\partial n = -(E/n)\,\Psi'(E; n, s, p)$ by (16). On an open interval of positions on which ω_s is continuous, it is also continuously differentiable in s, with $\partial\Psi/\partial s = -\omega_s(s)\,\ell(n\omega_s(s)/E)$, and likewise in p. Because $1 - \Psi'(E^\star; n, s, p) > 0$ (Step 1 of the proof of Theorem 1), the implicit function theorem applies to $E = \Psi(E; n, s, p)$ and gives $\partial E^\star/\partial x = M\,\partial\Psi/\partial x$, for $x = n$ at every (s, p) and for $x = s$ or $x = p$ wherever the profile is continuous on a neighborhood of the marginal position. With $\Psi' M = M - 1 = \alpha$, this gives the derivatives of $E^\star$, and $\tau^\star = n/E^\star$ gives $\partial \ln \tau^\star/\partial \ln n = 1 - \partial \ln E^\star/\partial \ln n = 1 + \alpha$ and $\partial \ln \tau^\star/\partial x = -\partial \ln E^\star/\partial x$ for $x = s, p$. The arrows of Figure 3 are the partial derivatives of Equation (3) with scale, noise and precision in logs: $\ln \tau = \ln n - \ln E$ gives 1 and -1; $\partial \ln e/\partial \ln \tau = \tau e_\tau/e$, which is $-\Psi'$ at $\tau^\star$ by Step 1 of the proof of Theorem 1; at $\tau^\star$, $\partial \ln e/\partial s = -\omega_s(s)\,\ell_s/E^\star$ and $\partial \ln e/\partial p = -\omega_p(p)\,\ell_p/E^\star$, the arrow that Section 3.3 adds; and scope moves the upper limit of the integral that defines value, which adds $\omega_s(s)\,\ell(\tau\omega_s(s))$ at fixed precision.

(ii). The fraction learned of each weight, $\ell(\tau^\star\omega)$, depends on scale and on post-prediction scope only through $\tau^\star$, with elasticity $1 - \ell$. Differentiating $\mathcal{V} = \int_0^s \omega\,\ell(\tau^\star\omega)\,du$ under the integral, which the bounded derivatives in (16) permit because $E^\star \geq \underline{E}(s, p) > 0$, and using Lemma 2(c), the effect of precision on value at fixed scope is

$$\Gamma \equiv \int_0^s \omega\,\ell(\tau^\star\omega)\big(1 - \ell(\tau^\star\omega)\big)\,du = \frac{E^\star}{n}\int_0^s \ell(\tau^\star\omega)^2\,du = E^\star\Psi'_s = \frac{\alpha_s E^\star}{M}, \tag{17}$$

which is the arrow from precision to value in Figure 3. The chain rule with (i) gives

$$\frac{\partial \mathcal{V}}{\partial \ln n} = M\Gamma = \alpha_s E^\star, \qquad \frac{\partial \mathcal{V}}{\partial p} = \Gamma\,\frac{\partial \ln \tau^\star}{\partial p} = \omega_p(p)\,\ell_p\,\alpha_s.$$

Scope also moves the upper limit of the integral, which adds $\omega_s(s)\,\ell_s$, so $\partial\mathcal{V}/\partial s = \omega_s(s)\,\ell_s(1 + \alpha_s)$. □

Averages. For a function g on $[0, 1]$, write

$$\langle g(\ell)\rangle \equiv \frac{\int_0^s \ell\big(\tau^\star\omega_s(u)\big)^2\, g\big(\ell(\tau^\star\omega_s(u))\big)\,du + \int_0^p \ell\big(\tau^\star\omega_p(u)\big)^2\, g\big(\ell(\tau^\star\omega_p(u))\big)\,du}{\int_0^s \ell\big(\tau^\star\omega_s(u)\big)^2\,du + \int_0^p \ell\big(\tau^\star\omega_p(u)\big)^2\,du}$$

for the average of g of the fraction learned over the sample, each covariate weighted by its contribution to the slope Ψ', and $\langle g(\ell)\rangle_s$ for the average over the covariates observed for the new individual, which keeps only the first integral in the numerator and in the denominator.

Write $\tilde{\ell} \equiv \langle \ell \rangle$, $\bar{\ell}_s \equiv \langle \ell \rangle_s$ and $\bar{\ell} \equiv \alpha\langle 1-\ell \rangle$, and $\Psi'_s \equiv \tau^\star v_\tau(\mathcal{S})/E^\star$, so that $\alpha_s = (1+\alpha)\Psi'_s$. Then $\varepsilon_{\mathcal{S}} = 2\bar{\ell}_s$, $\varepsilon_{\mathcal{T}} = 2\tilde{\ell}$ and $\alpha(2-\varepsilon_{\mathcal{T}}) = 2\bar{\ell}$ (Steps 1 and 2 below). Write $\kappa(s) \equiv \int_0^s \omega_s^3 / \int_0^s \omega_s^2 - \int_0^s \omega_s^2$ for the *threshold*, so that Inequality (8) reads $\int_0^p \omega_p^2 > \kappa(s)$.

Returns to scale. Throughout the next results, write $\underline{E} \equiv \underline{E}(s,p) > 0$; for a precision τ and a signal ω, write $\ell \equiv \ell(\tau\omega)$, which is the fraction learned at $\tau = \tau^\star$, and $\psi \equiv \omega(1-\ell) = \omega/(1+\tau\omega)$ for the unlearned signal of a weight, so that $\tau\psi = \ell$ and $\partial\psi/\partial\tau = -\psi^2$. Write $\int_C f(\omega)\,du \equiv \int_0^s f\big(\omega_s(u)\big)\,du + \int_0^p f\big(\omega_p(u)\big)\,du$ for an integral over all the covariates in the sample, so that $e(\tau; s, p) = \underline{E} + \int_C \psi\,du$.

Step 1: the value of precision. At fixed scope, write $F(\tau) \equiv v(\tau, s) = \int_0^s \omega\ell\,du$. Since $\partial(\omega\ell)/\partial\tau = \omega^2(1-\ell)^2 = \psi^2$ and $\partial\psi^2/\partial\tau = -2\psi^3$, differentiating under the integral gives $v_\tau = F'(\tau) = \int_0^s \psi^2 du > 0$, because $\int_0^s \omega\,du > 0$, and $v_{\tau\tau} = F''(\tau) = -2\int_0^s \psi^3 du < 0$. Since $\psi = \ell/\tau$, $F'' = -(2/\tau)\int_0^s \psi^2\ell\,du = -(2/\tau)\langle\ell\rangle_s F'$, because ψ^2 is proportional to ℓ^2; and $F' = \Gamma/\tau = E^{\star 2}\Psi'_s/n$ by (17).

Step 2: the marginal value of an observation and the loop. By Lemma 3(i), $\partial\tau^\star/\partial n = M\tau^\star/n = M/E^\star$, so $\partial\mathcal{V}/\partial n = F'(\tau^\star)\,M/E^\star = \alpha_s E^\star/n$, which is Equation (5), and $\partial E^\star/\partial n = E^\star(1-M)/n = -\alpha E^\star/n$. Write $\Psi'' \equiv \partial^2\Psi(E^\star; n, s, p)/\partial E^2$ for the curvature of Ψ at learning noise. By Lemma 2(d), $\Psi'' = -\frac{2}{nE}\int_C \ell^2(1-\ell)\,du < 0$: lower noise speeds up the learning of every weight. By Lemma 2(c), $\Psi' = \frac{1}{E}\int_C \omega\ell(1-\ell)\,du$ at every noise E, and by Lemma 2(a), $\partial\ell/\partial n = \ell(1-\ell)/n$ at fixed noise, so $\partial\Psi'/\partial n = \frac{1}{nE}\int_C \omega\ell(1-\ell)(1-2\ell)\,du = \frac{1}{n^2}\int_C \ell^2(1-2\ell)\,du$: at given noise, each weight contributes its speed, $\omega\ell(1-\ell)/E = \ell^2/n$, times $(1-2\ell)/n$. Since $\partial\alpha/\partial\Psi' = 1/(1-\Psi')^2 = M^2$, the chain rule along $E^\star(n,s,p)$ splits the change in the amplification into a shift of the map at given noise and a move along it, through lower noise:

$$\frac{\partial\alpha}{\partial n} = M^2\Big[\underbrace{\frac{\partial\Psi'}{\partial n}}_{\substack{\text{at given}\\ \text{noise } (\pm)}} + \underbrace{\Psi''\frac{\partial E^\star}{\partial n}}_{\text{lower noise } (+)}\Big]. \tag{18}$$

At $E = E^\star$,

$$\frac{\partial\alpha}{\partial n} = \frac{M^2}{n^2}\Big[2\alpha\int_C \ell^2(1-\ell)\,du + \int_C \ell^2(1-2\ell)\,du\Big] = \frac{M\alpha}{n}\Big[2\alpha\langle 1-\ell\rangle + 1 - 2\langle\ell\rangle\Big],$$

using $\int_C \ell^2 du = n\Psi'$ and $M\Psi' = \alpha$. The bracket equals $2M\langle 1-\ell\rangle - 1$, which is positive if and only if $\langle\ell\rangle < 1 - 1/(2M) = (1+\Psi')/2$. As $n \to \infty$, $\tau^\star \to \infty$ and $\ell \to 1$ wherever $\omega > 0$, so $\langle\ell\rangle \to 1$ by dominated convergence and the bracket tends to -1, while $\alpha = M\Psi'$ and $\Psi' \le (s+p)/n \to 0$.

Step 3: returns to scale. Since $\partial(1/E^\star)/\partial n = \alpha/(nE^\star)$, differentiating $\partial\tau^\star/\partial n = M/E^\star$ gives $\partial^2\tau^\star/\partial n^2 = M\alpha/(nE^\star) + (1/E^\star)\,\partial\alpha/\partial n$, and the chain rule splits the curvature of value in scale

into three terms:

$$\frac{\partial^2 \mathcal{V}}{\partial n^2} = \underbrace{v_{\tau\tau}\Big(\frac{\partial \tau^\star}{\partial n}\Big)^2}_{\substack{\text{value of}\\ \text{precision } (-)}} + \underbrace{v_\tau\, M\, \frac{\alpha}{nE^\star}}_{\text{impact } (+)} + \underbrace{v_\tau\, \frac{1}{E^\star}\, \frac{\partial \alpha}{\partial n}}_{\text{amplification } (\pm)} . \tag{19}$$

Without the loop, $\alpha = 0$, and the last two terms vanish. By Step 2

$$\frac{\partial^2 \tau^\star}{\partial n^2} = \frac{M\alpha}{nE^\star}\Big[1 + 2\alpha\langle 1-\ell\rangle + 1 - 2\langle\ell\rangle\Big] = \frac{2M^2\alpha\langle 1-\ell\rangle}{nE^\star} = \frac{2\bar{\ell}}{\tau^\star}\Big(\frac{\partial \tau^\star}{\partial n}\Big)^2 > 0,$$

with $\bar{\ell} = \alpha\langle 1-\ell\rangle < \alpha$ and $\tilde{\ell} = \langle\ell\rangle$. With Step 1 and $\bar{\ell}_s = \langle\ell\rangle_s$, this gives the curvature of value in scale,

$$\frac{\partial^2 \mathcal{V}}{\partial n^2} = \frac{2}{\tau^\star}\, v_\tau\Big(\frac{\partial \tau^\star}{\partial n}\Big)^2\big(\bar{\ell} - \bar{\ell}_s\big) = \frac{1}{\tau^\star}\, v_\tau(\mathcal{S})\Big(\frac{\partial \tau^\star}{\partial n}\Big)^2\Big(\underbrace{\alpha(2-\varepsilon_{\mathcal{T}})}_{\substack{\text{acceleration}\\ \text{of precision}}} - \underbrace{\varepsilon_{\mathcal{S}}}_{\substack{\text{decay of the}\\ \text{value of precision}}}\Big) : \tag{20}$$

a one percent increase in precision raises the precision that one more observation adds by $\alpha(2-\varepsilon_{\mathcal{T}})$ percent and lowers the value of precision for the weights the firm applies by $\varepsilon_{\mathcal{S}}$ percent. The full and the reflected value share the sample, and so the acceleration, so Equation (20) holds for each with $\mathcal{T}$ or $\mathcal{P}$ in place of $\mathcal{S}$. Steps 2 and 3 prove Proposition 1(i) and Equation (20). Differentiating $\mathcal{V} = v(\tau^\star, \mathcal{T}) - v(\tau^\star, \mathcal{P})$ (Corollary 1) twice in n gives Proposition 2(i), with $\tau^\star_{nn} > 0$ by Step 3, $v_\tau(\mathcal{T}) - v_\tau(\mathcal{P}) = v_\tau(\mathcal{S}) > 0$ and $v_{\tau\tau}(\mathcal{T}) - v_{\tau\tau}(\mathcal{P}) = v_{\tau\tau}(\mathcal{S}) < 0$ by Step 1; at $p = 0$, $\tilde{\ell} = \bar{\ell}_s$ and $\bar{\ell} = \alpha(1-\bar{\ell}_s)$. They also prove the claims of Section 4.1 on $\bar{\ell}$. The multiplier rises with scale in small samples and falls in large ones by the continuity of the bracket in Step 2 and its limits, 1 as $n \to 0$, because $\alpha \to 0$ and $\langle\ell\rangle \to 0$ (below), and -1 as $n \to \infty$; and returns to scale increase only if $\bar{\ell}_s < \alpha$ by Equation (20), since $\bar{\ell} < \alpha$. As $n \to 0$, $E^\star \to 1$ because $\Psi(E; n, s, p) \to 1$ for every $E > 0$, so $\tau^\star \to 0$ and $\ell/\tau^\star \to \omega$ uniformly. Hence $F' \to \int_0^s \omega^2 du$, $F'' \to -2\int_0^s \omega^3 du$, $\partial\tau^\star/\partial n \to 1$, $\alpha/n \to Q$, $\langle\ell\rangle \to 0$ and $\partial^2\tau^\star/\partial n^2 \to 2Q$, which gives the limit of $\partial^2\mathcal{V}/\partial n^2$ in Section 4.1.

Step 4: a representation for the proofs below. Multiplying the fixed point $E^\star = \Psi(E^\star; n, s, p)$ by $\tau^\star = n/E^\star$ and using $\tau\omega(1-\ell) = \ell$ gives $n = C(\tau^\star)$, where

$$C(\tau) \equiv \underline{E}\tau + \mathrm{df}(\tau), \qquad \mathrm{df}(\tau) \equiv \int_C \ell(\tau\omega)\, du, \tag{21}$$

is the scale at which the firm attains precision τ. The term df counts one degree of freedom for each weight in the sample, of either kind, weighted by the fraction of it that the firm has learned. Since $\partial\ell/\partial\tau = \omega(1-\ell)^2$, $C'(\tau) = \underline{E} + \int_C \omega(1-\ell)^2 du \geq \underline{E}$, and differentiating $n = C(\tau^\star)$ gives $\partial\tau^\star/\partial n = 1/C'$, so $\tau^\star C'(\tau^\star) = n/M$ by Step 2. Let

$$\bar{\psi}_F(\tau) \equiv \frac{\int_0^s \psi^2 \cdot \psi\, du}{F'(\tau)}, \qquad \bar{\psi}_C(\tau) \equiv \frac{\underline{E}\cdot 0 + \int_C \omega(1-\ell)^2 \cdot \psi\, du}{C'(\tau)}.$$

The first averages the unlearned signal of the weights that the firm applies, each weighted by its term in F'. The second averages it over the weights in the sample, each weighted by its term in C', and over the signal outside the sample, weighted by $\underline{E}$, at $\psi = 0$. Since $\tau\psi = \ell$, $\tau\bar{\psi}_F = \langle\ell\rangle_s = \bar{\ell}_s$. Multiplying the numerator and the denominator of $\tau\bar{\psi}_C$ by τ and using $\tau\omega(1-\ell) = \ell$,

$$\tau\,\bar{\psi}_C = \frac{\int_C \ell^2(1-\ell)\,du}{\underline{E}\tau + \int_C \ell(1-\ell)\,du} = \frac{n\Psi'\langle 1-\ell\rangle}{n/M} = \alpha\,\langle 1-\ell\rangle = \bar{\ell},$$

because $\tau C'(\tau) = n/M$, $n\Psi' = \int_C \ell^2 du$ and $M\Psi' = \alpha$. By Step 3, $\partial^2\mathcal{V}/\partial n^2$ has the sign of $D \equiv \bar{\psi}_C - \bar{\psi}_F$.

Proof of Proposition 2(ii) when $\mathcal{P} = \varnothing$. At $p = 0$, $\langle\ell\rangle = \langle\ell\rangle_s = \bar{\ell}_s$, so by Step 3 and $\alpha = M\Psi'$, $\bar{\ell} - \bar{\ell}_s = \alpha(1-\bar{\ell}_s) - \bar{\ell}_s = M(\Psi' - \bar{\ell}_s)$, so $\partial^2\mathcal{V}/\partial n^2$ has the sign of $\Psi' - \bar{\ell}_s$. By Equation (21), $n = \underline{E}\tau^\star + \int_0^s \ell\,du$, and $n\Psi' = \int_0^s \ell^2 du$ (Lemma 2(c)), so $\Psi' < \bar{\ell}_s$ if and only if

$$\Big(\int_0^s \ell^2 du\Big)^2 < \Big(\underline{E}\tau^\star + \int_0^s \ell\,du\Big)\int_0^s \ell^3 du.$$

By the Cauchy–Schwarz inequality, $\big(\int_0^s \ell^2 du\big)^2 \le \int_0^s \ell\,du \int_0^s \ell^3 du$, and $\underline{E}\tau^\star \int_0^s \ell^3 du > 0$ because $\underline{E} > 0$, $\tau^\star > 0$ and the profile is positive. Hence $\partial^2\mathcal{V}/\partial n^2 < 0$ at every n. Lemma 4(i) below reaches the same conclusion by another route. □

Lemma 4 (Scale in small and large samples)**.**

(i) Returns to scale diminish at every n if every covariate in the sample has at least as much signal as every post-prediction covariate in the sample, in particular if $p = 0$.

(ii) The threshold is positive, $\kappa(s) > 0$. If $\int_0^p \omega_p^2 > \kappa(s)$, there is $\hat{n} > 0$ such that returns to scale increase at every $n < \hat{n}$; if the reverse strict inequality holds, there is $\hat{n} > 0$ such that they diminish at every $n < \hat{n}$.

(iii) There is $n^\dagger > 0$ such that returns to scale diminish at every $n > n^\dagger$.

Proof. (i). The average $\bar{\psi}_C$ runs over three groups: the weights that the firm applies, the post-prediction weights and the signal outside the sample. On the weights that the firm applies, $\bar{\psi}_C$ weights ψ by $\omega(1-\ell)^2$ and $\bar{\psi}_F$ by $\psi^2 = \omega\cdot\omega(1-\ell)^2$; since ψ increases with ω, Chebyshev's integral inequality makes the $\bar{\psi}_C$-weighted average of ψ over these weights at most $\bar{\psi}_F$. On the post-prediction weights, every value of ψ is at most every value on the weights that the firm applies, because ψ increases with ω and these covariates have at most as much signal, and hence at most $\bar{\psi}_F$; without post-prediction covariates, there are no such weights. The signal outside the sample has positive weight $\underline{E}/C'$ and $\psi = 0 < \bar{\psi}_F$. Hence $\bar{\psi}_C < \bar{\psi}_F$ at every $n > 0$.

(ii). As $n \to 0$, $\tau \to 0$ and $\psi \to \omega$ uniformly. Hence $\bar{\psi}_F \to \int_0^s \omega^3 du / \int_0^s \omega^2 du$ and $\bar{\psi}_C \to \int_C \omega^2 du / \big(\underline{E} + \int_C \omega\, du\big) = \int_0^s \omega_s(u)^2 du + \int_0^p \omega_p(u)^2 du$, so $D \to \int_0^p \omega_p(u)^2 du - \kappa(s)$, which is positive if and only if $\int_0^p \omega_p^2 > \kappa(s)$. Both averages are continuous in n, so if the limits differ, there is $\hat{n} > 0$ such that $\bar{\psi}_C - \bar{\psi}_F$ has the sign of their difference at every $n < \hat{n}$. By the Cauchy–Schwarz inequality, $\big(\int_0^s \omega^2\big)^2 \le \int_0^s \omega \int_0^s \omega^3$, so $\int_0^s \omega^3 / \int_0^s \omega^2 \ge \int_0^s \omega^2 / \int_0^s \omega > \int_0^s \omega^2$, because $0 < \int_0^s \omega < 1$; hence $\kappa(s) > 0$. Since $\omega_p > 0$, the map $p \mapsto \int_0^p \omega_p(u)^2 du$ is continuous and strictly increasing from zero, so it crosses $\kappa(s)$ at a unique $\hat{p} \in (0, \bar{p})$ when $\int_0^{\bar{p}} \omega_p(u)^2 du > \kappa(s)$, and $\int_0^p \omega_p^2 > \kappa(s)$ if and only if $p > \hat{p}$.

(iii). As $n \to \infty$, $\tau \to \infty$ and $\ell(\tau^\star \omega) \to 1$ wherever $\omega > 0$, so $\langle \ell \rangle_s \to 1$ by dominated convergence, while $\alpha\langle 1 - \ell \rangle = \tau \bar{\psi}_C \le \int_C \ell(1-\ell) du / (\underline{E}\tau) \le (s+p)/(4\underline{E}\tau) \to 0$. Hence there is $n^\dagger > 0$ such that $\partial^2 \mathcal{V} / \partial n^2 < 0$ at every $n > n^\dagger$.

□

Proportional growth of the factors. Let x be scope or post-prediction scope, s or p. The proofs below divide second derivatives by marginal values, which leaves their signs unchanged. The marginal covariate of kind x has signal $\omega_x(x)$ and fraction learned ℓ_x, and it raises precision at the rate $r_x \equiv \partial \ln \tau^\star / \partial x = M \omega_x(x) \ell_x / E^\star$ (Lemma 3(i)). Inside integrals, ℓ stands for $\ell(\tau^\star \omega)$. Since $\ell(x) = x/(1+x)$, ℓ and $1-\ell$ have elasticities $1-\ell$ and $-\ell$ with respect to $\tau^\star$. The quantity Γ of (17) depends on post-prediction scope only through $\tau^\star$; each term of its integrand has elasticity $1 - 2\ell$, and the terms are proportional to ℓ^2 (Lemma 2(c)), the weights of $\langle \cdot \rangle_s$. Hence

$$\frac{\partial \ln \Gamma}{\partial p} = r_p \langle 1 - 2\ell \rangle_s, \qquad \frac{\partial \Gamma}{\partial s} = \omega_s(s) \ell_s (1 - \ell_s) + r_s \Gamma \langle 1 - 2\ell \rangle_s, \tag{22}$$

where the first term of $\partial \Gamma / \partial s$ comes from the upper limit of the integral. Since $M = 1/\big(1 - \Psi'(E^\star; n, s, p)\big)$ and the new covariate adds ℓ_x^2/n to the slope at fixed noise,

$$\frac{\partial \ln M}{\partial x} = M\Big[\frac{\ell_x^2}{n} + \frac{\partial \Psi'(E^\star; n, s, p)}{\partial E} \frac{\partial E^\star}{\partial x}\Big] = r_x (1 - \ell_x) + 2 r_x \alpha \langle 1 - \ell \rangle = r_x \big[(1 - \ell_x) + 2\bar{\ell}\big], \tag{23}$$

where the first term uses Lemma 2(c), the second uses Lemma 2(d), Lemma 3(i) and $M\Psi' = \alpha$, and the last equality uses $\bar{\ell} = \alpha\langle 1 - \ell \rangle$ (Step 3 of the returns to scale). In the same way, the value of lower noise, $\Psi'_s = \frac{1}{n} \int_0^s \ell^2 du$, has elasticity $2\langle 1 - \ell \rangle_s = 2(1 - \bar{\ell}_s)$ with respect to $\tau^\star$, and only scope adds a term, $\ell_s^2/n = r_s(1 - \ell_s)/M = r_s \Psi'_s (1 - \ell_s)/\alpha_s$. Hence

$$\frac{\partial \ln \Psi'_s}{\partial x} = 2 r_x (1 - \bar{\ell}_s) + \mathbf{1}_{\{x = s\}} \frac{r_s (1 - \ell_s)}{\alpha_s}. \tag{24}$$

Finally, the learned signal of the marginal covariate of either kind changes with its own scope through its signal and through precision, and with the other scope through precision only. By

Lemma 2(b) and Lemma 3(i),

$$\frac{d\ln(\omega_x(x)\ell_x)}{dx} = (2-\ell_x)\,\frac{\omega_x'(x)}{\omega_x(x)} + r_x(1-\ell_x), \qquad \frac{\partial\ln(\omega_p(p)\ell_p)}{\partial s} = r_s(1-\ell_p). \tag{25}$$

Proof of Proposition 4. Value depends on the covariates in the sample only through integrals over their positions, so reordering the covariates of each kind leaves it unchanged. Order them so that the covariate at u comes last. It is then the marginal covariate of kind x, with $x = s$ if $u \le \bar{s}$ and $x = p$ otherwise, $\omega_x(x) = \omega(u)$ and $\ell_x = \ell_u$, and $\mathcal{V}_u = \partial\mathcal{V}/\partial x$ by Lemma 3(ii).

(i). By Lemma 3(ii) and (17), $\partial\mathcal{V}/\partial n = (M/n)\Gamma$, so (22) and (23) give

$$\frac{\partial^2\mathcal{V}}{\partial n\,\partial x} = \frac{M}{n}\Big[\Gamma\,\frac{\partial\ln M}{\partial x} + \frac{\partial\Gamma}{\partial x}\Big] = \frac{\partial\mathcal{V}}{\partial n}\,r_x\big(2-\ell_x+2\bar{\ell}-2\bar{\ell}_s\big) + \mathbf{1}_{\{x=s\}}\,\frac{M}{n}\,\omega_s(s)\ell_s(1-\ell_s).$$

Since $r_x = M\omega_x(x)\ell_x/E^\star$ and $n\,\partial\mathcal{V}/\partial n = \alpha_s E^\star$, the last term equals $(\partial\mathcal{V}/\partial n)\,r_s(1-\ell_s)/\alpha_s$. With $\varepsilon_{\mathcal{S}} = 2\bar{\ell}_s$ and $\alpha(2-\varepsilon_{\mathcal{T}}) = 2\bar{\ell}$, this is Equation (9), and its last term, $2(\bar{\ell}-\bar{\ell}_s)$, has the sign of $\partial^2\mathcal{V}/\partial n^2$ by (20). The other terms in parentheses sum to more than one, because $\ell_x < 1$, so the cross effect is negative only if $2(\bar{\ell}-\bar{\ell}_s) < -1$, which requires $\partial^2\mathcal{V}/\partial n^2 < 0$ and $\varepsilon_{\mathcal{S}} = 2\bar{\ell}_s > 1 + 2\bar{\ell} \ge 1$.

(ii). As $n \to \infty$, $\tau^\star = n/E^\star \ge n \to \infty$, so every fraction learned tends to one, because the profiles are positive. Hence $\bar{\ell}_s \to 1$ by dominated convergence, and $\bar{\ell} \le \alpha = M - 1 \to 0$, because $\Psi'(E^\star; n, s, p) \le (s+p)/n$. For $x = p$, the parentheses of (9) therefore tend to $2-1-2 = -1$. For $x = s$, the unlearned signal in the sample is at most $(s+p)E^\star/n$, so $E^\star \to \underline{E} \equiv \underline{E}(s,p)$, and $n(1-\ell_s) = n/(1+\tau^\star\omega_s(s)) \to \underline{E}/\omega_s(s)$. On $[0,s]$, $0 \le \omega\,\ell\,n(1-\ell) \le E^\star \le 1$, and the integrand tends to $\underline{E}$, so $n\Gamma \to s\,\underline{E}$ by dominated convergence, and $n\alpha_s = n^2(\partial\mathcal{V}/\partial n)/E^\star = nM\Gamma/E^\star \to s$. Hence $(1-\ell_s)/\alpha_s \to \underline{E}/(s\,\omega_s(s))$, and the parentheses tend to $\underline{E}/(s\,\omega_s(s)) - 1$, which has the sign of $\underline{E} - s\,\omega_s(s)$. Since $\underline{E} = 1 - \int_{\mathcal{T}}\omega$ and $s = |\mathcal{S}|$, this gives (ii). □

Any part of the covariates, in the limit. For the next results, the sample contains no post-prediction covariates. For a measurable $g : [0,\bar{s}] \to [0,1]$, read $g(u)$ as the fraction of the covariates near position u that the firm observes, in its sample and for the new individual, and let

$$\Psi_g(E) \equiv 1 - \int g\,\omega\,\ell\Big(\frac{n\omega}{E}\Big)du, \qquad \Psi_g'(E) = \frac{1}{n}\int g\,\ell\Big(\frac{n\omega}{E}\Big)^2 du,$$

with integrals over $[0,\bar{s}]$, in which ω stands for $\omega_s(u)$. Fix $\bar{g}$ with $\int\bar{g}\,\omega < 1$, let $E_0 \equiv 1 - \int\bar{g}\,\omega > 0$, and consider $g \le \bar{g}$. Since $\omega\,\ell(n\omega/E) = n\omega^2/(n\omega+E)$ is decreasing and convex in E, Ψ_g is nondecreasing and concave, with $\Psi_g(0) = 1 - \int g\,\omega \ge E_0$ and $\Psi_g \le 1$. It therefore has a unique fixed point $E^\star(g) \in [E_0, 1]$, and concavity gives $1 - \Psi_g'(E^\star(g)) \ge \Psi_g(0)/E^\star(g) \ge E_0$.

Write

$$M(g) \equiv \frac{1}{1-\Psi_g'\big(E^\star(g)\big)}, \qquad \mu(u;g) \equiv \omega_s(u)\,\ell\Big(\frac{n\omega_s(u)}{E^\star(g)}\Big)\,M(g),$$
$$\mathcal{V}(g) \equiv \int g\,\omega\,\ell\Big(\frac{n\omega}{E^\star(g)}\Big)\,du = 1 - E^\star(g).$$

For a set $\mathcal{S}$ of positions that is a union of J disjoint intervals, $\mathcal{V}(\mathbf{1}_\mathcal{S})$ is the limit of the value of observing, in the sample and for the new individual, the covariates whose positions lie in $\mathcal{S}$, those whose cells lie in $\mathcal{S}$. Lemma 1 holds for every set of covariates in the sample, and Step 5 of the proof of Theorem 1 uses the signal only through the L^1 convergence of its step function. Let $\omega_\mathcal{S}$ be the profile ω_s restricted to $\mathcal{S}$ and translated, interval by interval, onto $[0, |\mathcal{S}|]$, so that $\int_0^{|\mathcal{S}|} f(\omega_\mathcal{S})\,du = \int_\mathcal{S} f(\omega)\,du$ for every bounded f. List the covariates in $\mathcal{S}$ interval by interval. On each interval, their step function is that of their order translated by a multiple of $1/K$, and at most $2J$ cells straddle an endpoint of $\mathcal{S}$, so the two translations differ by at most $2J/K$. The L^1 distance between this step function and $\omega_\mathcal{S}$ therefore vanishes as $K \to \infty$, because translation is continuous in L^1. Step 5 applies with $\omega_\mathcal{S}$ in place of ω_s, $|\mathcal{S}|$ in place of s and no post-prediction covariates, and gives the limit $\mathcal{V}(\mathbf{1}_\mathcal{S})$.

Lemma 5 (Combining covariates in the limit)**.** *Let $g, g', \eta : [0, \bar{s}] \to [0, 1]$ be measurable with $g \le g'$ and $g' + \eta \le \bar{g}$. Then:*

(i) the value that η adds integrates the marginal values along the path from g to $g+\eta$: $\mathcal{V}(g+\eta) - \mathcal{V}(g) = \int_0^1 \int \eta(u)\,\mu(u; g + t\eta)\,du\,dt$;

(ii) more covariates lower learning noise and raise the multiplier, $E^\star(g') \le E^\star(g)$ and $M(g') \ge M(g)$, so $\mu(u; g') \ge \mu(u; g)$ for every u;

(iii) the value that η adds rises with the covariates already held, $\mathcal{V}(g' + \eta) - \mathcal{V}(g') \ge \mathcal{V}(g + \eta) - \mathcal{V}(g)$;

(iv) if, in addition, $\int (g' - g)\,\omega > 0$, then $E^\star(g') < E^\star(g)$ and $\mu(u; g') > \mu(u; g)$ for every u with $\omega_s(u) > 0$;

(v) there is $L > 0$, which depends only on E_0, such that $|\mu(u; g') - \mu(u; g)| \le L\,\omega_s(u) \int |g' - g|\,\omega$ for every u.

Proof. Write $\rho(E; \omega) \equiv \omega\,\ell(n\omega/E)$ for the learned signal at noise E.

(i). The function $(E, t) \mapsto E - \Psi_{g+t\eta}(E)$ is continuously differentiable, with derivative $1 - \Psi'_{g+t\eta}(E) \ge E_0$ in E at the fixed point and $\int \eta\,\rho(E; \omega)\,du$ in t. The implicit function theorem gives $dE^\star(g + t\eta)/dt = -M(g + t\eta) \int \eta\,\rho(E^\star(g + t\eta); \omega)\,du$, and $\mathcal{V} = 1 - E^\star$ gives (i).

(ii). Since $v \ge 0$, $\Psi_{g'} \le \Psi_g$, so $\Psi_{g'}(E^\star(g)) \le E^\star(g)$. A concave function $\Psi_{g'}$ with $\Psi_{g'}(0) > 0$ lies above the diagonal below its fixed point and below it above, so $E^\star(g') \le E^\star(g)$. Since

$\ell(n\omega/E)$ decreases in E and $g' \geq g$, $\Psi'_{g'}(E^\star(g')) \geq \Psi'_g(E^\star(g')) \geq \Psi'_g(E^\star(g))$, so $M(g') \geq M(g)$; and v decreases in E.

(iii). Apply (i) to both differences and (ii) to $g' + t\eta \geq g + t\eta$.

(iv). Since $\rho(E;\omega) > 0$ whenever $\omega > 0$ and $E > 0$, $\Psi_{g'}(E) < \Psi_g(E)$ for every $E > 0$, so $\Psi_{g'}(E^\star(g)) < E^\star(g)$, and the argument of (ii) gives $E^\star(g') < E^\star(g)$. Since $\rho(E;\omega)$ strictly decreases in E wherever $\omega > 0$, and $M(g') \geq M(g)$ by (ii), $\mu(u;g') > \mu(u;g)$ wherever $\omega_s(u) > 0$.

(v). Write $\epsilon \equiv \int |g' - g|\,\omega$. Since $0 \leq \rho(E;\omega) \leq \omega$, $|\Psi_{g'}(E) - \Psi_g(E)| \leq \epsilon$ at every E. The function $\Phi_{g'}(E) \equiv E - \Psi_{g'}(E)$ is convex, at most $-E_0$ at $E = 0$ and zero at $E^\star(g') \leq 1$, so the chord from $(0, \Phi_{g'}(0))$ to $(E^\star(g'), 0)$ has slope at least E_0. Below $E^\star(g')$, convexity puts $\Phi_{g'}$ below this chord; above $E^\star(g')$, it puts $\Phi_{g'}$ above its tangent at $E^\star(g')$, whose slope is at least the chord's. Hence $|E - \Psi_{g'}(E)| \geq E_0|E - E^\star(g')|$ for every $E \geq 0$, and evaluating at $E = E^\star(g)$ gives $|E^\star(g') - E^\star(g)| \leq \epsilon/E_0$. On $[E_0, 1]$, $|\partial\rho(E;\omega)/\partial E| \leq \omega/(4E_0)$, $\ell(n\omega/E)^2 \leq n\omega/E_0$ and, by Lemma 2(d), $|\partial\Psi'_g/\partial E| \leq 2/E_0^2$. Hence $|\Psi'_{g'}(E) - \Psi'_g(E)| \leq \epsilon/E_0$, $|\Psi'_{g'}(E^\star(g')) - \Psi'_g(E^\star(g))| \leq C\epsilon$ and, since both multipliers are at most $1/E_0$, $|M(g') - M(g)| \leq C\epsilon$, with C depending only on E_0. Since $v \leq \omega$ and $M \leq 1/E_0$, the bound on $\mu = v\,M$ follows. □

Groups of post-prediction covariates. Now let the sample contain the first s covariates and the post-prediction covariates whose positions lie in a set $\mathcal{P}$, a union of finitely many disjoint intervals of $[0, \bar{p}]$. The argument above, applied to the post-prediction covariates, with $\omega_{\mathcal{P}}$ in place of ω_p and $|\mathcal{P}|$ in place of p, shows that value converges to $\mathcal{V}(\mathcal{P})$, the value of Theorem 1 with that post-prediction profile.

For the next lemma, let the sample contain the first s covariates and the post-prediction covariates in a group $\mathcal{P}$ that leaves out the signal $\underline{E}_{\mathcal{P}} \equiv 1 - \int_0^s \omega_s - \int_{\mathcal{P}} \omega_p > 0$. Write Q_2, Q_3 and Q_4 for the integrals of ω_s^2, ω_s^3 and ω_s^4 over $[0, s]$, $\hat{Q}_2 \equiv Q_2 + \int_{\mathcal{P}} \omega_p^2$ and $\hat{Q}_3 \equiv Q_3 + \int_{\mathcal{P}} \omega_p^3$ for the same integrals over the whole sample, and $I_s \equiv \int_0^s du/\omega_s(u)$.

Lemma 6 (Groups in small and large samples)**.**

(i) As $n \to 0$, $\mathcal{V}(\mathcal{P}) = nQ_2 + n^2(Q_2\hat{Q}_2 - Q_3) + n^3(2Q_2\hat{Q}_2^2 - Q_2\hat{Q}_3 - 2Q_3\hat{Q}_2 + Q_4) + O(n^4)$.

(ii) As $n \to \infty$, $n\big[\int_0^s \omega_s - \mathcal{V}(\mathcal{P})\big] \to s\,\underline{E}_{\mathcal{P}}$.

(iii) If the signals of the covariates in the sample are bounded away from zero, then, as $n \to \infty$,

$$\mathcal{V}(\mathcal{P}) = \int_0^s \omega_s - \frac{s\,\underline{E}_{\mathcal{P}}}{n} - \frac{s\,(s + |\mathcal{P}|)\,\underline{E}_{\mathcal{P}}}{n^2} + \frac{I_s\,\underline{E}_{\mathcal{P}}^2}{n^2} + O(n^{-3}).$$

The bounds in the O-terms depend only on $\bar{\omega}$, $s + |\mathcal{P}|$, *a positive lower bound on* $\underline{E}_{\mathcal{P}}$ *and, in (iii), the lower bound on signal.*

Proof. Write E for the learning noise, which lies in $[\underline{E}_{\mathcal{P}}, 1]$, $x \equiv n\omega/E$ and $\int_C$ for an integral over all the covariates in the sample.

(i). Since $x \le n\bar{\omega}/\underline{E}_{\mathcal{P}}$, $\ell(x) = x - x^2 + x^3 + O(x^4)$ uniformly. Hence $1 - E = \int_C \omega\,\ell = n\hat{Q}_2/E - n^2\hat{Q}_3/E^2 + O(n^3)$, so $1 - E = O(n)$, $1/E = 1 + (1-E) + (1-E)^2 + O(n^3)$, and substituting twice gives $1/E = 1 + n\hat{Q}_2 + n^2(2\hat{Q}_2^2 - \hat{Q}_3) + O(n^3)$. Substituting this into $\mathcal{V} = nQ_2/E - n^2Q_3/E^2 + n^3Q_4/E^3 + O(n^4)$ gives (i).

(ii). Since $\omega(1-\ell(x)) = \omega E/(E+n\omega) \le E/n$, $E - \underline{E}_{\mathcal{P}} = \int_C \omega(1-\ell) \le (s+|\mathcal{P}|)/n \to 0$. On $[0,s]$, $n\omega(1-\ell) = n\omega E/(E+n\omega)$ is at most one and tends to $\underline{E}_{\mathcal{P}}$ wherever $\omega > 0$, so dominated convergence gives $n\int_0^s \omega(1-\ell) \to s\,\underline{E}_{\mathcal{P}}$, which is (ii).

(iii). Let $\omega_{\min} > 0$ bound from below the signals in the sample. Since $x \ge n\omega_{\min}$, $1-\ell(x) = 1/(1+x) = 1/x - 1/x^2 + O(x^{-3})$, so $\omega(1-\ell) = E/n - E^2/(n^2\omega) + O(n^{-3})$ uniformly in the sample. Hence $E = \underline{E}_{\mathcal{P}} + (s+|\mathcal{P}|)E/n + O(n^{-2})$, so $E = \underline{E}_{\mathcal{P}}\big(1 + (s+|\mathcal{P}|)/n\big) + O(n^{-2})$, and

$$\int_0^s \omega_s - \mathcal{V}(\mathcal{P}) = \int_0^s \omega(1-\ell) = \frac{sE}{n} - \frac{I_s E^2}{n^2} + O(n^{-3}) = \frac{s\,\underline{E}_{\mathcal{P}}}{n} + \frac{s\,(s+|\mathcal{P}|)\,\underline{E}_{\mathcal{P}}}{n^2} - \frac{I_s\,\underline{E}_{\mathcal{P}}^2}{n^2} + O(n^{-3}). \quad \square$$

Proof of Proposition 3. (i). By Lemma 1, with $\mathcal{S}$ as the group of covariates in the sample and of covariates observed for the new individual, $V_N(\mathcal{S}) = \frac{1}{K}\sum_{k\in\mathcal{S}} \omega_k\,\mathbb{V}\big(m_k(N,\mathcal{S})\big)$, where $m_k(N,\mathcal{S})$ is the posterior mean of covariate k's weight when the firm observes $\mathcal{S}$ for N individuals. With $\mathcal{S}_1 \cup \mathcal{S}_2$, the firm observes everything it observes with $\mathcal{S}_i$ alone, so the monotonicity argument in the proof of Lemma 1 gives $\mathbb{V}\big(m_k(N,\mathcal{S}_1\cup\mathcal{S}_2)\big) \ge \mathbb{V}\big(m_k(N,\mathcal{S}_i)\big)$ for $k \in \mathcal{S}_i$, $i = 1,2$. Splitting the sum over $\mathcal{S}_1 \cup \mathcal{S}_2$ into its parts over $\mathcal{S}_1$ and over $\mathcal{S}_2$ gives $V_N(\mathcal{S}_1\cup\mathcal{S}_2) \ge V_N(\mathcal{S}_1) + V_N(\mathcal{S}_2)$, which is the claim.

(ii). Let $\mathcal{S}_1$ grow to $\mathcal{S}_1'$, disjoint from $\mathcal{S}_2$, with $\int_{\mathcal{S}_1'\setminus\mathcal{S}_1}\omega > 0$ and $\int_{\mathcal{S}_1'\cup\mathcal{S}_2}\omega < 1$, and write $g \equiv \mathbf{1}_{\mathcal{S}_1}$, $g' \equiv \mathbf{1}_{\mathcal{S}_1'}$, $\eta \equiv \mathbf{1}_{\mathcal{S}_2}$ and $\bar{g} \equiv g' + \eta$. By Lemma 5(i), the value that $\mathcal{S}_2$ adds rises by

$$\int_0^1 \int \eta(u)\,\big[\mu(u; g' + t\eta) - \mu(u; g + t\eta)\big]\,du\,dt,$$

and by Lemma 5(iv), applied to $g + t\eta \le g' + t\eta$, the integrand is positive wherever $\eta(u)\,\omega_s(u) > 0$. Since $\int_{\mathcal{S}_2}\omega > 0$, the increase is positive. With $\mathcal{S}_1 = \varnothing$ growing to $\mathcal{S}_1$, the value that $\mathcal{S}_2$ adds to $\mathcal{S}_1$ therefore exceeds $\mathcal{V}(\mathcal{S}_2)$: the two groups exhibit economies of scope.

(iii). Write $\mathcal{S} \equiv \mathcal{S}_1 \cup \mathcal{S}_2$, $g \equiv \mathbf{1}_{\mathcal{S}}$ and $\bar{g} \equiv g$, so that $\int \bar{g}\,\omega < 1$. For $i \ne j$, Lemma 5(i) with $g - \mathbf{1}_{\mathcal{S}_i}$ in place of g and $\eta = \mathbf{1}_{\mathcal{S}_i}$ gives

$$\mathcal{V}(\mathcal{S}) - \mathcal{V}(\mathcal{S}_j) = \int_0^1 \int_{\mathcal{S}_i} \mu\big(u; g - (1-t)\mathbf{1}_{\mathcal{S}_i}\big)\,du\,dt < \int_{\mathcal{S}_i} \mu(u; g)\,du,$$

where the inequality follows from Lemma 5(iv), because $\mathcal{S}_i$ has positive signal. Summing over $i = 1,2$ and using $\int_{\mathcal{S}} \mu(u; g)\,du = M(g)\,\mathcal{V}(\mathcal{S}) = (1+\alpha)\,\mathcal{V}(\mathcal{S})$ gives $2\mathcal{V}(\mathcal{S}) - \mathcal{V}(\mathcal{S}_1) - \mathcal{V}(\mathcal{S}_2) < (1+\alpha)\,\mathcal{V}(\mathcal{S})$, which is (iii). $\square$

Lemma 7 (Economies of scope at the ends)**.** *Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be disjoint groups of prediction covariates with $\int_{\mathcal{S}_1\cup\mathcal{S}_2}\omega < 1$ and $\omega_s > 0$ almost everywhere on $\mathcal{S}_1 \cup \mathcal{S}_2$. Write a_i, b_i and q_i for*

the length, signal and squared signal of S_i, G for the economies of scope, and α and $\mathcal{V}$ for the amplification and the value of $S_1 \cup S_2$.

(i) As $n \to 0$, $G/n^2 \to 2q_1q_2$ and $G/(\alpha\mathcal{V}) \to 2q_1q_2/(q_1+q_2)^2 \le 1/2$.

(ii) As $n \to \infty$, $nG \to a_1b_2 + a_2b_1$ and $G/(\alpha\mathcal{V}) \to (a_1b_2+a_2b_1)/\big[(a_1+a_2)(b_1+b_2)\big] < 1$.

Proof. The limit $\mathcal{V}(S)$ is the value of a sample with profile ω_S on $[0, |S|]$ and no post-prediction covariates, so Lemma 6 applies with ω_S in place of ω_s, $|S|$ in place of s and $\mathcal{P} = \varnothing$; its bounds hold uniformly over the three groups. Length, signal and the integrals $\int_S \omega^k$ are additive over disjoint groups; write $a \equiv a_1 + a_2$, $b \equiv b_1 + b_2$ and $q \equiv q_1 + q_2$. With $g \equiv \mathbf{1}_{S_1 \cup S_2}$, $1 + \alpha = M(g)$, so $\alpha = \Psi'/(1-\Psi')$ with $\Psi' \equiv \Psi'_g(E^\star(g)) = \frac{1}{n}\int g\, \ell(n\omega/E^\star)^2 du$.

(i). By Lemma 6(i), $\mathcal{V}(S) = nq_S + n^2(q_S^2 - c_S) + O(n^3)$ for a group with squared signal q_S and $c_S \equiv \int_S \omega^3$, so the terms in nq_S and n^2c_S cancel in G, and $G = n^2\big[(q_1+q_2)^2 - q_1^2 - q_2^2\big] + O(n^3) = 2n^2q_1q_2 + O(n^3)$; also $\mathcal{V}/n \to q$. Since $E^\star \in [1-b, 1]$, $n\omega/E^\star \to 0$ uniformly, so $\ell(n\omega/E^\star) = n\omega + o(n)$ uniformly, $\Psi'/n \to q$ and $\alpha/n \to q$. Hence $G/(\alpha\mathcal{V}) \to 2q_1q_2/q^2$, which is at most $1/2$ because $4q_1q_2 \le q^2$.

(ii). By Lemma 6(ii), $n\big[\int_S \omega - \mathcal{V}(S)\big] \to |S|\big(1 - \int_S \omega\big)$ for each group. Signal cancels in G, so $nG \to -\big[a(1-b) - a_1(1-b_1) - a_2(1-b_2)\big] = a_1b_2 + a_2b_1$, and $\mathcal{V} \to b$. Since $E^\star \le 1$ and $\omega > 0$ almost everywhere on the support of g, $n\omega/E^\star \to \infty$ there, so dominated convergence gives $n\Psi' \to a$; since $\Psi' \to 0$, $n\alpha \to a$. Hence $G/(\alpha\mathcal{V}) \to (a_1b_2 + a_2b_1)/(ab)$, which is below one because $ab = a_1b_2 + a_2b_1 + a_1b_1 + a_2b_2$ and $a_ib_i > 0$. □

Remaining parts. Proposition 2(ii) is Lemma 4(ii)–(iii) and, for $\mathcal{P} = \varnothing$, Lemma 4(i).

C Proofs for Section 5

Two facts recur; ω stands for $\omega_s(u)$ in integrals, and $p = 0$ unless stated otherwise. First, $\mathcal{V}(\cdot, s)$ is continuously differentiable and strictly concave on $(0, \infty)$ (Proposition 2(ii)), and $\mathcal{V}(v, s) \to 0$ as $v \to 0$, because $\omega\,\ell(v\omega/E) \le v\omega^2/\underline{E}(s)$. For $x \in [0, n]$, let $f(x) \equiv \mathcal{V}(n, s) - \mathcal{V}(n-x, s)$, the value lost when a holding of scale x leaves. Then $f(0) = 0$ and f is strictly convex, so $f(x+y) > f(x) + f(y)$ for $x, y > 0$ with $x + y \le n$. Second, in the economy of covariates, $W(\mathcal{H}) = \mathcal{V}(\mathbf{1}_{S_\mathcal{H}})$ in the notation of Lemma 5, with $S_\mathcal{H} \equiv \bigcup_{h\in\mathcal{H}} S_h$, $\bar{g} \equiv \mathbf{1}_{[0,s]}$ and $E_0 = \underline{E}(s)$; write $g_{\text{all}} \equiv \mathbf{1}_{[0,s]}$, so that $\mu(u; g_{\text{all}}) = \omega\,\ell(\tau^\star\omega)M$.

Proof of Proposition 5. (i). Since $\Delta_h = f(n_h) = \int_{n-n_h}^{n} \partial\mathcal{V}(v, s)/\partial v\, dv$ and $\partial\mathcal{V}/\partial v$ strictly decreases, $\Delta_h > n_h\, \partial\mathcal{V}(n, s)/\partial n$, and summing gives $\sum_h \Delta_h > n\,\partial\mathcal{V}/\partial n = -\partial E^\star/\partial \ln n = \alpha E^\star$. For the upper bound, order the holders and let $v_h \equiv n_1 + \cdots + n_h$. Then $\mathcal{V}(n, s) = \sum_h \big[\mathcal{V}(v_h, s) - \mathcal{V}(v_{h-1}, s)\big]$, and since $v_{h-1} \le n - n_h$, strict concavity makes each increment at least $f(n_h) = \Delta_h$, strictly for $h = 1$ because $m \ge 2$.

(ii). Let $p > 0$ and $n < \hat{n}$. By the definition of $\hat{n}$, $\mathcal{V}(\cdot, s, p)$ is strictly convex on $(0, \hat{n})$, and $\mathcal{V}(v, s, p) \to 0$ as $v \to 0$. The argument of (i) applies with the inequalities reversed. Since

$\partial\mathcal{V}(v,s,p)/\partial v$ strictly increases on $[n-n_h,n]$, $\Delta_h < n_h\,\partial\mathcal{V}(n,s,p)/\partial n$, and summing gives $\sum_h \Delta_h < n\,\partial\mathcal{V}/\partial n = \alpha_s E^\star$ (Lemma 3(ii)). Since $v_{h-1} \le n-n_h$, strict convexity makes each increment $\mathcal{V}(v_h,s,p)-\mathcal{V}(v_{h-1},s,p)$ at most Δ_h, strictly for $h=1$, so $\mathcal{V}(n,s,p) < \sum_h \Delta_h$. □

Proof of Proposition 6. Order the holders and let $\mathcal{S}_{<h} \equiv \bigcup_{j<h} \mathcal{S}_j$. Then $\mathcal{V} = \sum_h \left[\mathcal{V}(\mathcal{S}_{<h}\cup\mathcal{S}_h) - \mathcal{V}(\mathcal{S}_{<h})\right]$, and by Lemma 5(iii), each increment is at most $\mathcal{V}([0,s]) - \mathcal{V}([0,s]\setminus\mathcal{S}_h) = \Delta_h$, strictly for $h=1$ by Lemma 5(i) and (iv), because $[0,s]\setminus\mathcal{S}_1$ has positive signal when $m \ge 2$. For the upper bound, Lemma 5(i) with $g = g_{\text{all}} - \mathbf{1}_{\mathcal{S}_h}$ and $\eta = \mathbf{1}_{\mathcal{S}_h}$ gives

$$\Delta_h = \int_0^1 \int_{\mathcal{S}_h} \mu\big(u; g_{\text{all}} - (1-t)\mathbf{1}_{\mathcal{S}_h}\big)\,du\,dt < \int_{\mathcal{S}_h} \mu(u; g_{\text{all}})\,du, \tag{26}$$

where the inequality follows from Lemma 5(iv), because $\mathcal{S}_h$ has positive signal. Summing over holders, $\sum_h \int_{\mathcal{S}_h} \omega\,\ell(\tau^\star\omega) M\,du = M\mathcal{V} = (1+\alpha)\mathcal{V}$. □

Proof of Proposition 7. (i)–(ii). By the mean value theorem, $\left|\Delta_h - n_h\,\partial\mathcal{V}(n,s,p)/\partial n\right| \le n_h\,\varpi(\max_j n_j)$, where ϖ is the modulus of continuity of $\partial\mathcal{V}/\partial n$ on $[n/2,n]$. Summing, the gap between $\sum_h \Delta_h$ and $n\,\partial\mathcal{V}/\partial n$ is at most $n\,\varpi(\max_j n_j) \to 0$, and $n\,\partial\mathcal{V}/\partial n = \alpha_s E^\star$ by Lemma 3(ii), which equals $\alpha E^\star$ when $p=0$. Without post-prediction covariates, $\mathcal{V} = R^2$ and $E^\star = 1-R^2$, so the share is $\alpha(1-R^2)/R^2$, and it is below one by Proposition 5(i). When Inequality (8) holds, $\alpha_s E^\star > \mathcal{V}$ at $n < \hat{n}$ by Proposition 5(ii).

(iii). By (26) and Lemma 5(v), $\left|\Delta_h - \int_{\mathcal{S}_h} \mu(u; g_{\text{all}})\,du\right| \le L\big(\int_{\mathcal{S}_h}\omega\big)^2$, so $\left|\sum_h \Delta_h - (1+\alpha)\mathcal{V}\right| \le L\int_0^s \omega\,\max_h \int_{\mathcal{S}_h}\omega \to 0$.

(iv). Every holder of observations faces all covariates, so its marginal contribution is that of the economy of observations; every holder of covariates faces all observations, so its marginal contribution is that of the economy of covariates. By (i) and (iii), the sum converges to $\alpha E^\star + (1+\alpha)\mathcal{V} = \mathcal{V} + \alpha(E^\star+\mathcal{V}) = \mathcal{V}+\alpha$, because $E^\star+\mathcal{V}=1$; and $\alpha = (\alpha/R^2)\mathcal{V}$. □

Proof of Corollary 2. By Proposition 7(i), (iii) and (iv), holders receive φ times the shares $\alpha(1-R^2)/R^2$, $1+\alpha$ and $1+\alpha/R^2$ of value. Payments do not exceed value if and only if these products are at most one. The first is below one for every $\varphi \le 1$ by Proposition 7(i), and the other two give the thresholds. Since $\alpha > 0$ and $R^2 < 1$, $1+\alpha/R^2 > 1+\alpha > 1$, which orders the thresholds. □

Proof of Corollary 3. A merger of holders 1 and 2 leaves $W(\mathcal{H}_{\text{all}})$ and the marginal contribution of every other holder unchanged, so the firm's payoff changes by φ times the fall in the merged holders' marginal contribution.

(i). The merged holder's marginal contribution is $f(n_1+n_2) > f(n_1)+f(n_2)$. A single holder of all observations has marginal contribution $\mathcal{V}$, which exceeds the atomistic sum $\alpha E^\star$ by the factor $\mathcal{V}/(\alpha E^\star) = R^2/[\alpha(1-R^2)]$.

(ii). Let $p > 0$ and $n < \hat{n}$. The value lost when a holding of scale x leaves, $\mathcal{V}(n, s, p) - \mathcal{V}(n - x, s, p)$, is zero at $x = 0$ and strictly concave in x on $[0, n]$, because $\mathcal{V}(\cdot, s, p)$ is strictly convex on $(0, \hat{n})$, so the merged holder's marginal contribution is below $\Delta_1 + \Delta_2$.

(iii). With $\mathcal{G} \equiv [0, s] \setminus (\mathcal{S}_1 \cup \mathcal{S}_2)$, the merged holder's marginal contribution minus $\Delta_1 + \Delta_2$ is

$$-\Big\{ \big[W(\mathcal{G} \cup \mathcal{S}_1 \cup \mathcal{S}_2) - W(\mathcal{G} \cup \mathcal{S}_2) \big] - \big[W(\mathcal{G} \cup \mathcal{S}_1) - W(\mathcal{G}) \big] \Big\} < 0$$

by Lemma 5(i) and (iv), because $\mathcal{S}_2$ has positive signal. A single holder of all covariates has marginal contribution $\mathcal{V}$, the fraction $1/(1 + \alpha)$ of the atomistic sum, so the firm gains $\varphi\alpha\mathcal{V}$. □

Proof of Proposition 8. (i). The payments sum to $\lambda \int_0^s \omega\, \ell(\tau^\star \omega)\, du = \lambda \mathcal{V}$.

(ii). A coalition that does not contain the firm is worth zero, and payments are nonnegative. For a coalition of the firm with the holders in $\mathcal{H}$, the allocation gives it $\mathcal{V} - \lambda \int_{[0,s] \setminus \mathcal{S}_\mathcal{H}} \omega\, \ell(\tau^\star \omega)\, du$, which must be at least $W(\mathcal{H})$. By Lemma 5(i), with $g = \mathbf{1}_{\mathcal{S}_\mathcal{H}}$ and $\eta = g_{\text{all}} - g$, $\mathcal{V} - W(\mathcal{H}) = \int_0^1 \int \eta\, \mu(u; g + t\eta)\, du\, dt$. By Lemma 5(ii), $E^\star(g + t\eta) \geq E^\star(g_{\text{all}})$, so $\omega\, \ell\big(n\omega / E^\star(g + t\eta)\big) \leq \omega\, \ell(\tau^\star \omega)$, and $M(g + t\eta) \geq 1$. Hence $\mathcal{V} - W(\mathcal{H}) \geq \int_{[0,s] \setminus \mathcal{S}_\mathcal{H}} \omega\, \ell(\tau^\star \omega)\, du \geq \lambda \int_{[0,s] \setminus \mathcal{S}_\mathcal{H}} \omega\, \ell(\tau^\star \omega)\, du$.

(iii). Since $\int_{\mathcal{S}_h} \omega\, \ell(\tau^\star \omega)\, du = \int_{\mathcal{S}_h} \mu(u; g_{\text{all}})\, du/(1 + \alpha)$, the sum is at most $\lambda/(1 + \alpha)$ times the bound in the proof of Proposition 7(iii), which vanishes. □

D Proofs for Section 6

The assumptions and notation of Appendix B apply.

Proof of Proposition 9 and Online Appendix Proposition OA.5. Let x be scope or post-prediction scope, s or p, and $\omega_x(x)$ and ℓ_x the signal and the fraction learned of the marginal covariate of that kind. By Section 3.3, $\partial E^\star / \partial x = -M \omega_x(x) \ell_x < 0$. The strength of the loop, $\Psi'(E; n, s, p) = \frac{1}{n} \int_C \ell(n\omega/E)^2 du$, gains the term $\ell_x^2 / n = \omega_x(x) \ell_x (1 - \ell_x)/E$ (Lemma 2(c)) at the upper limit of the integral over positions of kind x, and $\Psi'_s = \frac{1}{n} \int_0^s \ell^2 du$ gains ℓ_s^2 / n when $x = s$ only. By Lemma 2(a), $\partial \Psi'_s / \partial E = -\frac{2}{nE} \int_0^s \ell^2 (1 - \ell)\, du < 0$, and $\Psi'' < 0$ (Step 2 of the returns to scale). Since $\partial \alpha / \partial \Psi' = M^2$, the chain rule along $E^\star(n, s, p)$ gives, for $x = s, p$,

$$\frac{\partial \alpha}{\partial x} = M^2 \Big[\underbrace{\frac{\omega_x(x) \ell_x (1 - \ell_x)}{E^\star}}_{\substack{\text{own} \\ \text{speed } (+)}} + \underbrace{\Psi'' \frac{\partial E^\star}{\partial x}}_{\text{lower noise } (+)} \Big] > 0, \tag{27}$$

which is the first claim of Proposition OA.5(ii), and

$$\frac{\partial \Psi'_s}{\partial p} = \frac{\partial \Psi'_s}{\partial E} \frac{\partial E^\star}{\partial p} > 0, \qquad \frac{\partial \Psi'_s}{\partial s} = \frac{\omega_s(s) \ell_s (1 - \ell_s)}{E^\star} + \frac{\partial \Psi'_s}{\partial E} \frac{\partial E^\star}{\partial s} > 0.$$

Hence $\alpha_s = M\Psi'_s$, which equals α at $p = 0$, rises with both kinds of scope:

$$\frac{\partial \alpha_s}{\partial x} = \Psi'_s \frac{\partial \alpha}{\partial x} + M \frac{\partial \Psi'_s}{\partial x} > 0.$$

Proposition OA.5(i) is Lemma 3(ii). The learned signal $\omega_y(y)\ell(n\omega_y(y)/E)$ of the marginal covariate of kind y changes with its own position through its signal, with $\partial(\omega\ell)/\partial\omega = \ell(2-\ell)$ by Lemma 2(b), and with noise, with $\partial(\omega\ell)/\partial E = -\ell^2/n$ by (16). Hence

$$\frac{\partial\big(\omega_p(p)\ell_p\big)}{\partial p} = \underbrace{\ell_p(2-\ell_p)\,\omega'_p(p)}_{\substack{\text{signal of the}\\ \text{next covariate } (\pm)}} + \underbrace{\frac{\omega_p(p)\ell_p(1-\ell_p)}{E^\star}\Big(-\frac{\partial E^\star}{\partial p}\Big)}_{\substack{\text{learned}\\ \text{better } (+)}}, \tag{28}$$

the same holds with s in place of p, and $\partial\big(\omega_p(p)\ell_p\big)/\partial s = \big(\omega_p(p)\ell_p(1-\ell_p)/E^\star\big)(-\partial E^\star/\partial s) > 0$. The product rule, applied to Lemma 3(ii), gives the curvatures

$$\frac{\partial^2 \mathcal{V}}{\partial p^2} = \underbrace{\frac{\partial \Psi'_s}{\partial p}\,\omega_p(p)\ell_p\,M}_{\substack{\text{value of}\\ \text{lower noise } (+)}} + \underbrace{\Psi'_s\,\frac{\partial\big(\omega_p(p)\ell_p\big)}{\partial p}\,M}_{\text{impact } (\pm)} + \underbrace{\Psi'_s\,\omega_p(p)\ell_p\,\frac{\partial \alpha}{\partial p}}_{\text{amplification } (+)}, \tag{29}$$

$$\frac{\partial^2 \mathcal{V}}{\partial s^2} = \underbrace{(1+\alpha_s)\,\frac{\partial\big(\omega_s(s)\ell_s\big)}{\partial s}}_{\substack{\text{direct effect}\\ \text{and impact } (\pm)}} + \underbrace{\frac{\partial \Psi'_s}{\partial s}\,\omega_s(s)\ell_s\,M}_{\substack{\text{value of}\\ \text{lower noise } (+)}} + \underbrace{\Psi'_s\,\omega_s(s)\ell_s\,\frac{\partial \alpha}{\partial s}}_{\text{amplification } (+)}, \tag{30}$$

and the cross effect of the two kinds of scope,

$$\frac{\partial^2 \mathcal{V}}{\partial s\,\partial p} = \underbrace{\frac{\partial \Psi'_s}{\partial s}\,\omega_p(p)\ell_p\,M}_{\substack{\text{value of}\\ \text{lower noise } (+)}} + \underbrace{\Psi'_s\,\frac{\partial\big(\omega_p(p)\ell_p\big)}{\partial s}\,M}_{\text{impact } (+)} + \underbrace{\Psi'_s\,\omega_p(p)\ell_p\,\frac{\partial \alpha}{\partial s}}_{\text{amplification } (+)} > 0, \tag{31}$$

In Equations (29) and (30), every term is positive except the part of the impact term due to the signal of the next covariate, $\Psi'_s M\,\ell_p(2-\ell_p)\,\omega'_p(p)$ and $(1+\alpha_s)\,\ell_s(2-\ell_s)\,\omega'_s(s)$, which proves the second claim of Proposition OA.5(ii).

Proposition 9(i). Since $\partial(\omega\ell)/\partial\tau = \omega^2(1-\ell)^2$ and $\tau^\star_x = (\tau^\star/E^\star)(-\partial E^\star/\partial x)$, the second term of (28) equals $\omega_x(x)^2(1-\ell_x)^2\tau^\star_x$, and $\tau^\star_x > 0$ by Lemma 3(i). In (29) and (30), the terms on the value of lower noise and on the amplification sum to $\omega_x(x)\ell_x\,\partial\alpha_s/\partial x > 0$, and the remaining term is $(\mathbf{1}_{\{x=s\}} + \alpha_s)\,\partial(\omega_x(x)\ell_x)/\partial x$. Substituting (28) gives Equation (11).

Proposition 9(ii). The marginal value of scope, $(1+\alpha_s)\,\omega_s(s)\ell_s$, depends on post-prediction scope through α_s and through precision, with $\partial(\omega_s(s)\ell_s)/\partial p = \omega_s(s)^2(1-\ell_s)^2\tau^\star_p$. Hence

$$\frac{\partial^2 \mathcal{V}}{\partial s\,\partial p} = (1+\alpha_s)\,\omega_s(s)^2(1-\ell_s)^2\,\tau^\star_p + \omega_s(s)\,\ell_s\,\frac{\partial \alpha_s}{\partial p},$$

in which $\tau_p^\star > 0$ by Lemma 3(i) and $\partial\alpha_s/\partial p > 0$. Equation (31) gives the same cross effect from the marginal value of post-prediction scope.

For the cross effects with scale, $v_\tau = \int_0^s \omega^2(1-\ell)^2 du$ gains $v_{\tau s} = \omega_s(s)^2(1-\ell_s)^2$ at the upper limit when $x = s$ and depends on x otherwise through $\tau^\star$, and $\partial\tau^\star/\partial n = M/E^\star$ depends on x through $E^\star$ and M. The product rule, applied to Equation (5), gives

$$\frac{\partial^2 \mathcal{V}}{\partial n\,\partial x} = \mathbf{1}_{\{x=s\}} v_{\tau s} \frac{\partial \tau^\star}{\partial n} + \underbrace{v_{\tau\tau} \frac{\partial \tau^\star}{\partial x} \frac{\partial \tau^\star}{\partial n}}_{\substack{\text{value of}\\ \text{precision } (-)}} + \underbrace{v_\tau M \frac{-\partial E^\star/\partial x}{E^{\star 2}}}_{\text{impact } (+)} + \underbrace{v_\tau \frac{1}{E^\star} \frac{\partial \alpha}{\partial x}}_{\text{amplification } (+)}.$$

Since $v_{\tau\tau} = -(2\bar{\ell}_s/\tau^\star)v_\tau$ and $\partial\tau^\star/\partial x = (\tau^\star/E^\star)(-\partial E^\star/\partial x)$, the term on the value of precision and the term on the impact sum to $Mv_\tau(-\partial E^\star/\partial x)(1-2\bar{\ell}_s)/E^{\star 2}$. For $x = p$, this gives Equation (OA.3); for $x = s$, it gives

$$\frac{\partial^2 \mathcal{V}}{\partial n\,\partial s} = \underbrace{v_{\tau s} \frac{\partial \tau^\star}{\partial n}}_{\substack{\text{own weight}\\ \text{applied } (+)}} + \underbrace{v_\tau(1+\alpha) \frac{-\partial E^\star/\partial s}{E^{\star 2}} \left(1 - 2\bar{\ell}_s\right)}_{\substack{\text{impact and}\\ \text{value of precision } (\pm)}} + \underbrace{v_\tau \frac{1}{E^\star} \frac{\partial \alpha}{\partial s}}_{\text{amplification } (+)}. \tag{32}$$

a covariate adds a weight that the firm learns from every further observation, lower noise raises the impact of an observation and lowers the value of precision, and the covariate raises the amplification. The middle term is nonnegative if $\bar{\ell}_s \le 1/2$, and the other terms are positive, which proves the sufficient condition of Corollary OA.2(v).

Dividing Equations (29) and (30) by the marginal values, and using (23), (24) and (25),

$$\frac{\partial^2 \mathcal{V}/\partial x^2}{\partial \mathcal{V}/\partial x} = (2 - \ell_x) \frac{\omega_x'(x)}{\omega_x(x)} + r_x B_x, \tag{33}$$

where

$$B_p \equiv 2(1-\bar{\ell}_s) + 2(1-\ell_p) + 2\bar{\ell},$$
$$B_s \equiv (1-\ell_s) + \frac{\alpha_s}{1+\alpha_s}\left[\frac{1-\ell_s}{\alpha_s} + 2(1-\bar{\ell}_s) + (1-\ell_s) + 2\bar{\ell}\right].$$

If $\omega_p(p) = \omega_s(s)$ and $\omega_p'(p)/\omega_p(p) = \omega_s'(s)/\omega_s(s)$, the two marginal covariates have the same fraction learned ℓ_x and the same rate r_x, and $B_p - B_s = \left[2\bar{\ell} + 2(1-\bar{\ell}_s)\right]/(1+\alpha_s) > 0$; so if $\partial^2\mathcal{V}/\partial s^2 > 0$, the right side of (33) is positive for $x = s$, hence for $x = p$, and $\partial^2\mathcal{V}/\partial p^2 > 0$.

Lemma 8 (Scope in small and large samples)**.**

(i) As $n \to 0$, $r_x \to 0$ and B_x stays bounded for $x = s, p$, and $\partial\mathcal{V}/\partial p \to 0$.

(ii) As $n \to \infty$, r_x stays bounded and $B_x \to 0$, while $\partial\mathcal{V}/\partial p \to 0$ and $\partial\mathcal{V}/\partial s \to \omega_s(s)$.

Proof. Recall that $E^\star \in [\underline{E}, 1]$, with $\underline{E} \equiv \underline{E}(s,p) > 0$, and $\omega \le \bar{\omega}$. The averages $\langle\cdot\rangle$ and $\langle\cdot\rangle_s$ of quantities in $[0,1]$ lie in $[0,1]$, $\bar{\ell} = \alpha\langle 1-\ell\rangle \le \alpha$ and $\alpha_s \le \alpha$, so every term of B_x is nonnegative and $B_x \le 5 + 2\alpha$.

(i). As $n \to 0$, every fraction learned is at most $n\bar{\omega}/\underline{E}$, so $\omega_x(x)\ell_x \leq n\bar{\omega}^2/\underline{E}$ and $\Psi'(E^\star; n, s, p) \leq n(s+p)\bar{\omega}^2/\underline{E}^2$. Hence $M \to 1$, $\alpha \to 0$, $r_x \to 0$, $B_x \leq 5 + 2\alpha$ stays bounded, and $\partial\mathcal{V}/\partial p = \alpha_s\omega_p(p)\ell_p \to 0$.

(ii). As $n \to \infty$, $\tau^\star = n/E^\star \geq n \to \infty$, so $1 - \ell_x = 1/(1 + \tau^\star\omega_x(x)) \leq 1/(n\omega_x(x)) \to 0$, because $\omega_x(x) > 0$, and $\ell \to 1$ wherever $\omega > 0$, so $1 - \bar{\ell}_s = \langle 1 - \ell \rangle_s \to 0$ by dominated convergence, because the profiles are positive. Since $\Psi'(E^\star; n, s, p) \leq (s+p)/n$, $M \to 1$, and α, α_s and $\bar{\ell}$ tend to zero. Hence $B_x \to 0$, because the term $(1-\ell_s)/(1+\alpha_s)$ of B_s is at most $1-\ell_s$, while $r_x = M\omega_x(x)\ell_x/E^\star \leq M\bar{\omega}/\underline{E}$ stays bounded. Finally, $\partial\mathcal{V}/\partial p = \alpha_s\omega_p(p)\ell_p \leq \alpha\bar{\omega} \to 0$ and $\partial\mathcal{V}/\partial s = \omega_s(s)\ell_s(1+\alpha_s) \to \omega_s(s)$. □

Proof of Corollaries 4(i)–(ii), OA.2(iii)–(iv) and OA.3. Let x be p or s. Corollary OA.3 is the comparison that follows Equation (33). Since $\partial\mathcal{V}/\partial x > 0$, $\partial^2\mathcal{V}/\partial x^2$ has the sign of the right side of Equation (33), in which $r_x B_x > 0$.

Signal that does not decline. If $\omega_x'(x) \geq 0$, the first term is nonnegative, so $\partial^2\mathcal{V}/\partial x^2 > 0$.

Declining signal. If $\omega_x'(x) < 0$, the first term is at most $\omega_x'(x)/\omega_x(x) < 0$, because $2-\ell_x \geq 1$, while $r_x B_x \to 0$ as $n \to 0$ and as $n \to \infty$ (Lemma 8). Hence $\partial^2\mathcal{V}/\partial x^2 < 0$ at all sufficiently small and all sufficiently large n. □

A group of each kind. Let $\mathcal{S}_2$ and $\mathcal{P}_2$ be groups of prediction and post-prediction covariates outside the sample, each a finite union of intervals of positive length. Order the covariates of each kind so that these groups come last, at the positions $(s_1, s_2]$ and $(p_1, p_2]$. Value is continuous in scope and post-prediction scope, and its partial derivatives exist except at the finitely many positions at which a profile jumps, so the value that $\mathcal{S}_2$ adds when the firm also holds $\mathcal{P}_2$ exceeds the value it adds without $\mathcal{P}_2$ by

$$\int_{s_1}^{s_2} \left[\frac{\partial\mathcal{V}}{\partial s}(n, u, p_2) - \frac{\partial\mathcal{V}}{\partial s}(n, u, p_1) \right] du = \int_{s_1}^{s_2} \int_{p_1}^{p_2} \frac{\partial^2\mathcal{V}}{\partial s\, \partial p}(n, u, v)\, dv\, du > 0$$

by Proposition 9(ii). The two groups are therefore complements at every scale.

References

Acemoğlu, D., A. Makhdoumi, A. Malekian, and A. Özdağlar (2022). "Too Much Data: Prices and Inefficiencies in Data Markets." *American Economic Journal: Microeconomics* 14 (4): 218–56.

Aridor, G., Y.-K. Che, and T. Salz (2023). "The Effect of Privacy Regulation on the Data Industry: Empirical Evidence from GDPR." *RAND Journal of Economics* 54 (4): 695–730.

Arrieta-Ibarra, I., L. Goff, D. Jiménez-Hernández, J. Lanier, and E. G. Weyl (2018). "Should We Treat Data as Labor? Moving beyond "Free"." *AEA Papers and Proceedings* 108: 38–42.

Ausubel, L. M. and P. Milgrom (2006). "The Lovely but Lonely Vickrey Auction." In *Combinatorial Auctions*. Ed. by P. Cramton, Y. Shoham, and R. Steinberg. Cambridge, MA: MIT Press, 17–40.

Bahri, Y., E. Dyer, J. Kaplan, J. Lee, and U. Sharma (2024). "Explaining Neural Scaling Laws." *Proceedings of the National Academy of Sciences* 121 (27): e2311878121.

Bajari, P., V. Chernozhukov, A. Hortaçsu, and J. Suzuki (2019). "The Impact of Big Data on Firm Performance: An Empirical Investigation." *AEA Papers and Proceedings* 109: 33–37.

Baumol, W. J., J. C. Panzar, and R. D. Willig (1982). *Contestable Markets and the Theory of Industry Structure*. New York: Harcourt Brace Jovanovich.

Bergemann, D., A. Bonatti, and T. Gan (2022). "The Economics of Social Data." *RAND Journal of Economics* 53 (2): 263–96.

Bergemann, D., A. Bonatti, and A. Smolin (2018). "The Design and Price of Information." *American Economic Review* 108 (1): 1–48.

Bordelon, B., A. Atanasov, and C. Pehlevan (2024). "A Dynamical Model of Neural Scaling Laws." *International Conference on Machine Learning*, 4345–82.

Börgers, T., Á. Hernando-Veciana, and D. Krähmer (2013). "When Are Signals Complements or Substitutes?" *Journal of Economic Theory* 148 (1): 165–95.

Bria (2025). *Bria Secures $40M in Series B to Drive Fair Gen AI Usage for Enterprises*. Press release. URL: https://www.intelcapital.com/bria-secures-40m-in-series-b-to-drive-fair-gen-ai-usage-for-enterprises/.

Buchanan, J. M. and Y. J. Yoon (2000). "Symmetric Tragedies: Commons and Anticommons." *Journal of Law and Economics* 43 (1): 1–13.

Campbell, J., A. Goldfarb, and C. Tucker (2015). "Privacy Regulation and Market Structure." *Journal of Economics & Management Strategy* 24 (1): 47–73.

Carballa-Smichowski, B., N. Duch-Brown, S. Höcük, P. Kumar, B. Martens, J. Mulder, and P. Prüfer (2025). "Economies of Scope in Data Aggregation: Evidence from Health Data." *Information Economics and Policy* 71: 101146.

Chetty, R. (2009). "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics* 1: 451–88.

Choi, J. P., D.-S. Jeon, and B.-C. Kim (2019). "Privacy and Personal Data Collection with Information Externalities." *Journal of Public Economics* 173: 113–24.

Coase, R. H. (1960). "The Problem of Social Cost." *Journal of Law and Economics* 3: 1–44.

Collard-Wexler, A., G. Gowrisankaran, and R. S. Lee (2019). ""Nash-in-Nash" Bargaining: A Microfoundation for Applied Work." *Journal of Political Economy* 127 (1): 163–95.

Competition and Markets Authority (2020). *Online Platforms and Digital Advertising Market Study, Appendix I: Search Quality and Economies of Scale*. Tech. rep. London: Competition and Markets Authority.

Cournot, A. A. (1838). *Recherches sur les principes mathématiques de la théorie des richesses*. Paris: Hachette.

Crémer, J., Y.-A. de Montjoye, and H. Schweitzer (2019). *Competition Policy for the Digital Era*. Final report. European Commission, Directorate-General for Competition.

Dasaratha, K., J. Ortner, and C. Zhu (2025). *Markets for Models*. arXiv:2503.02946.

de Cornière, A. and G. Taylor (2025). "Data and Competition: A Simple Framework." *RAND Journal of Economics* 56 (4): 494–510.

Digital Competition Expert Panel (2019). *Unlocking Digital Competition*. Report of the Digital Competition Expert Panel, chaired by Jason Furman. London: HM Treasury.

Dobriban, E. and S. Wager (2018). "High-Dimensional Asymptotics of Prediction: Ridge Regression and Classification." *Annals of Statistics* 46 (1): 247–79.

Dosis, A. and W. Sand-Zantman (2023). "The Ownership of Data." *Journal of Law, Economics, and Organization* 39 (3): 615–41.

European Commission (2020). *A European Strategy for Data*. Communication COM(2020) 66 final. European Commission.

European Commission (2022). *Data Act: Commission Proposes Measures for a Fair and Innovative Data Economy*. Press release IP/22/1113. 23 February 2022.

European Parliament and Council (2016). *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data (General Data Protection Regulation)*. Official Journal of the European Union, L 119, 1–88.

European Parliament and Council (2023). *Regulation (EU) 2023/2854 on Harmonised Rules on Fair Access to and Use of Data (Data Act)*. Official Journal of the European Union, L 2023/2854.

European Union (2022). *Digital Markets Act, Regulation (EU) 2022/1925*. Official Journal of the European Union, L 265.

Farboodi, M. and L. Veldkamp (2026). "A Model of the Data Economy." *Review of Economic Studies*. Forthcoming.

Ghorbani, A. and J. Zou (2019). "Data Shapley: Equitable Valuation of Data for Machine Learning." *Proceedings of the 36th International Conference on Machine Learning*. Ed. by K. Chaudhuri and R. Salakhutdinov. Vol. 97. Proceedings of Machine Learning Research. PMLR, 2242–51.

Grossman, S. J. and O. D. Hart (1986). "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration." *Journal of Political Economy* 94 (4): 691–719.

Gu, Y., L. Madio, and C. Reggiani (2022). "Data Brokers Co-opetition." *Oxford Economic Papers* 74 (3): 820–39.

Hagiu, A. and J. Wright (2023). "Data-Enabled Learning, Network Effects, and Competitive Advantage." *RAND Journal of Economics* 54 (4): 638–67.

Hart, O. and J. Moore (1990). "Property Rights and the Nature of the Firm." *Journal of Political Economy* 98 (6): 1119–58.

Heller, M. A. (1998). "The Tragedy of the Anticommons: Property in the Transition from Marx to Markets." *Harvard Law Review* 111 (3): 621–88.

Hestness, J. et al. (2017). "Deep Learning Scaling is Predictable, Empirically." arXiv:1712.00409.

Horn, H. and A. Wolinsky (1988). "Bilateral Monopolies and Incentives for Merger." *RAND Journal of Economics* 19 (3): 408–19.

Ichihashi, S. (2021). "The Economics of Data Externalities." *Journal of Economic Theory* 196: 105316.

Johnson, G. A., S. K. Shriver, and S. G. Goldberg (2023). "Privacy and Market Concentration: Intended and Unintended Consequences of the GDPR." *Management Science* 69 (10): 5695–721.

Jones, C. I. and C. Tonetti (2020). "Nonrivalry and the Economics of Data." *American Economic Review* 110 (9): 2819–58.

Lemley, M. A. and C. Shapiro (2007). "Patent Holdup and Royalty Stacking." *Texas Law Review* 85: 1991–2049.

Lerner, J. and J. Tirole (2004). "Efficient Patent Pools." *American Economic Review* 94 (3): 691–711.

Liang, A. and X. Mu (2020). "Complementary Information and Learning Traps." *Quarterly Journal of Economics* 135 (1): 389–448.

Llanes, G. and S. Trento (2011). "Anticommons and Optimal Patent Policy in a Model of Sequential Innovation." *The B.E. Journal of Economic Analysis & Policy* 11 (1).

Menezes, F. and R. Pitchford (2004). "A Model of Seller Holdout." *Economic Theory* 24 (2): 231–53.

Mézard, M., G. Parisi, and M. A. Virasoro (1987). *Spin Glass Theory and Beyond*. Singapore: World Scientific.

Montiel Olea, J. L., P. Ortoleva, M. M. Pai, and A. Prat (2022). "Competing Models." *Quarterly Journal of Economics* 137 (4): 2419–57.

Moscarini, G. and L. Smith (2002). "The Law of Large Demand for Information." *Econometrica* 70 (6): 2351–66.

Prüfer, J. and C. Schottmüller (2021). "Competing with Big Data." *Journal of Industrial Economics* 69 (4): 967–1008.

Radner, R. and J. E. Stiglitz (1984). "A Nonconcavity in the Value of Information." In *Bayesian Models in Economic Theory*. Ed. by M. Boyer and R. E. Kihlstrom. Amsterdam: North-Holland, 33–52.

Rhodes, A., J. Zhou, and J. Zhou (2025). *Digital Ecosystems and Data Regulation*. Working Paper 25-1621. Toulouse School of Economics.

Ricardo, D. (1817). *On the Principles of Political Economy and Taxation*. London: John Murray.

Salazar, D. (2024). "This New AI Answer Engine Plans to Pay Media Companies for Their Content." *Fast Company*.

Schäfer, M. (2025). "When Should We Expect Non-Decreasing Returns from Data in Prediction Tasks?" arXiv:2503.03602.

Schäfer, M. and G. Sápi (2023). "Complementarities in Learning from Data: Insights from General Search." *Information Economics and Policy* 65: 101063.

Shapley, L. S. (1971). "Cores of Convex Games." *International Journal of Game Theory* 1 (1): 11–26.

Silverstein, J. W. and Z. Bai (1995). "On the Empirical Distribution of Eigenvalues of a Class of Large Dimensional Random Matrices." *Journal of Multivariate Analysis* 54 (2): 175–92.

Stigler Committee on Digital Platforms (2019). *Final Report*. George J. Stigler Center for the Study of the Economy and the State, University of Chicago Booth School of Business.

United States v. Google LLC (2024). *Memorandum Opinion*. 747 F. Supp. 3d 1 (D.D.C. 2024).

United States v. Google LLC (2025a). *Final Judgment*. No. 20-cv-3010 (APM) (D.D.C. Dec. 5, 2025).

United States v. Google LLC (2025b). *Memorandum Opinion on Remedies*. No. 20-cv-3010 (APM) (D.D.C. Sept. 2, 2025).

Valavi, E., J. Hestness, N. Ardalani, and M. Iansiti (2022). *Time and the Value of Data*. arXiv:2203.09118.

Vapnik, V. and A. Vashist (2009). "A New Learning Paradigm: Learning Using Privileged Information." *Neural Networks* 22 (5–6): 544–57.

Varian, H. (2019). "Artificial Intelligence, Economics, and Industrial Organization." In *The Economics of Artificial Intelligence: An Agenda*. Ed. by A. Agrawal, J. Gans, and A. Goldfarb. Chicago: University of Chicago Press, 399–419.

Wicksteed, P. H. (1894). *An Essay on the Co-ordination of the Laws of Distribution*. London: Macmillan.